

User manual

ToolCommander® – MueTec

Version 4.17.3

Confidential

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Version History

1 Version History

Beginning with current version of the document the history contains statements to all modifications.

Version	Release	Author	Date	Modification
0.8	12/14/2018	P. Herrmann	12/14/2018	• First version
0.9		T. Köhler	12/20/2018	• Add missing components
0.10		T. Köhler	01/14/2019	• Correct spellings and wrong descriptions • Rename "Transport Module" in "Transfer Module" • Correct Chapters ↗ Aligner Messages, ↗ Transfer Module Messages.
1.0		T. Köhler	01/26/2019	• add Chapters ↗ Installation, ↗ User Groups, ↗ User Rights. • new right "Auto Initialize After Login"
1.1		T.Köhler	07/05/2019	• add Chapters ↗ Operator GUI, ↗ Configuration. • delete obsolete chapters • correct chapters ↗ Administrator GU, ↗ Load Port Module State, ↗ Production Dialog • Update changes on components Loadport, Job Creator Dialog, Machine, Hostinterface, Transfer Module
1.2		T. Köhler	09/25/2019	Update for Rembrandt II • update parameter in chapter ↗ Configuration, ↗ Loadport Parameters

Version	Release	Author	Date	Modification
				add new chapter ➔ Transfer Module Initialization
1.3.		T.Köhler	04/09/2020	new chapter ➔ Transfermodul Wafer Absense Error
1.4		T. Köhler	04/30/2020	update chapter ➔ Transfermodul Wafer Absense Error
1.5		T. Köhler	08/20/2020	update parameter in chapter ➔ Configuration,
4.0		H.Hoffmann	04/15/2021	update to software version 4.0
4.0.0.30		T. Kaiber	09/20/2021	update to software version 4.0.0.30
4.0.0.32		H.Hoffmann	10/21/2021	added Parameter "UnloadDelay" in chapter ➔ Configuration
4.0.0.43		T. Köhler	03/17/2022	add description of parameter "SpeedReduction" in chapter ➔ Station configuration
4.0.0.46		T.Kaiber	07/13/2022	add new reader configurations in machine parameter ➔ Machine parameters
4.1.0.0		T. Köhler	09/12/2022	adaption for Spector 5500: - update transfer module parameter ➔ Transfer Module Parameter - add description of alignment correction and camera aligner parameter in chapter ➔ Station configuration - add description of new parameter in ➔ MueTecApplication.ini

Version	Release	Author	Date	Modification
				- add new components ➔ Camera Aligner and ➔ Flipper - adapt chapter ➔ Installation because of modified installer sequence correction - remove no more existing Read Mode from ➔ ID Reader Parameters - add description of Wafer ID Reader Read Mode to ➔ Machine Parameters
4.2.0.0		H. Hoffmann	11/04/2022	- add description of new parameter in ➔ MueTecApplication.ini - removed parameters in ➔ MueTecApplication.ini - update chapter ➔ Job Creator Dialog
4.2.3.0		H. Hoffmann	11/30/2022	described E84 timeout Alarms for Fortrend Loadports ➔ Device Generated Alarms
4.2.3.0		P. Hunt	12/01/2022	adaptation for Argos 150 HT: - description of the ➔ Machine structure - Extension with tool configuration specific visualizations and values in ➔ Loadport Description, ➔ Loadport Mapping, ➔ Loadport Parameters, ➔ Loadport Alarms - Extension with tool configuration specific visualizations and values in ➔ Transfer Module Description, ➔ Transfer Module Initialization, ➔ Transfer Module Parameters, ➔ Transfer Module Manual, ➔ Transfer Module Alarms

Version	Release	Author	Date	Modification
				Extension with tool configuration specific visualizations and values in Camera Aligner Description , Camera Aligner Alarms
4.2.4.0		H. Hoffmann	12/07/2022	add value "SingleMask" to parameter CassetteTypeName General
4.6.0.0		T.Kaiber	13/02/2024	- add Parameter "UseWafer200ReadingPosition" and "TuningParameterForExternalSensor" in chapter Configuration - Adaptation of the description for "Reading Angle 200mm" and "Reading Angle 200mm" Machine parameters
4.17.0.0		T.Kaiber	14/08/2025	- described Hermos LFM4x parameters - Added setting for monitoring the protrusion sensor for load ports in the configuration

Version: Version of this document formatted <major_version>.<minor_version>

Release: Release date of the version of the document - empty if the version is not released.

Author: If there are several authors, the ones are named here that has inserted the modifications of a version into the document. If there is only one editor the field can remain free.

Date: Date of the modification.

Modification: Short description of the all modifications for this release.

2

Introduction

2 Introduction

2.1 General

This manual describes the control and visualization software, developed by MueTec.

2.2 Machine Structure

The machine consists of the following modules:

- up to 3 Load Port Modules (configurable, depending on the tool configuration)
- up to 2 Transfer Modules with Aligner (Aligner is only for tool configuration **Rembrandt/DaVinci** available, 2 Transfer Modules only for tool configuration **Argos 150 HT**)
- 1 Aligner (configurable, depending on the tool configuration)
- up to 2 Process Module (configurable, depending on the tool configuration)
- up to 2 Camera Aligners (configurable, 2 Camera Aligners only for tool configuration **Argos 150 HT**)
- Alignment Correction Station and Flipper (only for tool configuration **Spector 5500**)

3

Referenced Standards and Documents

3 Referenced Standards and Documents

3.1 ToolCommander Documents

- ToolCommander® User Manual

3.2 Applied Standards

Standard number	Standard name	Comment
E5-0707	SEMI Equipment communications standard 2 message content (SECS-II)	part of FabLink
E10-0304E	Specification for definition and measurement of equipment reliability, availability, and maintainability (RAM)	part of μ TFM and project implementation
E30-0307	Generic model for communications and control of manufacturing equipment (GEM)	part of FabLink
E37-0303	High-speed SECS message services (HSMS) Generic services	part of FabLink
E39-0703	Object services standard: concepts, behavior, and services	part of FabLink
E40-0307	Standard for processing management	part of FabLink and ToolCommander
E87-0707	Specification for carrier management (CMS)	part of ToolCommander and project implementation
E90-0707	Specification of substrate tracking	part of ToolCommander and project implementation
E94-1106	Specification for control job management	part of ToolCommander and FabLink

E95-1101 Reapproved 0307	Specification for human interface for semiconductor manufacturing equipment	part of ToolCommander and project implementation
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Table 1 – Applied SEMI Standards

3.3 Abbreviations

Abbreviation	Meaning
AMHS	Automatic Material Handling System
EPT	Equipment Performance Tracking (SEMI E116)
FFU	Fan Filter Unit
FOUP	Front Opening Unified Pod
GB	Giga Byte
GHz	Giga Hertz
GUI	Graphical User Interface
HDD	Hard Disk Drive
I/ O	Input/ Output
LED	Light Emitting Diode
LP	Load Port
LPM	Load Port Module
MB	Mega Byte
OEE	Over-all Equipment Efficiency
PDO	Pod Door Opener

PLC	Programmable Logic Controller.
PM	Process Module
PMC	Process Module Controller
RAM	Reliability, Availability and Maintainability (SEMI E10)
SECS	SEMI Equipment Communications Standard
SEMI®	Semiconductor Equipment and Materials International
SQL	Structured Query Language
TC	ToolCommander® (software product of AIS Automation Dresden GmbH)
TM	Transfer Module
TMC	Transfer Module Controller
XML	Extensible Markup Language

Table 2 – Abbreviations

4

Software Installation

4 Software Installation

4.1 Prerequisites

SQL-Server Database

The installer assumes the presence of an installed SQLServer 2014 or SQLServer Express 2014. This is checked and the installer cancels the installation if necessary. The SQLServer requires the .Net 3.5 Framework.

.Net 3.5

Although .Net 3.5 is available in Windows10 under Features and can be selected, it may not install. There is a workaround for this:

- The Windows10 installation files in the correct version (DVD, ISO image) are required.
- Open the command prompt (administrator mode) via Start -> right mouse button



- Enter the following line ("Drive" must be replaced by the drive letter of the Windows installation files):

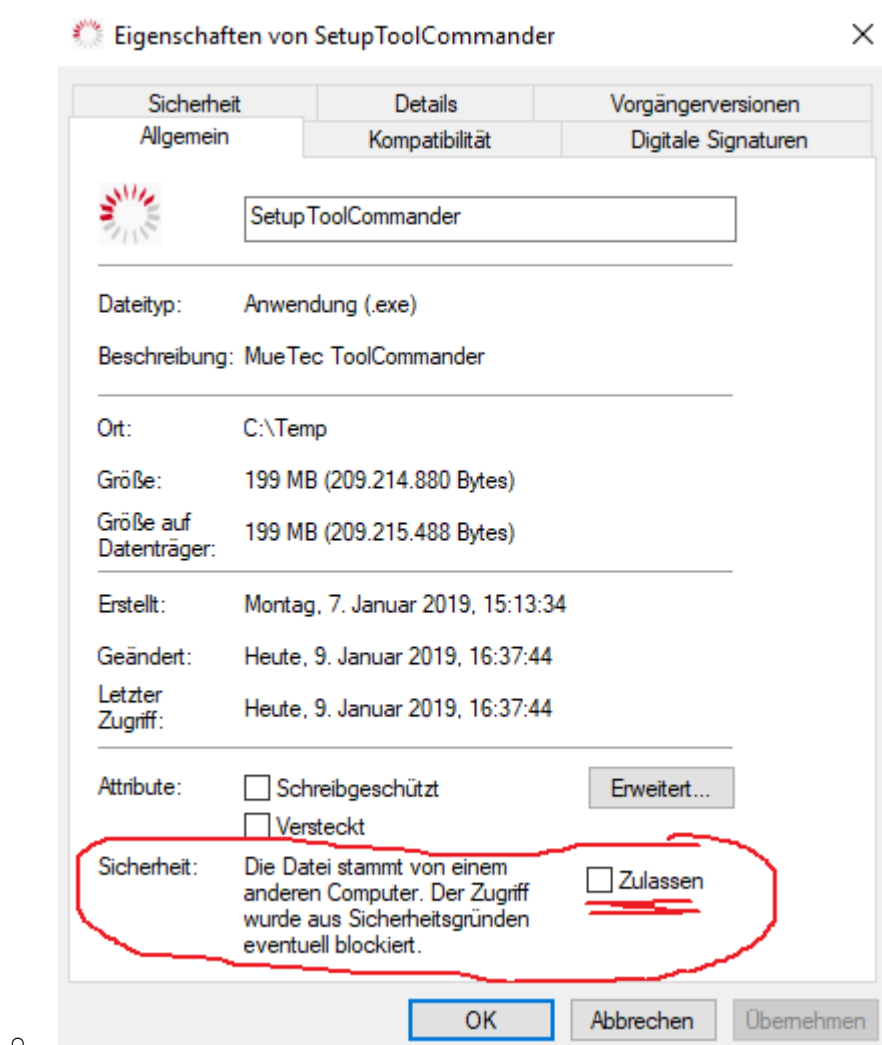
```
dism /online /enable-Feature /Featurename:NetFx3 /All /Source:Drive:  
\sources\sxs /Limitaccess
```

- after that the installation of .Net Framework 3.5 for Windows 10 will take place.

Windows Defender

If the installer was downloaded from the Internet (e.g. FTP), the file is automatically marked with a flag. The Windows Defender SmartScreen will then try to verify whether the application comes from a trustworthy source. There are two options for avoiding this:

- Disable Defender SmartScreen in Windows
- Removing the marker from the file after download
 - Right mouse button -> Properties -> General -> Security: Click on the "Allow" check mark.



4.2 Installation

Settings

Description	Path
Installation path	C:\AIS
File path database	C:\AIS\ToolCommander\Database\Data
Backup path	C:\AIS\Backup

Table 3 – Predefined paths

Installation of required software packages

Before installing the control software, the system checks whether the required software packages have already been installed on the computer. If this is not the case, the software is installed independently. Here is a list of the corresponding packages:

- Microsoft Visual Redistributable 2012
- Microsoft Visual Redistributable 2013
- Microsoft .Net Framework 4.6.1
- SQL Express CE Datenbank
- CodeMeterRuntime64 LizenzDongel Software

The default settings are simply used during installation.

Installation of the control software

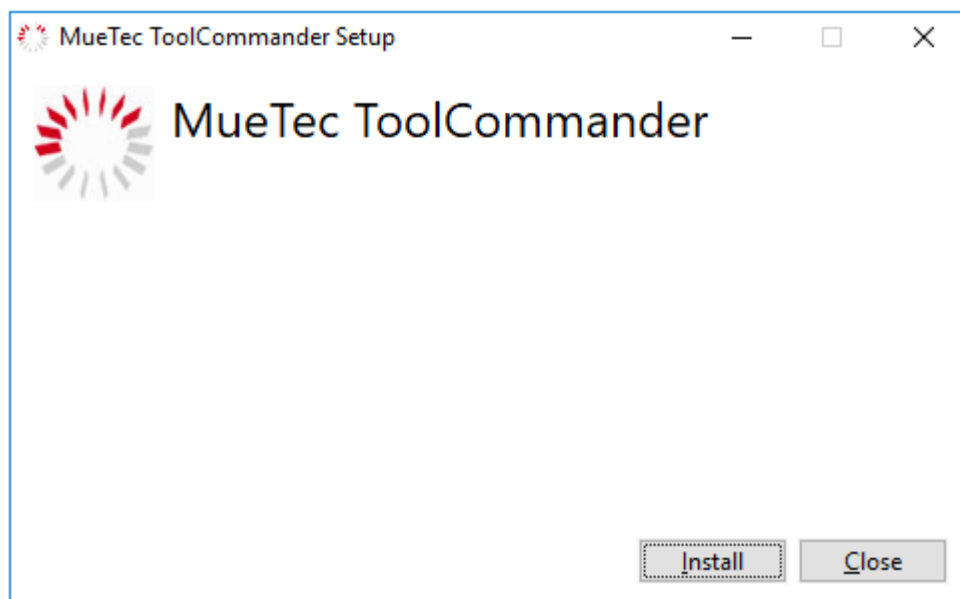


Figure 1 – Installer Start Window

The installer offers two installation modes - **Update** and **New**.

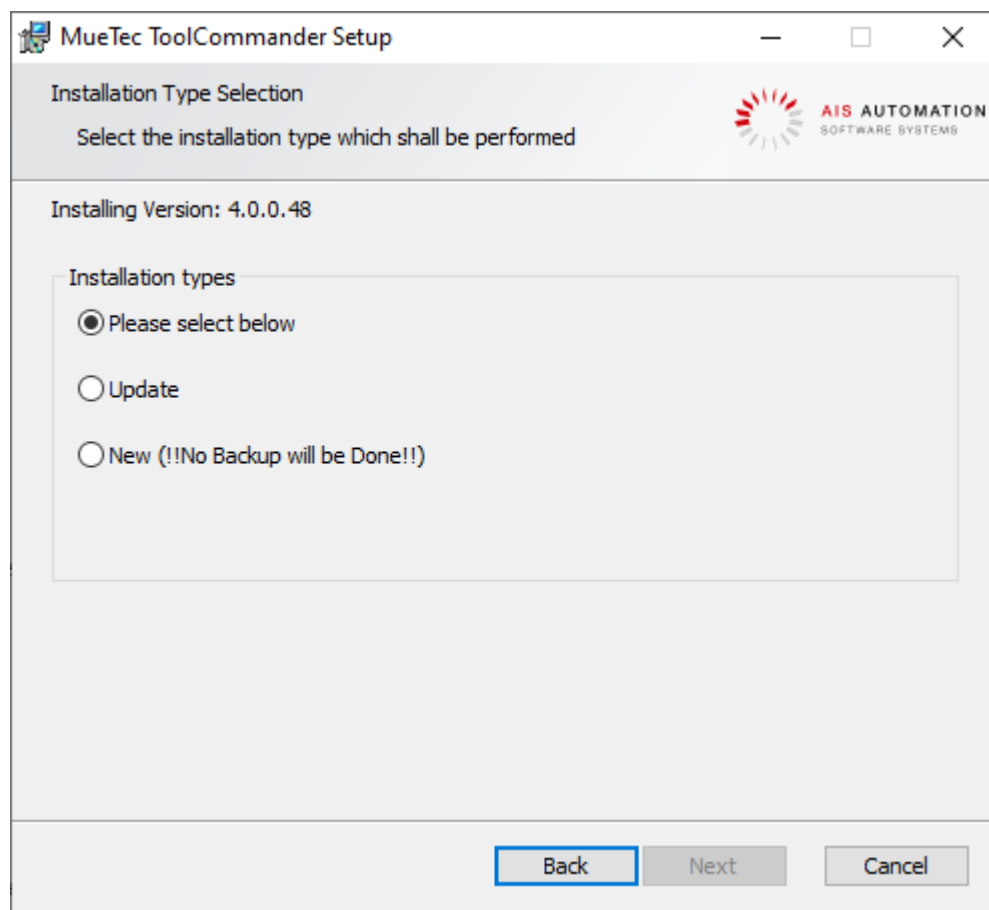


Figure 2 – Installer Mode Selection

The following is an overview of the differences between the two modes:

	Update	New-Installation
Backup software and database	yes	no
Initial recovery of the database	no	yes
Installing the Configuration Files	no	yes
Configuring the database in configuration files	no	yes
Writing the Recipe Template	no	yes
Overwriting the Configuration for the Application Starter	no	yes

Table 4 – Differences Update and New-Installation

In the installation mode **New** the connection to the database is configured. In installation mode **Update** the connection settings to the database will be shown and can be corrected:

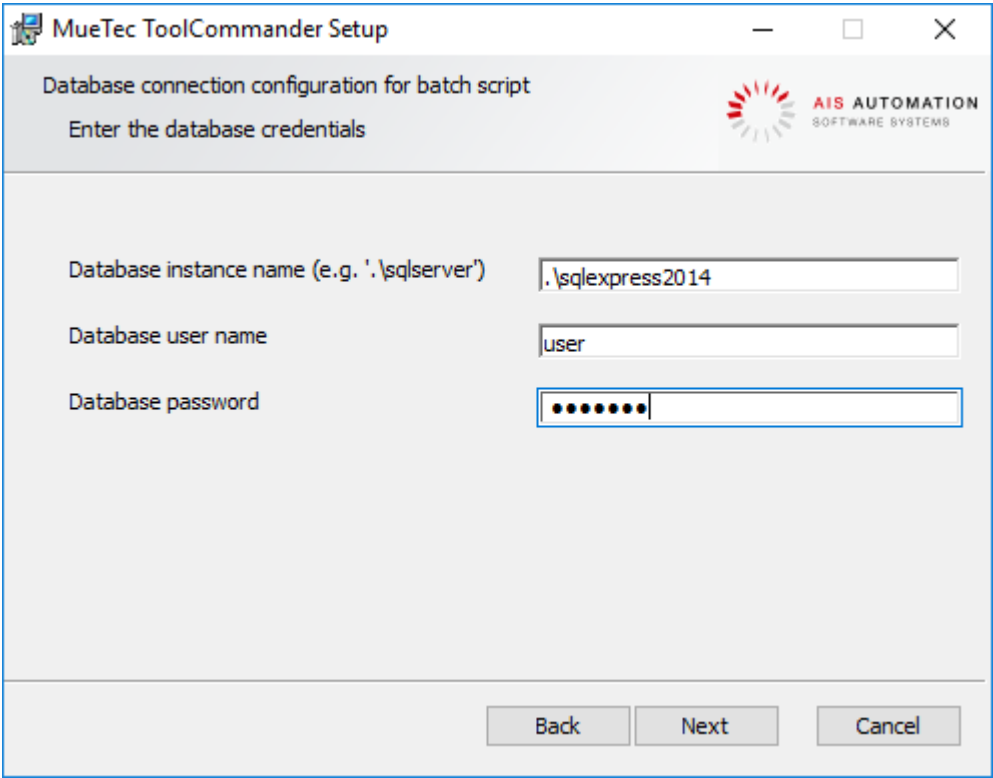


Figure 3 – Installer Configuration Database Connection

The **New** installation mode is described in the section "Installation mode - New".

Before the **Update**, the selected installation type and the new version are displayed



Figure 4 – Installer Summary

During the installation of the software, the state of the installation is displayed:

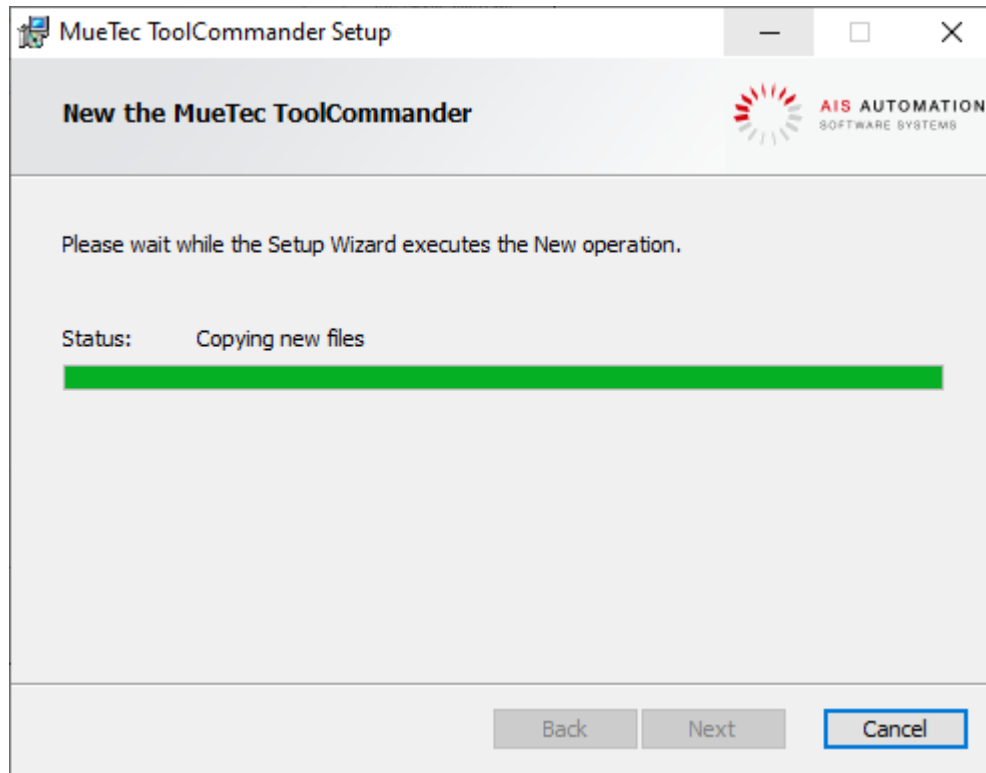


Figure 5 – Installer State

At the end of the software update, the System Configurator is started and the summary of the current system configuration is displayed.

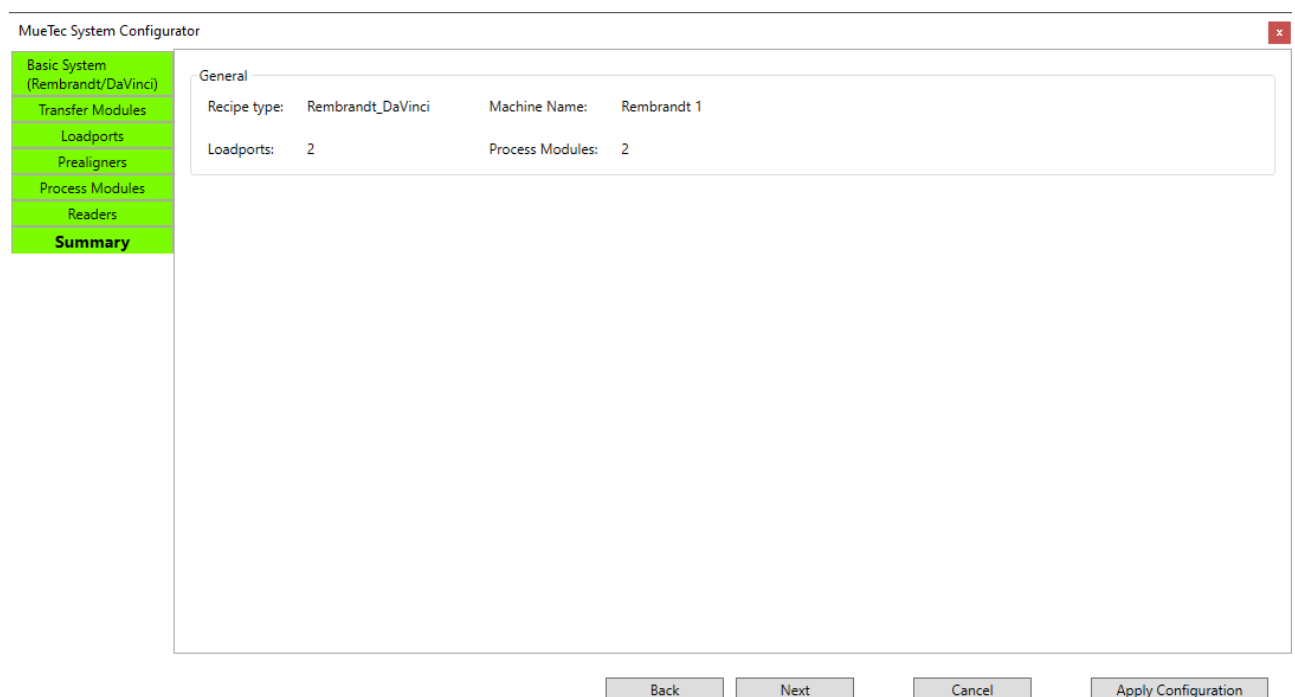


Figure 6 – System Configurator - Summary

If the configuration is confirmed with "**Apply Configuration**", the database update starts. After that the installation is finished.

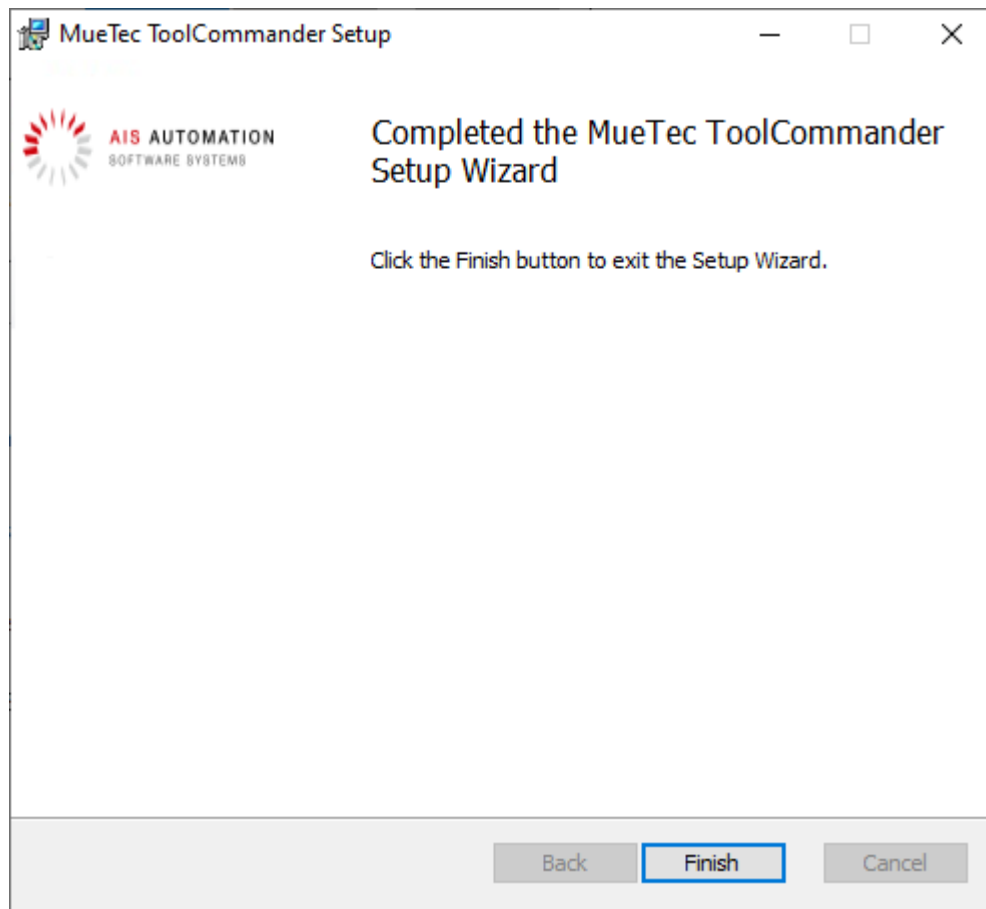


Figure 7 – Installation mode Update - Completed

Installation mode New

In **New** installation mode, the connection to the database is configured first.

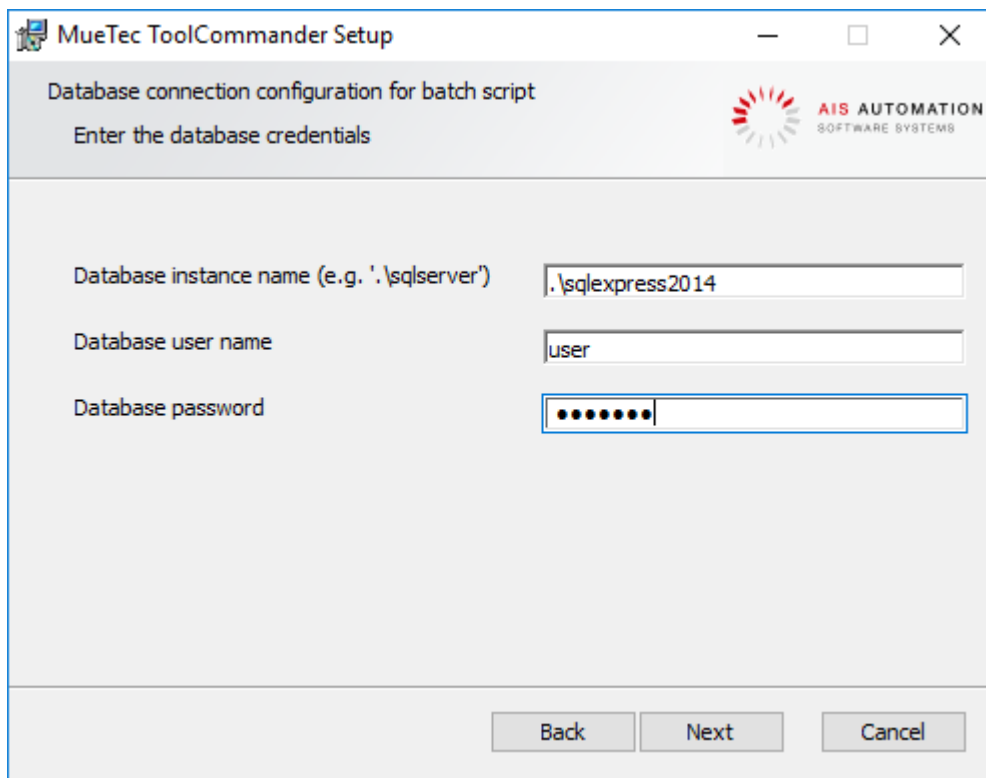


Figure 8 – Installer Configuration database connection

Afterwards, the installation still needs to be confirmed.

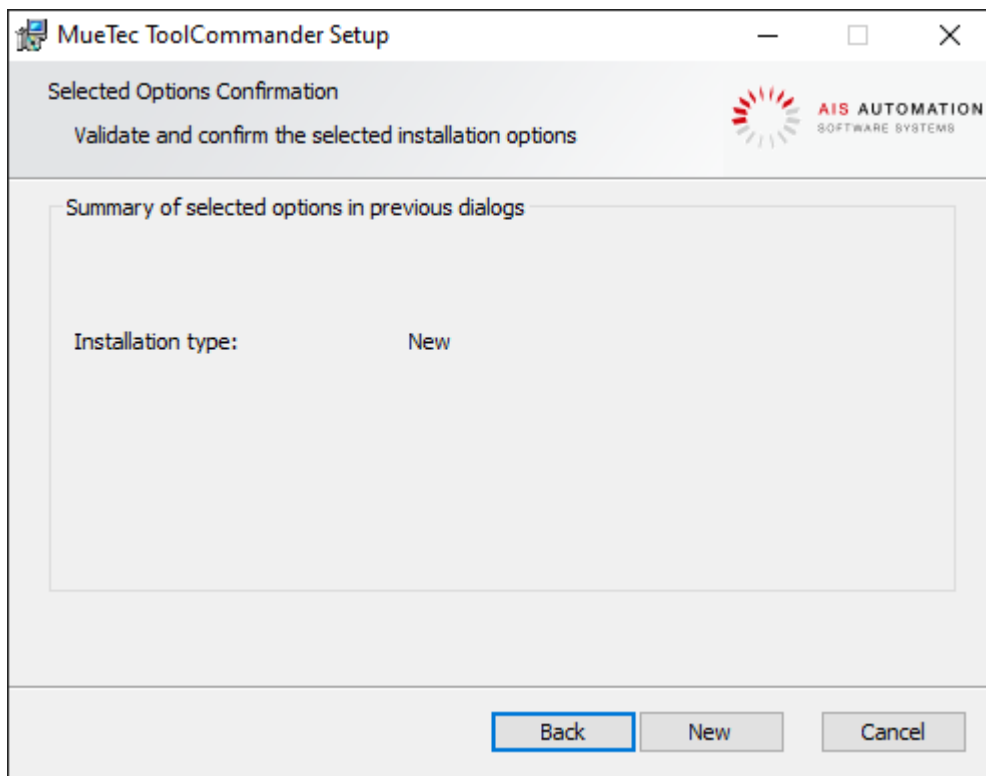
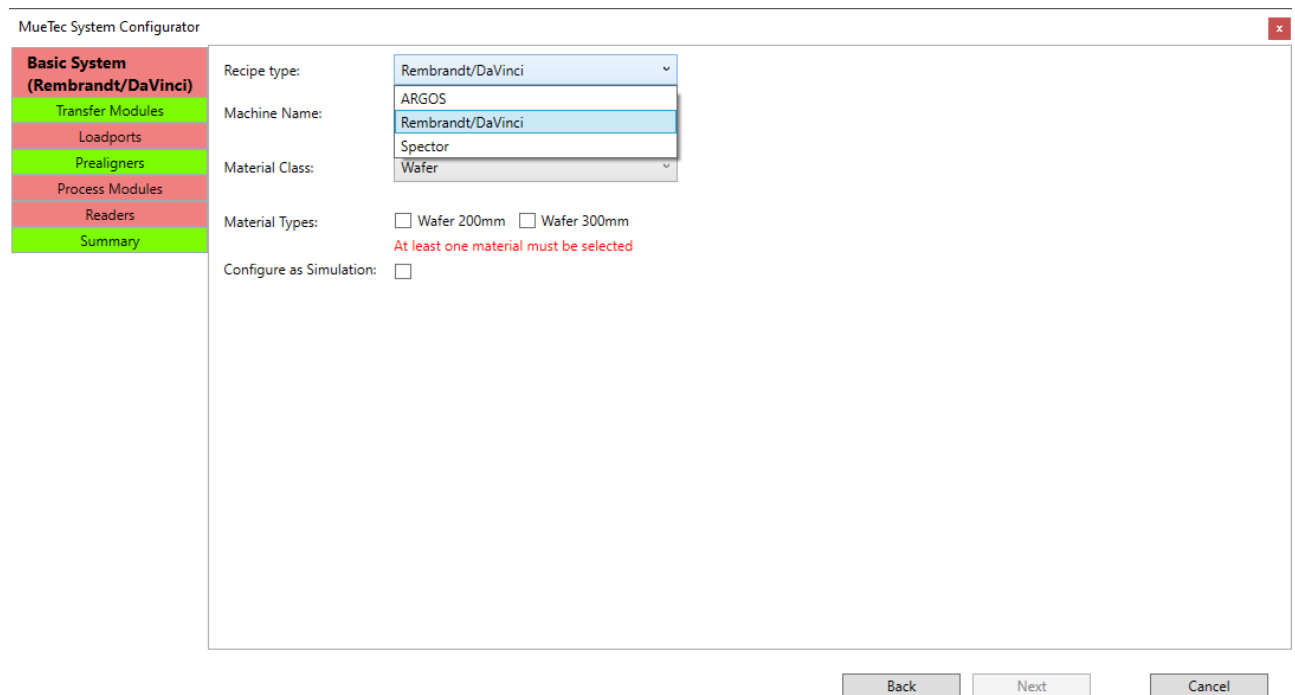


Figure 9 – Installation mode New - Confirmation

During the installation of the software, the system configurator is started to configure the machine.



The screenshot shows the 'MueTec System Configurator' window. On the left is a sidebar with a tree view containing the following items: 'Basic System (Rembrandt/DaVinci)' (highlighted in red), 'Transfer Modules' (green), 'Loadports' (green), 'Prealigners' (green), 'Process Modules' (red), 'Readers' (red), and 'Summary' (green). The main area displays configuration options for the selected 'Basic System'.

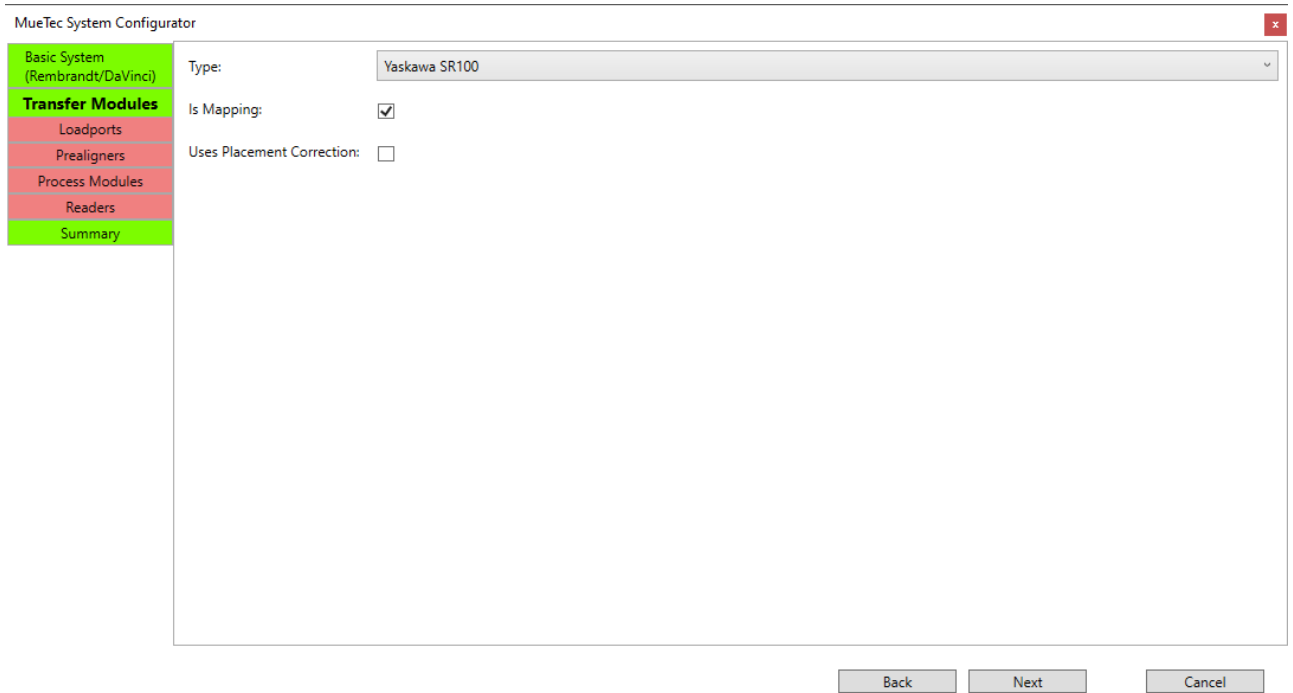
Recipe type:	Rembrandt/DaVinci
Machine Name:	ARGOS
	Rembrandt/DaVinci
	Spector
Material Class:	Wafer
Material Types:	☐ Wafer 200mm ☐ Wafer 300mm
	At least one material must be selected
Configure as Simulation:	☐

At the bottom right of the window are three buttons: 'Back', 'Next', and 'Cancel'.

Figure 10 – System Configurator - Basic system configuration

The Recipe type (**ARGOS**, **Rembrandt/DaVinci** or **Spector**) defines the tool configuration. The tool configuration defines the type of inspection and the **Rembrandt/DaVinci** tool configuration features an aligner. The **Spector** machine type has a camera aligner and an alignment correction station.

After setting the basic system, the transfer modules can be configured.



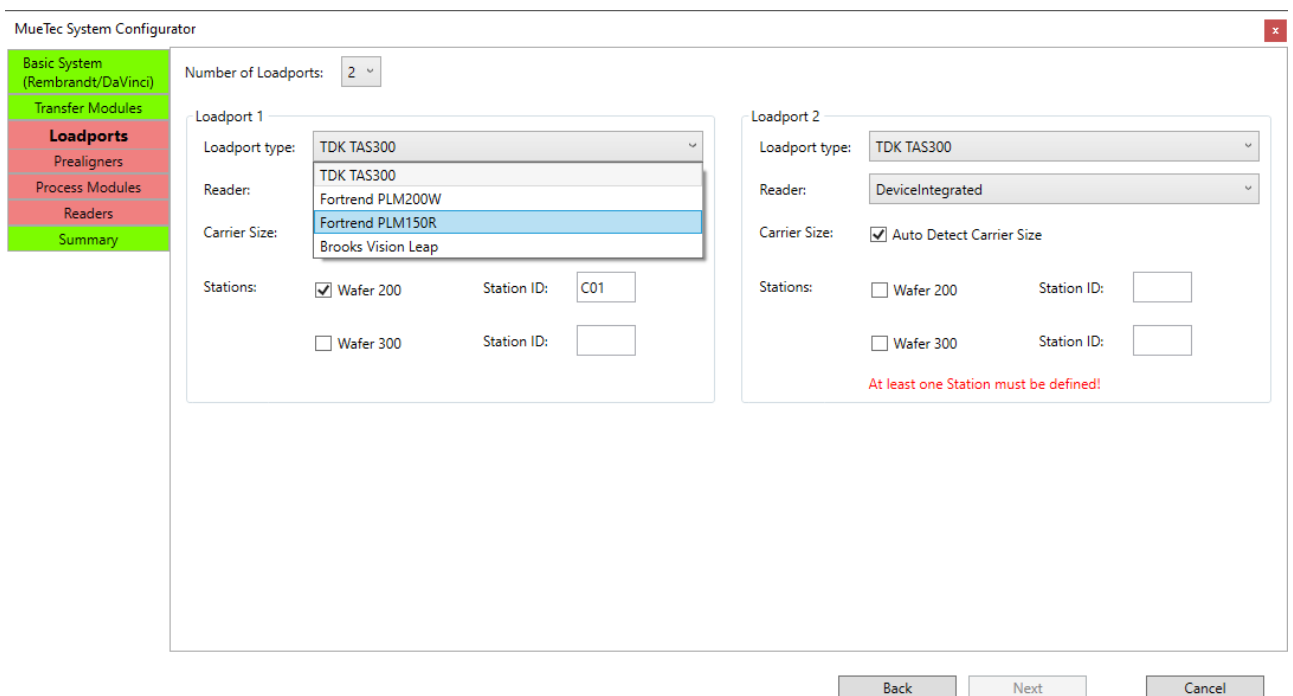
The screenshot shows the 'MueTec System Configurator' window with the 'Transfer Modules' tab selected in the left sidebar. The main area contains the following fields:

- Type: Yaskawa SR100 (dropdown menu)
- Is Mapping: ☒
- Uses Placement Correction: ☐

At the bottom right, there are three buttons: 'Back', 'Next', and 'Cancel'.

Figure 11 –System Configurator - Transfer-Module Configuration

Depending on the selected number of loadports, the individual loadports are now configured.



The screenshot shows the 'MueTec System Configurator' window with the 'Loadports' tab selected in the left sidebar. The main area contains the following fields:

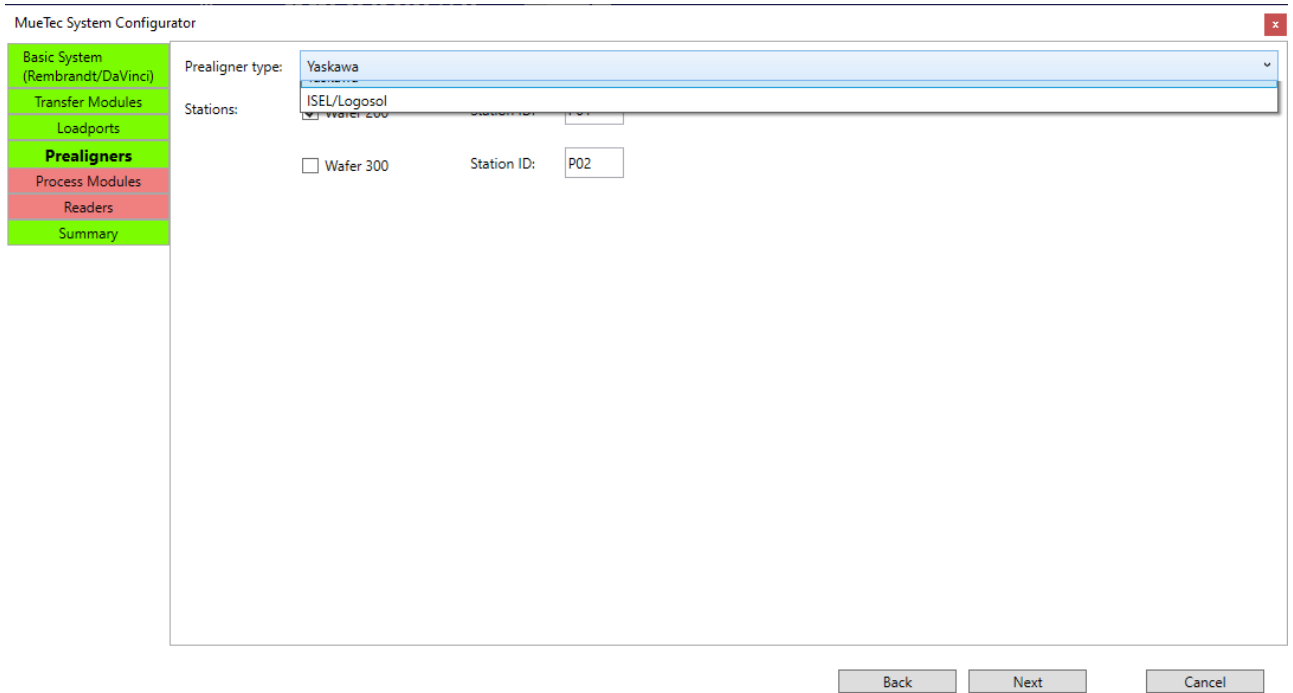
- Number of Loadports: 2 (dropdown menu)
- Loadport 1 configuration:
 - Loadport type: TDK TAS300 (dropdown menu)
 - Reader: Fortrend PLM200W (dropdown menu)
 - Carrier Size: Fortrend PLM150R (dropdown menu)
 - Stations: ☒ Wafer 200 (Station ID: C01), ☐ Wafer 300 (Station ID:)
- Loadport 2 configuration:
 - Loadport type: TDK TAS300 (dropdown menu)
 - Reader: DeviceIntegrated (dropdown menu)
 - Carrier Size: ☒ Auto Detect Carrier Size
 - Stations: ☐ Wafer 200 (Station ID:), ☐ Wafer 300 (Station ID:)

A red error message at the bottom right of the Loadport 2 section states: 'At least one Station must be defined!'.

At the bottom right, there are three buttons: 'Back', 'Next', and 'Cancel'.

Figure 12 –System Configurator - Loadport Configuration

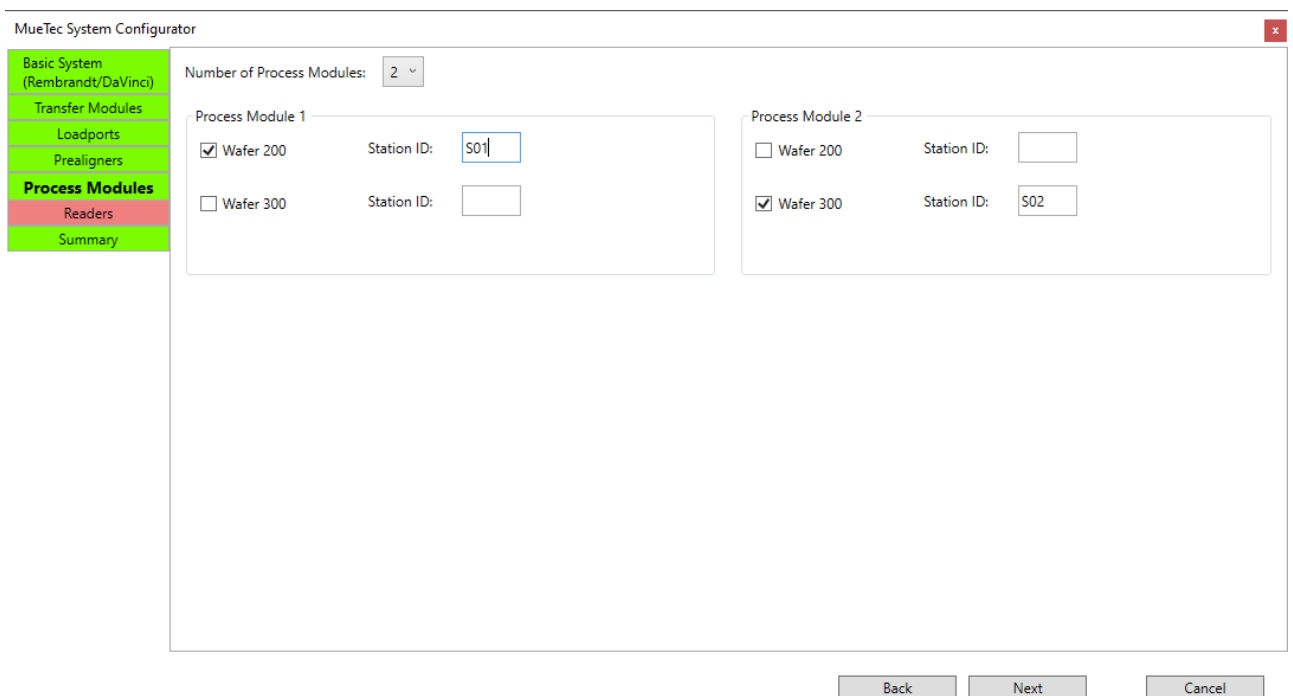
If the Recipe type "Rembrandt" is selected, the next step is to configure the prealigner.



The screenshot shows the 'MueTec System Configurator' window with the 'Prealigners' tab selected in the left sidebar. The main area displays the 'Prealigner type' as 'Yaskawa' and 'Stations' as 'ISEL/Logosol'. Below this, there are checkboxes for 'Wafer 200' (checked) and 'Wafer 300' (unchecked), and a 'Station ID' field with the value 'P02'. At the bottom right, there are 'Back', 'Next', and 'Cancel' buttons.

Figure 13 – System Configurator - Prealigner Configuration

Depending on the selected number of process modules, the individual process modules are now configured.



The screenshot shows the 'MueTec System Configurator' window with the 'Process Modules' tab selected in the left sidebar. The main area displays the 'Number of Process Modules' as '2'. Below this, there are two sections: 'Process Module 1' and 'Process Module 2'. Each section has checkboxes for 'Wafer 200' and 'Wafer 300', and a 'Station ID' field. In 'Process Module 1', 'Wafer 200' is checked and the 'Station ID' is 'S01'. In 'Process Module 2', 'Wafer 300' is checked and the 'Station ID' is 'S02'. At the bottom right, there are 'Back', 'Next', and 'Cancel' buttons.

Figure 14 – System Configurator - Process Module Configuration

The next step is to configure the reader.

MueTec System Configurator

Basic System (Rembrandt/DaVinci)

Transfer Modules

Loadports

Prealigners

Process Modules

Readers

Summary

Reader type: Keyence SR750

Reading position: Keyence SR750

Process Module

WID120

☐ Wafer 200 Station ID:

☐ Wafer 300 Station ID:

At least one Station must be defined!

Back Next Cancel

Figure 15 – System Configurator - Reader configuration

The next step displays a summary of the configuration.

MueTec System Configurator

Basic System (Rembrandt/DaVinci)

Transfer Modules

Loadports

Prealigners

Process Modules

Readers

Summary

General

Recipe type: Rembrandt_DaVinci Machine Name: Rembrandt 1

Loadports: 2 Process Modules: 2

Back Next Cancel Apply Configuration

Figure 16 – System Configurator - Summary

If the configuration is confirmed with "**Apply Configuration**", the database update and the update of the configuration files starts. After that the installation is finished.

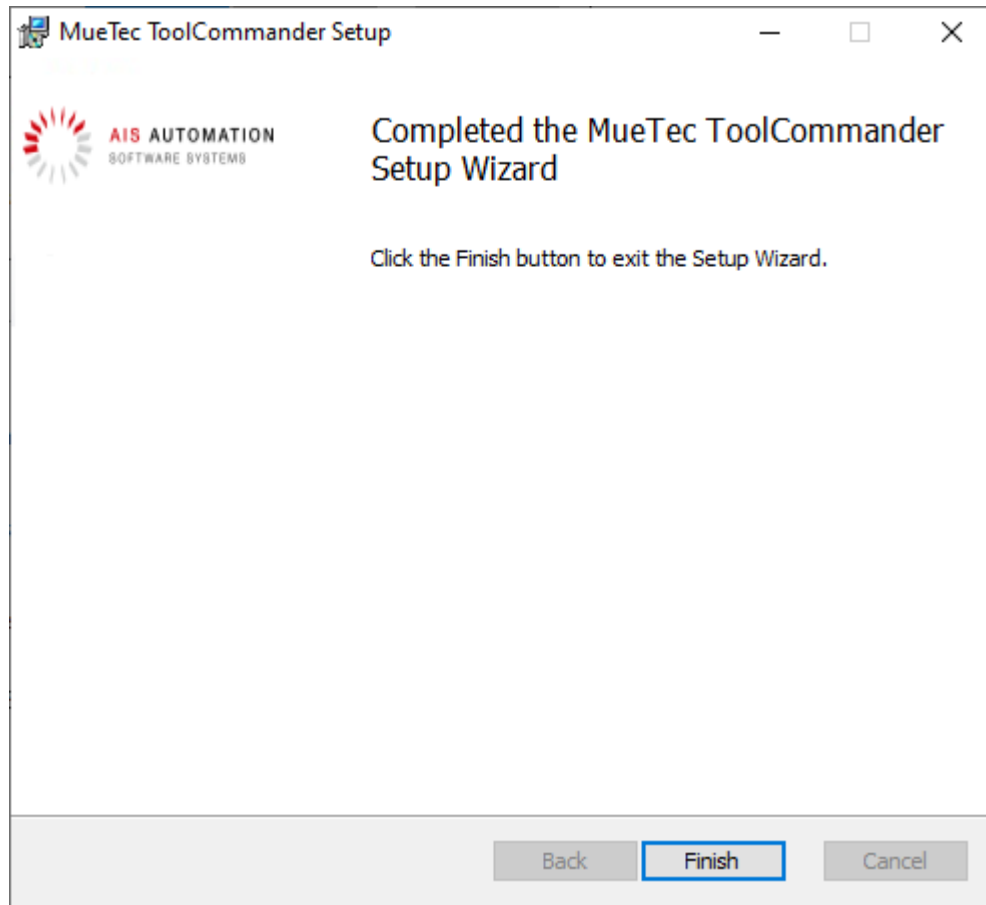


Figure 17 – Installation Mode New - Finished

Failure analysis

The two logfiles of the installation process start with "MueTec_ToolCommander_" and are written to the following directory: "%AppData%\..\Local\Temp\".

The log files of the System Configurator have the prefix "SystemConfigurator_" and are written to the directory "C:\AIS\ApplicationStarter\SystemConfigurator\logs\".

5

System Structure

5 System Structure

5.1 Database and Data Logging

The control system uses one database named "Muetec" that is hosted by a database server "SQL Server Express". The database backup is to be done manually by the user (it's recommended to back-up the whole system with a HDD image).

Database backup mechanism can be used to store and restore the contents of all used databases. This has to be done manually by the user. Database backup can be started with the file *Save_DB.bat* in the folder "{Installation Directory}\Database\Backup\". Restoring database from backup file can be started with the file *Restore_DB.bat* in the folder "{Installation Directory}\Database\Backup\". The default "Installation Directory" is "C:\AIS".

The production logs are written in the directory "{Installation Directory}\ToolCommander\ProductionLog\".

The log files for software trouble shooting are stored in the directory "{Installation Directory}\ToolCommander\VCTC_InternalLogging\{Current Year}\{Current Month}". "Current Year" is a sub folder with name of the current year e.g. "2019", and "Current Month" a sub folder of the year folder with the current month, e.g. "01." In case of errors the log files should be zipped and added to the error report.

All logged data can be backed up manually.

5.2 Used Software Products

Product	Version
VAC	5.3
ToolCommander®	4.0.110
FabLink® suite	6.5.0

Table 5 – Used Software

5.3 Main Functions

The control implements process control in automatic (recipe) mode and manual control in service mode. The main functions for the control systems are defined as follows:

- Communication to the periphery:

- Coupling of digital/analog out and input signals etc. EMO, door states via Modbus-TCP/IP
- Coupling of transfer module, load ports using device specific ASCII protocol via TC/IP or RS232
- Recipe management
 - Creation and administration of module recipes via a external recipe editor
 - Execution of recipes
- Technological and manual control of the individual stations
 - Manual operation of single components and modules (e.g.: transfer module, load port, chamber lamp, etc.)
 - Configuration of component with regard to permanently set properties
- Job Management
 - Creation and execution of control jobs and process jobs by using internal generated flow recipes
- Message management
 - Message generation, archiving and evaluation
- Process protocol
 - Logging of process values and messages
 - Batch-related logging of ControlJob ID, Process Job ID, Wafer ID
 - Batch-related logging of start and end time and the recipe used
- User and access right administration
 - Set up, edit and delete user names, rights and passwords
 - Evaluation of the user names with corresponding restriction of the operating options
- Display of Online- and Offline-Trends
- Visualization
 - Displaying of the current states of the transfer module, process modules, hardware aggregates and process values
 - Presentation of recipe and program execution

- SECS/GEM Host-interface with 300mm services
 - The communication and control with a SECS/GEM host over the FabLink® suite.

5.4 Operation Modes

The operation of the system takes place in different operating modes. The operating modes as well as the operation of the various functions of the machine, are through the integrated user management regulated. The operator must have the appropriate rights to operate the feature.

Following operation modes are realized from the control software:

1. Service-Mode:

- Administration of the user management
- Service operation of all digital and analog process signals, override in the I/O Diagnosis
- Machine configuration
 - configuration of technology-irrelevant machine settings (media monitoring, device parameters)
 - configuration of the machine components (limits, time monitoring)

2. Maintenance-Mode (Engineering mode):

- Operation of machine components and modules (transfer module, process module, etc.)
- Manual setpoint settings
- Manual operation of the machine

3. Production-Mode:

- Operating the machine in automatic, execution of recipe and Control Jobs
- Creation of recipes with the recipe editor

5.5 Power-up and Power-down Behavior

The control is switched on with the main switch of the system. The computer starts automatically and starts the control software automatically. A user logs on to the Windows operating system automatically. Processes that were active before a power failure, for example, do not restart automatically. All components and modules will be initialized when the user starts the initialize function in the operator screen. First the transport module will be initialized, thereafter the load ports and process modules.

Before switching off, all processes must be terminated. The computer can then be shut down using the "Exit program" operating function. The system should only be switched off, after the computer has been shut down completely, otherwise the database or operating system may be damaged.

All set parameters and recipes are retained after switching off, provided no inconsistency occurs (e.g. due to failure of the UPS, or if no UPS is used due to an unexpected interruption of the power supply).

5.6 Security/ Safety Functions

5.6.1 User Rights

The control system uses user rights to enable or disable manual functions for the logged on user. Execution of manual functions, changing parameters or recipes is only possible if the user has the appropriate right. There is a default user so that rights can be specified for case that no user is logged on. For more detailed information about rights and password protection see ToolCommander® User Manual Chapter 10 "User and Rights Management".

The following user rights can be assigned to user groups in the system.

User right	Description
Administrate	Allows to create, to change and to delete system user accounts and user groups
Allow Stop Job	Allows the stopping/canceling of process jobs.
Application Settings	Allows changing the general application settings.
Assign Operator Rights for Module Recipes	This right is not used in the control.

User right	Description
Auto Initialize After Login	If this right is set, the initializing routine off the machine will start after the user log in. The Initializing routine will only be started when the machine is not initialized.
Automatic Mode Operation	Allows to control the production functions of the system.
Change Energy Saving Mode	This right is not used in the control.
Change Parameters	Allows changing device parameters of components.
Control Fab Hostinterface	Allows to change the control mode (online remote, offline, etc.) of the host interface.
Delete Production Log	Allows to delete production logs.
Edit Flow Recipes	Allows to edit module recipes. In the system are module recipe and flow recipe the same.
Edit Module Recipes	Allows to create and to modify module recipes. If the right Edit Module Recipes (Expert Mode) is also supported, a user needs both rights, Edit Module Recipes and Edit Module Recipes(Expert Mode) , for creating and editing module recipes. This right is not used in the control.
Edit Module Recipes (Experte Mode)	Allows to create and modify module recipes in case the user has the right Edit Module Recipes , too. This right is not used in the control.
I/O Diagnosis	Allows to use the I/O diagnosis functions.

User right	Description
InstallComponent	This right is not used in the control.
Manual Operation	Grants access to manual operation dialogs of components.
Manually control Sequencer	Allows the manual control the execution of module recipes. This right is not used in the control.
Operator Start Screen	The operation allows you to create process jobs using the Job Creator Dialog .
Quit Application	Allows to exit the program and to shut down the computer.
User Administration	Allows to administrate users/ user groups that have the same user rights assigned as the user who gets this right.
VAC Diagnosis	Allows to use the VAC Diagnosis tool.
View Flow Recipes	Allows to view module recipes. In the system are module recipe and flow recipe the same.
View Parameters	On the components screen allows the user to display the parameters of components and to print them.
View Production Log	Allows viewing production logs.
Windows Access	Allows access to the operating system Windows.

Table 6 – User-Rights

5.6.1.1 User Groups

The creation of user groups and the assignment of rights can be individually designed. The following groups of users are only a proposal for setting up the user management of the machine control system. The user group "Administrators" is predefined and can not be modified. Users of this group have all rights.

Administrators (IT)

The users of this group should only manage the users on the system and change the software settings.

User right	Grant right
Administrate	yes
Allow Stop Job	no
Application Settings	yes
Auto Initialize After Login	no
Automatic Mode Operation	no
Change Parameters	no
Control Fab Hostinterface	no
Delete Production Log	no
Edit Flow Recipes	no
I/O Diagnosis	no
Manual Operation	no
Operator Start Screen	no
Quit Application	yes
User Administration	yes

User right	Grant right
VAC Diagnosis	no
View Flow Recipes	no
View Parameters	no
View Production Log	no
Windows Access	yes

Table 7 – User Rights User Group Administrators

Service (Experts, Maintenance)

The operators of this group should operate the system in production mode and the components in manual mode. In addition, they perform diagnostics on individual plant components and adjust parameters.

User right	Grant right
Administrate	no
Allow Stop Job	yes
Application Settings	no
Auto Initialize After Login	yes
Automatic Mode Operation	yes
Change Parameters	yes
Control Fab Hostinterface	yes
Delete Production Log	yes
Edit Flow Recipes	yes

User right	Grant right
I/O Diagnosis	yes
Manual Operation	yes
Operator Start Screen	yes
Quit Application	optional (in case the user needs to shut down the software)
User Administration	optional (in case the user can add other service users to the service group)
VAC Diagnosis	no
View Flow Recipes	yes
View Parameters	yes
View Production Log	optional
Windows Access	yes

Table 8 – User Rights User Group Service

Engineering

The operators in this group should be able to operate the system in production mode and change, test recipes.

User right	Grant right
Administrate	no
Allow Stop Job	yes
Application Settings	no
Auto Initialize After Login	yes

User right	Grant right
Automatic Mode Operation	yes
Change Parameters	no
Control Fab Hostinterface	optional (in case the user needs to change the host control mode)
Delete Production Log	optional (in case the user needs to delete the production log)
Edit Flow Recipes	yes
I/O Diagnosis	no
Manual Operation	optional (in case the user needs to test a recipe on the process module only)
Operator Start Screen	optional (in case the user needs to start processing without host or in offline mode)
Quit Application	optional (in case the user needs to shut down the software)
User Administration	optional (in case the user can add other service users to the service group)
VAC Diagnosis	no
View Flow Recipes	yes
View Parameters	yes
View Production Log	yes
Windows Access	no

Table 9 – User Rights User Group Engineering

Operators (Production)

The operators of this group should only operate the system in production mode.

User right	Grant right
Administrate	no
Allow Stop Job	optional (in case the user needs to stop a process job)
Application Settings	no
Auto Initialize After Login	yes
Automatic Mode Operation	yes
Change Parameters	no
Control Fab Hostinterface	optional (in case the user needs to change the host control mode)
Delete Production Log	optional (in case the user needs to delete the production log)
Edit Flow Recipes	no
I/O Diagnosis	no
Manual Operation	no
Operator Start Screen	optional (in case the user needs to start processing without host or in offline mode)
Quit Application	no
User Administration	no
VAC Diagnosis	no

User right	Grant right
View Flow Recipes	optional (in case the user needs check the recipes)
View Parameters	no
View Production Log	no
Windows Access	no

Table 10 – User Rights User Group Operator

5.6.2 Software Interlocks

Technological safety functions/ locks are integrated in the control software of the components. For details, see the "Components" chapter.

6

Configuration

6 Configuration

6.1 General

The configuration of the software is basically done over the database. But some parts of the software can be configured over specific configuration files.

File	Location\Description
Application.ini	C:\AIS\ApplicationStarter Contains the start up information for the software start.
MueTecApplication.ini	C:\AIS\ToolCommander Contains the general configuration of the application.
StationConfig.xml	C:\AIS\ToolCommander\Config Contains station configuration for the transport module.
BaseConfig.xml	\AIS\ToolCommander\Config Contains the connection parameters for the Modbus/TCP IP connection the B&R PLC.
ParameterConfig.xml	C:\AIS\ToolCommander\Config Contains the I/O channel configuration of the B&R PLC for the Modbus/TCP IP Connector.
TuningParameters.json	C:\AIS\ToolCommander\Config Contains the configuration of the tuning parameters which are available in the Job Creator dialog.

Table 11 – General Configuration: Overview Configuration Files

The **Application.ini** contains the startup information for the application starter (application.exe). There the startup order of the applications are defined.

The **MueTecApplication.ini** contains the general configuration of the application.

Section/Entry	Description
[General]	
AutoInitOnLogon	Defines whether the machine will be initialized, when an operator with the User-Right "Auto Initialize After Login" is logged on. If this option is set to "False", the machine will not be initialized after user logon. If this option is set to "True", the machine will be initialized after start, when an operator with the User-Right "Auto Initialize After Login" is logged on.
ManualCarrierTypeValidation	Defines whether the carrier type validation dialog is shown when the operator places a carrier on the load port. If this option is set to "False", the carrier type validation dialog will not be shown. If this option is set to "True", the carrier type validation dialog will be shown after the operator has placed a carrier on the load port. As default the value is set to "True".
MappingArm	Defines the mapping arm for mapping with the robot. The mapping arm must be free for the mapping. The parameter can be changed as follows: "0" the mapping is done with the elbow on the robot, "1" the mapping is done with the lower arm, "2" the mapping is done with the upper arm. If the mapping is supported by the loadport, this parameter is obsolete. As default the value is set to "2" (Mapping with the upper arm.).
InternalCommunicationPort	TCP/IP port for communication between Control and Visualization. As default the value is set to "2200".
LengthRecipeName	Determines how many characters the recipe name may contain. The default setting is 35 and the maximum number is limited to 80.
ShowPinkCarrierWarning	Determines whether an open carrier is surrounded by a pink frame in the overview screen.
[VisuApplication]	
RecipeConfigurationPath	Defines the path where the recipe information are stored. The default value is "C:\Temp".
RecipeExportPath	Defines the path for the recipe export. The default value is "C:\Temp\RecipeExportPath".

Section/Entry	Description
ExchangePossible	Defines whether the robot supports the exchange of wafer. As default the value is set to "True".
StepModeActive	Defines whether the scheduler is running in step mode or not. The step mode should be only activated for scheduler diagnosis. As default the value is set to "False".
UsePrepareNewMaterial	Defines whether the process module supports to send a prepare new material request. If the value is "True", new material is only transported to the process module after the message "PrepareNewMaterial". If the value is "False", new material will not be transported to the process module until the process is finished at the process module. As default the value is set to "True".
MaxPerformanceMode	Defines whether the max performance mode for wafer transfer is activated or not. If this option is set to "True," the transfer module picks the next wafer, without waiting of the process completion of the process module. As default the value is set to "False".
UseUpperBladeForLoader	Defines whether the wafers should be removed from the load port only with the upper arm of the robot. If tool configuration is Argos then the value must be "True" because the Wafer ID Reader can only read the ID on the upper arm. The default value is "False".
MoveToProcessModule	Defines whether the wafers should be moved during processing to the get position of the process module. If the value is set to "True", and the process of the wafer in the process module is not finished, the robot moves the next picked wafer to the get position of the process module. As default the value is set to "False".
TuningParametersConfigPath	Defines the path to a json file which defines allowed tuning parameters.
SkipSecondAlignmentIfReadingPositionAligner	Parameter is True: Checks the condition for skipping the second alignment step for turning back the material at the aligner for material whose ID is read at the aligner position. Parameter is False: Material whose ID is read at the aligner position always performs the second alignment step regardless of the angle or other conditions.

Section/Entry	Description
	The default value is "False".
SkipSecondAlignmentIf ReadingPositionExternal	Parameter is True: Checks the condition for skipping the second alignment step for turning back the material at the aligner for material whose ID is read at the reader position. Parameter is False: Material whose ID is read at the reader position always performs the second alignment step regardless of the angle or other conditions. The default value is "False".
ToleranceForAngleDifference	Unterhalb dieses Wertes gilt der Winkelvergleich als bestanden. The default value is "1". Berechnung der Bedingung: $\text{abs}(\text{Machine.ReadingAngle} - (\text{Recipe.AlignmentAngle} + \text{PM. WinkelKorrektur})) < \text{ToleranceForAngleDifference}$ Ist diese Bedingung erfüllt wird das Zurückdrehen übersprungen.
[LoadportStatePanel]	Defines the indicator mapping for the TDK Loadport LED Panel.
IndicatorReadyToLoad	Defines the indicator for the loadport transfer state "Ready To Load". If set to 0 no indicator will be set. The default value is "1".
IndicatorReadyToUnload	Defines the indicator for the loadport transfer state "Ready To Unload". If set to 0 no indicator will be set. The default value is "2".
IndicatorManual	Defines the indicator for the loadport access mode "Manual". If set to 0 no Indicator will be set. The default value is "3".
IndicatorAutomatic	Defines the indicator for the loadport access mode "Automatic". If set to 0 no Indicator will be set. The default value is "4".
[StateVariables]	
UpdateInterval	Defines the update interval in seconds at which the status variables are sent to the host after changes.
[ProcessModules]	

Section/Entry	Description
InitializeSuccessive	Defines whether the process modules are initialized one after the other. If the value is "0", the initialization of the process modules is started one after the other according to the time specified in the "InitializeDelayTime" parameter. If the value is "1", the system first waits until process module 1 is initialized and then initializes process module 2.
InitializeDelayTime	Defines the delay time in seconds between the initialization of process module 1 and process module 2. If the value is "0" and the parameter "InitializeSuccessive" = "0", the process modules are initialized simultaneously. If the value is greater than zero and the parameter "InitializeSuccessive" = "0", the time is waited after the start of the initialization of process module 1 until the initialization of process module 2 is started. If the value is greater than zero and the parameter "InitializeSuccessive" = "1", the time is waited after the initialization of process module 1 until the initialization of process module 2 is started.
[Loadport 1] .. [Loadport 4]	Defines loadport specific parameters.
CassetteTypeName	Sets the cassette type for TDK load ports. The settings FOUP, FOSB, RingFrame, SingleMask are supported. The default value is FOSB.
UnloadDelay	Defines the delay for the Unload command in milliseconds. If the value is greater than "0", then the defined time in milliseconds is waited before the Unload command is executed. The default value is "0".
SlotCount	Sets the default value for the number of slots of the loadport's carriers. This value is used when the UseDynamicSlotCount parameter is set to False. The default value is "0". The parameter is used with the Fortrend PLM200W and XML Loadport.
UseModbusCarrierPresenceDetection	Defines a boolean to determine whether carrier presence detection is done by Modbus. If set to True: Carrier presence detection will be determined by Modbus signals. If set to False: Carrier presence detection will be determined by Loadport
UseMappingInUnloadSequence	Defines a boolean to determine whether a mapping of the carrier is to be carried out before an unload. The Default is "False".

Section/Entry	Description
	If set to "True": The Mapping is activ. If set to "False": The Mapping is not activ.
StartLimitationCarrierId	Parameter for limiting a carrier ID read from the hardware. The value specifies the character from which the read value is to be used. The default value is "0".
ProtrusionSensorMode	The parameter defines how the protrusion sensor is to be monitored during robot movements in the load port. The default value is "0". The following settings are possible: "0": No monitoring "1": With the Pick "2": With the Place "3": With the Pick or Place If monitoring is active and the protrusion sensor is occupied, an alarm is triggered and the command is cancelled.
LengthLimitationCarrierId	Parameter for limiting a carrier ID read from the hardware. The value specifies how many characters of the read value are to be used. The default value is "0", in which case all characters after StartLimitationCarrierId are used. The calculation always starts from StartLimitationCarrierId.
CodingCarrier6Inch1Slot	A semicolon separated list with the coding of carriers with one slot and 6 Inch material.
CodingCarrier6Inch6Slots	A semicolon separated list with the coding of carriers with 6 slots and 6 Inch material.
CodingCarrier7Inch1Slot	A semicolon separated list with the coding of carriers with one slot and 7 Inch material.
CodingCarrier9Inch1Slot	A semicolon separated list with the coding of carriers with one slot and 9 Inch material.

Section/Entry	Description
CodingCarrier14Inch1 Slot	A semicolon separated list with the coding of carriers with one slot and 9 Inch material.
CodingCarrier8Inch13 Slots	A semicolon separated list with the coding of carriers with 13 slots and 8 Inch (200 mm) material.
CodingCarrier8Inch25 Slots	A semicolon separated list with the coding of carriers with 25 slots and 8 Inch (200 mm) material.
CodingCarrier12Inch1 3Slots	A semicolon separated list with the coding of carriers with 13 slots and 12 Inch (300 mm) material.
CodingCarrier12Inch2 5Slots	A semicolon separated list with the coding of carriers with 25 slots and 12 Inch (300 mm) material.
CodingCarrier15Inch1 3Slots	A semicolon separated list with the coding of carriers with 13 slots and 15 Inch (380 mm) material.
CodingCarrier15Inch2 5Slots	A semicolon separated list with the coding of carriers with 25 slots and 15 Inch (380 mm) material.
[Robot]	This section contains the parameters that apply to the robot.
EnableChangeVacuumLine	Determines whether, in the event of a wafer presence error (wafer presence confirmation timeout), the vacuum lines are switched over to wafer presence detection. This parameter is only taken into account by the control software for systems with a Yaskawa robot If the value is "True", when the robot sends a wafer presence confirmation timeout error during pick, the vacuum line is switched and the ring frame presence status of the robot is evaluated. If the value is "False", the vacuum line is not switched over in the event of a wafer presence confirmation timeout error.
[HostInterface]	This section contains the parameters for the communication between ToolCommander and Fablink. The values should not be changed.

Section/Entry	Description
TLKPort	Sets the port for communication between Fablink and the ToolCommander application. The default value is "3000".
TLKAddress	Sets the TCP/IP address for communication between Fablink and the ToolCommander application. The default value is "127.0.0.1".
[Aligner]	This section contains the parameters that apply to the aligner.
TuningParameterForExternalSensor	Defines the name of used tuning parameter. If the value is empty or not configured, the aligner uses its own notch detection.
[CameraAligner]	This section contains the parameters that apply to the Camera Aligner.
DeviationX	If the measured X value of the camera aligner exceeds this parameter, a warning is generated and the status of the material is set to Skipped.
DeviationY	If the measured Y value of the camera aligner exceeds this parameter, a warning is generated and the status of the material is set to Skipped.
DeviationTheta	If the measured Theta value of the camera aligner exceeds this parameter, a warning is generated and the status of the material is set to Skipped.
[HermosReader]	This section contains the parameters that apply to the Hermos Reader LFM.
AntennaCountStartsWith	Defines the offset for addressing the numbering of an antenna. The default value is "0". In the configuration of the reader, antenna 1 always uses the number 0.
TimeoutInSeconds	Defines the timeout for the reader's response.

Table 12 – General Configuration: MuetecApplication.ini

Note: If all 3 deviation settings for the CameraAligner are 0, no deviation is generally checked for the material. If the material changes to the Skipped status, it is immediately transported back to the load port.

Coding of carriers

The Info-Pads of the carriers are evaluated for the assignment of the carriers to the material and the carrier size. The info pad information is coded differently depending on the load port type and load port manufacturer:

Brooks Loadport:

Bit 0=1: Pad B is set

Bit 1=1: Pad A is set
 Bit 2=1: Pad C is set
 Bit 3=1: Pad D is set
 E.g. A carrier with info pad A has the coding 0010 = 2

TDK Loadport:

Bit 0=1: Pad A is set
 Bit 1=1: Pad B is set
 E.g. A carrier with info pad A has the coding 0001 = 1

Fortrend PLM150R:

1 = Single POD Type (1 Maske)
 2 = Multi POD Type (6 Masken)

Fortrend PLM200W:

1 = PodType1
 2 = PodType2
 3 = PodType3 (6")
 4 = PodType4 (8")

6.2 Station Configuration

The station configuration file StationConfig.xml in the path C:\AIS\ToolCommander\Config contains the station id mapping for stations of the transfer module.

```
<?xml version="1.0" encoding="utf-8" standalone="yes"?>
<TransferStationConfig>
  <Modules>
    <ProcessModule ID="1" PlacementCorrectionParameter="12" aferPresenceCheckAfterPlace="0">
      <TransferStation ID="S01" PutOffset ="2" MaterialType="WaferFrame 380"
SpeedReduction="0"/>
    </ProcessModule>
    <Loadport ID="3" SafePositionMin="-20.0" SafePositionMax="1.0"
WaferPresenceCheckAfterPlace="1">
      <MappingStation ID="C01" MaterialType=""/>
      <TransferStation ID="C01" PutOffset ="2" MaterialType="Wafer 200" SpeedReduction="1"/>
      <TransferStation ID="C02" PutOffset ="2" MaterialType="Wafer 300" SpeedReduction="1"/>
    </Loadport>
    <Aligner ID="8" WaferPresenceCheckAfterPlace="1">
      <TransferStation ID="P01" PutOffset ="2" MaterialType="Wafer 200" />
      <TransferStation ID="P01" PutOffset ="2" MaterialType="Wafer 300" />
    </Aligner>
    <ReaderModule ID="9">
      <TransferStation ID="S10" PutOffset ="2" MaterialType="Wafer 200" />
      <TransferStation ID="S03" PutOffset ="2" MaterialType="WaferFrame 380"/>
    </ReaderModule>
    <CameraAligner ID="10" >
      <TransferStation ID="S12" PutOffset ="2" MaterialType="Mask 6 Inch"/>
    </CameraAligner>
    <AlignmentCorrection ID="12" >
      <TransferStation ID="S12" PutOffset ="2" MaterialType="Mask 6 Inch">
        <AlignmentCorrectionStation ID="S11" ArmPosture ="Left" ArmLength="345"/>
      </TransferStation>
    </AlignmentCorrection>
  </Modules>
</TransferStationConfig>
```

The configuration is stored in XML format. Under the main tag TransferStationConfig there are the modules configured, for each module type there is a separate XML tag containing an ID. The ID is defined in the controller.

XML Tag	Description/Attribute
<ProcessModule>	Station configuration for process modules. Attributes: • ID: Internally used identifier, 1 or 2. • PlacementCorrectionParameter: Parameter for automatic placement correction with the Place command. The value must be greater than 0. The use of the parameter is enabled in the settings dialogue of the robot via the checkbox "Placement Correction". • WaferPresenceCheckAfterPlace: The parameter defines whether the vacuum sensor on the robot is checked after a place command. A value greater than 0 switches the function on. The value itself specifies the time that must elapse between switching on the vacuum and checking the sensor. If the sensor is occupied, an error is thrown.
<Loadport>	Station configuration for loadport modules. Attributes: • ID: Internally used identifier, 3, 4 or 5. • SafePositionMin and SafePositionMax: Definition of the station-specific safety range for the extension axis of the robot in mm. If it is outside the safety area, the carrier cannot be set up or removed at the loadport. • WaferPresenceCheckAfterPlace: The parameter defines whether the vacuum sensor on the robot is checked after a place command. A value greater than 0 switches the function on. The value itself specifies the time that must elapse between switching on the vacuum and checking the sensor. If the sensor is occupied, an error is thrown.
<Aligner>	Station configuration for aligner. • ID: Internally used identifier, : 8

XML Tag	Description/Attribute
	• WaferPresenceCheckAfterPlace: The parameter defines whether the vacuum sensor on the robot is checked after a place command. A value greater than 0 switches the function on. The value itself specifies the time that must elapse between switching on the vacuum and checking the sensor. If the sensor is occupied, an error is thrown.
<ReaderModule>	Station configuration for wafer ID reader. • ID: Internally used identifier : 9 • WaferPresenceCheckAfterPlace: The parameter defines whether the vacuum sensor on the robot is checked after a place command. A value greater than 0 switches the function on. The value itself specifies the time that must elapse between switching on the vacuum and checking the sensor. If the sensor is occupied, an error is thrown.
<CameraAligner>	Station configuration for the camera aligner. • ID: Internally used identifier : 10 • WaferPresenceCheckAfterPlace: The parameter defines whether the vacuum sensor on the robot is checked after a place command. A value greater than 0 switches the function on. The value itself specifies the time that must elapse between switching on the vacuum and checking the sensor. If the sensor is occupied, an error is thrown.
<AlignmentCorrection>	Station configuration for the alignment correction. • ID: Internally used identifier : 12 • WaferPresenceCheckAfterPlace: The parameter defines whether the vacuum sensor on the robot is checked after a place command. A value greater than 0 switches the function on. The value itself specifies the time that must elapse between switching on the vacuum and checking the sensor. If the sensor is occupied, an error is thrown.

Table 13 – Station Configuration: Assignment of XML-Tags to Modules

In each module, a transfer station (**TransferStation**) can be configured for each material type. The transfer stations contain the following information:

XML Attribute	Description
ID	Contains the station ID of the robot station. In the case of a Yaskawa robot, the station IDs are set as follows: • Loadport stations: "C01" - "C08" • Aligner station: "P01", "P02" • Process stations, measurement stations, reader stations: "S01"- "S12" The station IDs depend on the stations taught for transport on the robot.
PutOffset	Specifies an offset for placing wafers. This value is not evaluated for systems with a Yaskawa robot.
MaterialType	Specifies the material type for which the station ID is to be used. The following values can be entered in this attribute: "Wafer 200", "Wafer 300", "Wafer Frame 380", "Mask 6 Inch", "Mask 7.25 Inch", "Mask 9 Inch" or "Mask 14 Inch". If the value is empty, there is no material type specific differentiation of station IDs. The same station ID is always used in communication with the robot.
SpeedReduction	Indication of the speed reduction for pick and place commands. If the value is greater than Zero the pick/place commands will be executed in fast speed until the G1/P1-position (Ready Position) and with the lowest speed to the G4/P4-position (Final position). The feature is automatically deactivated if the attribute is missing or the value is lower than one.

Table 14 – Station Configuration: Attributes XML-Tag TransferStation

In addition to the transfer station, material specific mapping stations can be configured. These mapping stations only need to be configured if the mapping is done by the robot.

XML Attribute	Description
ID	Contains the station ID of the robot station. In the case of a Yaskawa robot, the station IDs are set as follows: • Loadport-Station: "C01" - "C08" The station IDs depend on the stations taught for transport on the robot.
MaterialType	Specifies the material type for which the care unit ID is to be used. The following values can be entered in this attribute: "Wafer 200", "Wafer 300", "Wafer Frame 380", "Mask 6 Inch", "Mask 7.25 Inch", "Mask 9 Inch" or "Mask 14 Inch". If the value is empty, there is no material type specific differentiation of stations. The same station ID is always used in communication with the robot.

Table 15 – Station Configuration: Attributes XML-Tag MappingStation

The parameters for the alignment correction can be configured via the XML element "AlignmentCorrectionStation" of the transfer station.

XML Attribute	Description
ID	Contains the station ID of the robot station on which the material (mask or wafer) is to be aligned. During the alignment correction, the station coordinates of the transfer station are read out and recalculated with the offset measured at the camera aligner before the material is placed in the station. After the calculation, the station coordinates of the specified robot station are overwritten with the new calculated station coordinates and the material is placed.
ArmPosture	Specification of the arm position "Left" or "Right" for which the station coordinates are to be read out and adjusted.
ArmLength	Specification of the length of the end effector in millimeters. This value is used to calculate the offset of the station coordinates.

Table 16 – Stationkonfiguration: Attribute XML-Tag AlignmentCorrectionStation

6.3 PLC Configuraton

The configuration of the PLC is located in the path C:\AIS\ToolCommander\Config.

The general communication parameters for the Modbus TCP/IP interface are setup in the file "BaseConfig.xml". This configuration file contains the TCP/IP-Address and the Port settings for the communication with the PLC.

```
<?xml version="1.0" encoding="utf-8" ?>
<BaseConfig>
  <IpAddress>127.0.0.1</IpAddress>
  <Port>502</Port>
  <SlaveId>1</SlaveId>
  <ReadCycle>400</ReadCycle>
</BaseConfig>
```

The assignment of the I/O channels to the Modbus reading areas (coils, discrete inputs, etc.) is defined in the file "ParameterConfig.xml".

```
<?xml version="1.0" encoding="utf-8"?>
<Configuration xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <GlobalSettings />
  <EqInstances />
  <ConnectionModels/>
  <MemoStructureModels>
    <MemoStructureModel Name="Default">
      <MessageStructures>
        <MessageStructure Name="Coils"
          ReadStartAddress="1"
          ReadLength="7" ==> Note: For the Fortrend OCSM Loadport, the
value must be changed to 13.
          SwitchBytes="0"
          ReadSource="COILS">
          <Channels>
            <Channel Name="STC_GREEN"
              Type="STATE"
              SrcFormat="BIT"
              DestFormat="BOOL"
              StartPos="0"
              BitPos="0" />
            <!-- ... -->
          </Channels>
        </MessageStructure>
        <MessageStructure Name="DiscreteInputs"
          ReadStartAddress="1"
          ReadLength="66" ==> Note: For the Fortrend OCSM Loadport,
the value must be changed to 75.
          ReadFormat="BYTE"
          SwitchBytes="0"
          ReadSource="DISCRETEINPUT">
          <Channels>
            <Channel Name="STAT_DOOR_1"
              Type="STATE"
              SrcFormat="BIT"
              DestFormat="BOOL"
              StartPos="0" BitPos="0" />
            <!-- ... -->
          </Channels>
        </MessageStructure>
      </MessageStructures>
    </MemoStructureModel>
  </MemoStructureModels>
</Configuration>
```

Changes in the assignment of the I/O channels should only be made in the channel mapping (<Channel>). The names of the I/O channels are defined via the controller. If changes are made, the controller may malfunction.

XML Attribute	Description
Name	This attribute contains the name of the channel. This is fixed in the controller and should not be changed.
Type	This attribute contains the channel type. A distinction is made between "STATE" for digital signals and "EQVAL" for analog signals.

XML Attribute	Description
SrcFormat	Specifies the format in the PLC. For digital signals the Src format is "BIT", for analog channels "INT8", "INT16", "INT32" or "FLOAT".
DestFormat	Specifies the target format in the controller. For digital signals the Dest format is "BOOL", for analog channels "INT8", "INT16", "INT32" or "FLOAT".
StartPos	Specifies the byte position beginning with "0" at which the value for the channel is located.
BitPos	Specifies the bit in which the value for the channel is located. In the case of digital signals via Modbus (coils or discrete inputs), the value should always be "0". With analog signals, the bit position is not evaluated..

Table 17 – PLC - Configuration: Attributes XML-Tag Channel

7

Online Visualization

7 Online Visualization

7.1 General

The GUI is based on the ToolCommander® framework, which is compliant to SEMI E95. The general functions, operating and display elements are described in detail in Chapter 2 of the ToolCommander® Framework Operation Manual (ToolCommander® - User manual). In the following only the machine specific online visualization will be described.

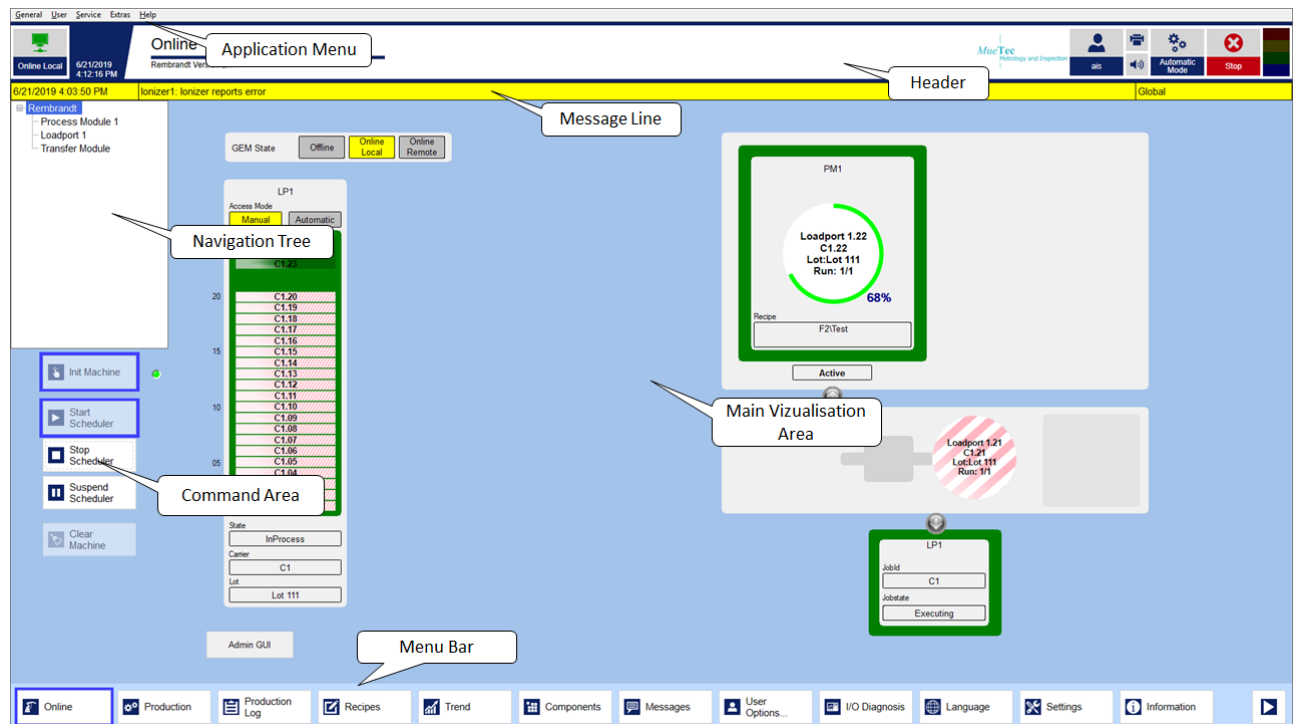


Figure 18 – Screen Layout

The number of visible Loadports and Process Modules depends on the tool configuration.

The online visualization can be selected via the button Online of the menu bar in the lower area of the screen. Depending on the selection in the navigation tree, a machine overview or a module overview screen is displayed in the main visualisation area. The online visualization shows details of parts of the machine. In the navigation tree are following detail views available:

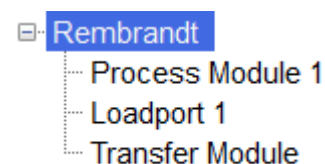


Figure 19 – Navigation Tree

- Online visualization Machine overview ➔ [Machine Overview](#)

- Process Module: Online visualization of the Process Module ➔ [Process Module Visualization](#)
- Loadport: Online visualization of the Loadport ➔ [Loadport Visualization](#)
- Transfer Module: Online visualization of the Transfer Module ➔ [Transfer Module Visualisation](#)

7.2 Machine Overview

7.2.1 Operator GUI

The Operator GUI of the machine view shows a systematic view of all modules, cassettes and substrates, which are on the machine.

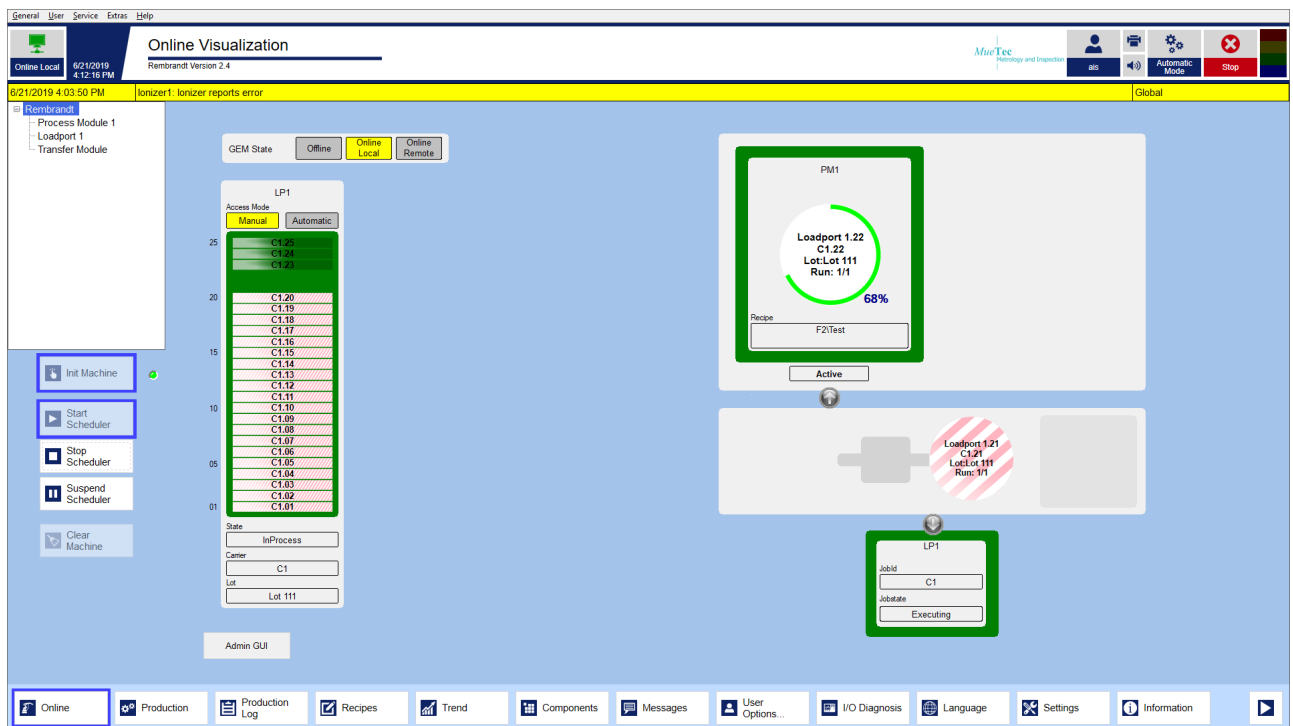


Figure 20 – Online Visualization Machine Overview Operator GUI

The state of the host interface connection is displayed in the upper left area of the screen labeled with **GEM State**. The background of the fields **Offline**, **Online Local**, **Online Remote** shows the current SECS/GEM connection status.

State	Description
Offline	The host interface is switched off or the Host is offline. In this state the background color of the field Offline is red. The background color of the fields Online Local and Online Remote is gray.
Online Local	The Host is connected to the equipment. The SECS/GEM operation mode was set to "Online Local". The background color of the field Online Local is yellow. The background color of the fields Offline and Online Remote is gray.

Online Remote	The Host is connected to the equipment. The SECS/GEM operation mode was set to "Online Remote". The background color of the field Online Remote is green. The background color of the fields Offline and Online Local is gray.
---------------	---

Table 18 – Machine Overview - Operator GUI: GEM Status

Below the **GEM State** area and the area of the transport module the state information of the loadport modules are shown. This areas are marked with a prefix "LP".

Information	Description	
Access mode	The background color of the fields Manual , Automatic indicates the current access mode of the load port.	
	Background color Manual is set to yellow, the background color Automatic is gray.	The access mode of the loadport is manual.
	Background color Manual is set to gray, the background color Automatic is gray.	The access mode of the loadport is set to automatic.
Loadport State	The current state of the loadport module is indicated by the background color.	
	Gray	The loadport module is in a undefined state, e.g. after application restart without initializing.
	Red	The loadport module is in error.
	Blue	The loadport module is initialized, has no error and no carrier is on the loadport.

	Light Green	The loadport module is initialized, has no error, a carrier was placed on the loadport but the carrier is not docked.
	Green	The loadport module is initialized, has no error, a carrier was placed on the loadport and the carrier is docked.
Slot map	In the loadport status information the actual slot map is shown. When a carrier was mapped 25 fields show which slot on the carrier is occupied. Is one slot occupied, the wafer number will be displayed in the slot field. The background color of the slot field shows the material state of the wafer placed in the slot. The assignment of the material state to the background color is described in chapter ↗ Substrate Information .	
State	In state field in the loadport information area the current processing state of the carrier is displayed. If no carrier is placed, this field is blank.	
Carrier	The ID read by the carrier ID reader is displayed in this field. If no carrier is placed, this field is blank.	
Lot	This field shows the lot id associated to the material of the carrier. If no lot is assigned to the material in the current carrier or no carrier is placed the field is empty. In the Host Communication Mode "Online Remote" the lot information is sent by the host. If the host does not sent the lot information, this field will remain blank. In the other Host Communication Modes the lot ID can be entered in the job creation dialog.(see ↗ Job Creator Dialog).	
JobId	In this field the ID of the current running control job for the carrier on the loadport is displayed. If no carrier is placed or no control job is assigned to the carrier, this field is blank.	

Jobstate	In this field the state of the current running control job for the carrier on the loadport is displayed. If no carrier is placed or no control job is assigned to the carrier, this field is blank.
----------	---

Table 19 – Machine Overview - Operator GUI: Loadport Information

Right from the **GEM State** area and above the area of the transport module the state information of the process modules are shown. This areas are marked with a prefix "PM".

Information	Description
Process Module State	The current state of the process module is indicated by the color of the border of the information area.
	Gray The module is not active
	Red The module is in error.
	Blinking: Green, Light Green The module is active, it is processing a wafer.
Wafer Status	When a wafer is placed on the module a wafer symbol shows the actual wafer information. A radial progress bar on the wafer symbols and a number field show the actual processing progress in percentage.
Recipe	This field shows the current selected recipe.

Table 20 – Machine Overview - Operator GUI: Process Modul Information

Below the process module information the state information of the transfer module is shown.

Information	Description
Transfer Module State	The current state of the transfer module is indicated by the background color of the robot.
	Gray The module is not active
	Red The module is in error.

	Green	The module is active, it is handling a wafer.
Wafer State	When a wafer is handled by the module a wafer symbol shows the actual wafer information.	
Handling State	Arrow symbols show the current handling of the transfer module to and from a process module or loadport.	
		No handling is active.
		The transfer of a wafer from the robot to or from a module is active.
	 blinking	The transfer of a wafer from the robot to or from a module is active. And one of the I/O channel "STAT_ACT_C1", "STAT_ACT_C2", "STAT_ACT_C3", "STAT_ACT_S1" or "STAT_ACT_S2" is active.
	 blinking	The transfer of a wafer from the robot to or from a module is not active. And one of the I/O channel "STAT_ACT_C1", "STAT_ACT_C2", "STAT_ACT_C3", "STAT_ACT_S1" or "STAT_ACT_S2" is active.

Table 21 – Machine Overview - Operator GUI: Transfer Modul Information

For the tool configuration **Rembrandt/DaVinci** in the state information area of the transfer module the state of the Aligner is shown.

Information	Description
Aligner State	The current state of the aligner module is indicated by the background color of the aligner.

	Gray	The module is not active
	Red	The module is in error.
	Green	The module is active, it is aligning a wafer.
Wafer Status	When a wafer is placed on the aligner a wafer symbol shows the actual wafer information.	

Table 22 – Machine Overview - Operator GUI: Aligner Information

A disc symbol indicates the material information of wafers.



Figure 21 – Machine Overview - Operator GUI: Wafer Material Information

A mask symbol indicates the material information of masks,

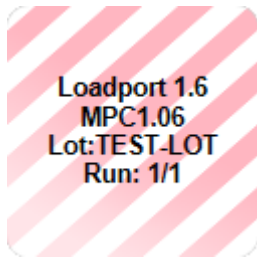


Figure 22 – Machine Overview - Operator GUI: Mask Material Information

Information	Description
Material State	The background color of the wafer symbol shows the current material state. The mapping of the material state to the background color is described in chapter Substrate Information .

Source	The first line in the material information field shows the source of the wafer. The name contains the source module and the source slot. E.g for a material (wafer or masks) from the first slot of the carrier on source module loadport 1 the source "Loadport 1.1" will be shown.
Material ID	The second line of the material information field shows the ID of the material.
Lot	The third line of the material information field shows the lot the wafer or mask belongs to. If no lot was set by the host or operator this field remains blank.
Run counter	The fourth line of the material information field shows the actual run counter and the maximal runs defined for the material. The maximal run counter depends on the cycle counter for a control job (see Job Creator Dialog) and the amount of consecutive process jobs for the material.

Table 23 – Machine Overview - Operator GUI: Material Information

The button **Admin GUI** switches to the Administrator GUI (see [Administrator GUI](#)) of the machine overview.

7.2.2 Administrator GUI

The Administrator GUI of the machine overview shows a systematic view of all modules, selected components and cassettes and substrates, which are actual on the handling system and/or in the process modules.

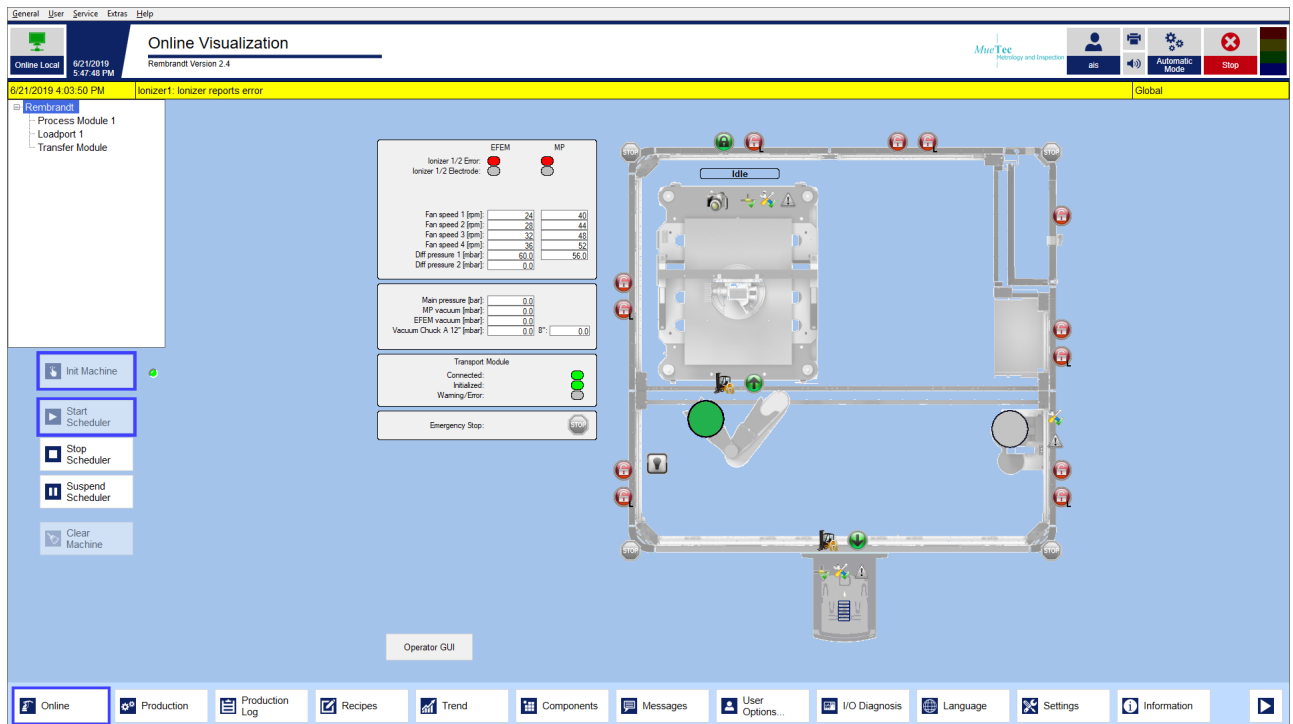


Figure 23 – Online Visualization Machine Overview: Administrator GUI

States of modules and components are displayed as following:

Information	Description	
Door state	For processing the cover of the machine needs to be closed. The state of the door sensors are displayed as following.	
		The door is opened.
Emergency Stop state		The door is closed.
	There are four emergency stop buttons on the machine. If one is pressed, the machine goes in stop. The status of the emergency stop button is displayed as following.	
		There is no emergency stop active.

		An emergency stop button is active.
Wafer state	If a wafer is inside the transfer module or in the process module a symbol shows the actual processing state.	
		The wafer is on the module but not processed.
		The wafer is processing in the module.
		The wafer is processed.
		The wafer was processed but with failure, or there is a material present mismatch in the material tracking system.
Inspection state	A symbol in the process module shows the actual state of the inspection. A click with the left mouse button opens the manual operation dialog for process module (see ↗ Process Module Manual Operation).	
		There is no inspection active.
		The inspection is in progress.
		The inspection is finished.
Reader State	A symbol shows the state of the substrate id reader. A click with the left mouse button opens the manual operation dialog for the substrate id scanner (see ↗ ID Reader Manual Operation). When no scanner is installed this symbol will not be shown.	
		The scanner is installed and has no failures.

		The scanner has warnings.
		The scanner is in error. E.g. connection error.
Service Lamp state	A symbols shows the actual state of the service lamp. A click with the left mouse button opens the manual operation dialog for the service lamp (see Service Lamp Manual Operation).	
		The lamp is off.
		The lamp is on.
Connection state	A symbol indicates the connection state of a module.	
		The module is not connected.
		The module is connected.
Initialization state	A symbol indicates the initialization state of a module.	
		The module is not initialized.
		The module is initialized.
Module/Component state	A symbol indicates the actual state of the module/component.	
		The module/component has no errors and no warnings.
		The module/component has at least one warning, but no error.
		The module/component has at least one error.
Transfer module handling state	Arrow symbols show the actual handling of the transfer module to and from a process module or loadport.	

		No handling is active.
		The transfer of a wafer from the robot to or from a module is active.
	 blinking	The transfer of a wafer from the robot to or from a module is active. And one of the I/O channel "STAT_ACT_C1", "STAT_ACT_C2", "STAT_ACT_C3", "STAT_ACT_S1" or "STAT_ACT_S2" is active.
	 blinking	The transfer of a wafer from the robot to or from a module is not active. And one of the I/O channel "STAT_ACT_C1", "STAT_ACT_C2", "STAT_ACT_C3", "STAT_ACT_S1" or "STAT_ACT_S2" is active.
Transport Module	In the area Transport Module additional state information to the transport module are shown.	
	Connected	The background color of the state field shows the connection state: • Green: The transport module is connected. • Gray: The transport module is not connected.
	Initialized	The background color of the state field shows the initializing state: • Green: The transport module is initialized.

		Gray: The transport module is not initialized.
	Warning/Error	The background color of the state field shows the error state: - Red: An error on the transport module is active. - Gray: The transport module has no errors.
Emergency Stop	In the area Emergency Stop the overall state of the emergency stop buttons are shown. Is one of the emergency stop buttons active the state changes as following	
		There is no emergency stop active.
		An emergency stop button is active.

Table 24 – Machine Overview - Administrator GUI: Status Information

In the upper right area of the Administrator GUI selected measurement values from the EFEM and measurement platform (MP) are shown:

	EFEM	MP
Ionizer 1/2 Error:		
Ionizer 1/2 Electrode:		
Fan speed 1 [rpm]:	24	40
Fan speed 2 [rpm]:	28	44
Fan speed 3 [rpm]:	32	48
Fan speed 4 [rpm]:	36	52
Diff pressure 1 [mbar]:	60.0	56.0
Diff pressure 2 [mbar]:	0.0	
Main pressure [bar]:	0.0	
MP vacuum [mbar]:	0.0	
EFEM vacuum [mbar]:	0.0	
Vacuum Chuck A 12" [mbar]:	0.0	
Vacuum Chuck B 12" [mbar]:	0.0	

Figure 24 – Online Visualization Machine Overview: Measurement Values

Measurement value	Description
Fan speed 1[rpm]	Shows the actual value of the speed of the fan 1 in the EFEM and Measurement platform.
Fan speed 2[rpm]	Shows the actual value of the speed of the fan 2 in the EFEM and Measurement platform.
Fan speed 3[rpm]	Shows the actual value of the speed of the fan 3 in the EFEM and Measurement platform.
Fan speed 4[rpm]	Shows the actual value of the speed of the fan 4 in the EFEM and Measurement platform.
Diff pressure 1 [mbar]	Shows the actual value of the difference pressure in the EFEM and in the Measurement platform.
Diff pressure 2 [mbar]	Shows the actual value of the second difference pressure sensor in the EFEM.
Main pressure [bar]	Shows the actual value of the main pressure sensor in the machine.
MP vacuum [mbar]	Shows the actual value of the vacuum in the measurement platform.
EFEM vacuum [mbar]	Shows the actual value of the vacuum in the EFEM.
Vacuum Chuck A 12" [mbar]	Shows the actual value of the vacuum on the 12-inch Chuck in process module 1.
Vacuum Chuck A 8" [mbar]	Shows the actual value of the vacuum on the 8-inch Chuck in process module 1 (Rembrandt only).
Vacuum Chuck B 12" [mbar]	Shows the actual value of the vacuum on the 12-inch Chuck in process module 2.
Vacuum Chuck B 8" [mbar]	Shows the actual value of the vacuum on the 8-inch Chuck in process module 2 (Rembrandt only).

Table 25 – Machine Overview - Administrator GUI: Measurement Values

The button **Operator GUI** switches to the Operator GUI (see [General](#)) of the machine overview.

7.2.3 Functions

The side bar of the machine overview and process module visualization contains several buttons. Following table lists the functions which can be executed with these buttons.

Online Visualization Functions	
Init Machine	Initializes all modules and components of the machine. First the robot will be initialized. After the robot is in home position the loadports and the process modules will be initialized parallel. After restart, the button is only operable if: all modules are connected to the devices they monitor, no emergency stop is active, and the host interface is started (if installed). The button is not operable if initialization is active, emergency stop is active, the scheduler is active, or the operator does not have the "Automatic Mode Operation" right. A LED symbols shows the state of the initialization
	red = not initialized, or emergency stop is active
	green = initialized
	yellow = one module or components is not initialized
	blue = initialized but one emergency stop alarm messages is still active
Start Scheduler	Starts the scheduler. Control and process jobs sent by the host or created by the Job Creator are started. The button can be pressed if the system is initialized, no emergency stop is active, the scheduler is not active, and the operator has the "Automatic Mode Operation" right.

Online Visualization Functions	
Suspend Scheduler	Suspends the scheduler. The button can be operated if the scheduler is active and the operator has the right "Automatic Mode Operation" right.
Stop Scheduler	Stops the scheduler. Current control jobs and process jobs are stopped. The button can be operated if the scheduler is active and the operator has the right "Automatic Mode Operation" right.
Clear Machine	Cancels all active process jobs and transports the material located in the machine to its origin slot in the origin carrier. The button can be operated if the system is initialized, no emergency stop is active and the operator has the right "Automatic Mode Operation".

Table 26 – Online Visualization Functions

7.3 Process Module Visualization

7.3.1 General

The visualization of the module developed by MueTec is displayed in the main area of the process module visualization.

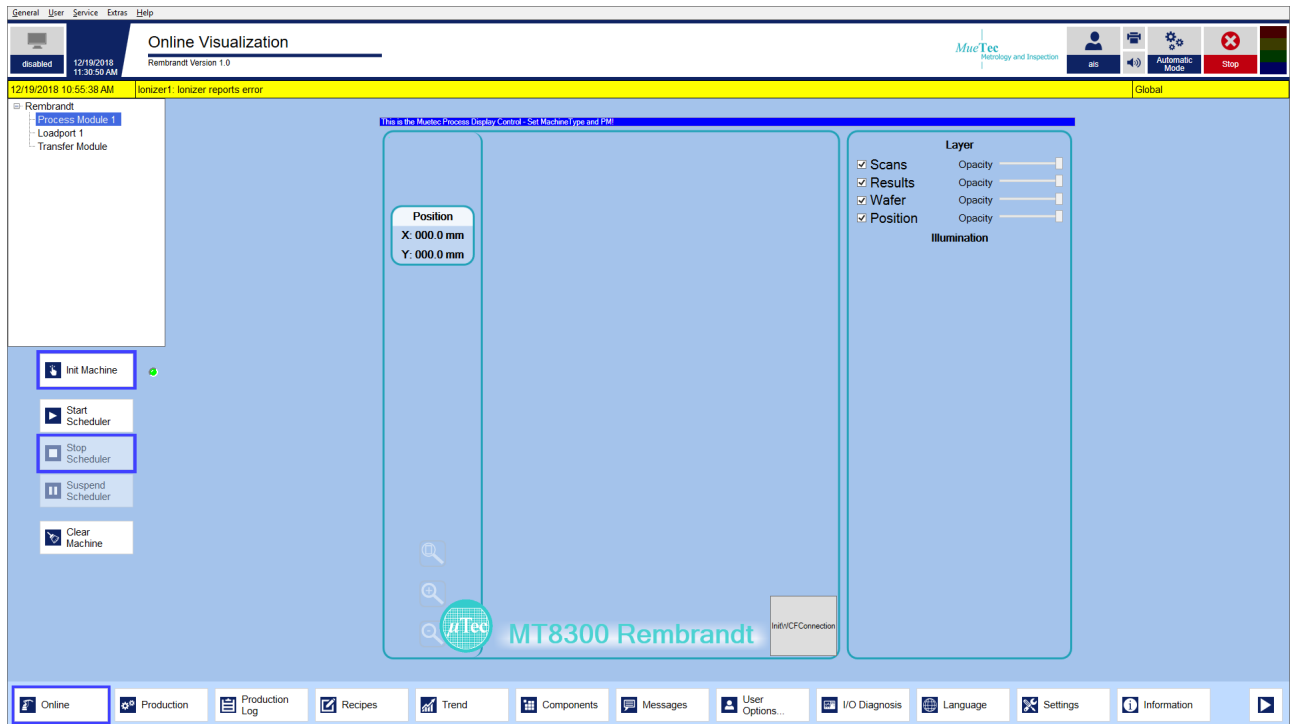


Figure 25 – Process Module Visualization

7.3.2 Functions

The side bar of the machine overview and process module visualization contains several buttons. Following table lists the functions which can be executed with these buttons.

Online Visualization Functions	
Init Machine	Initializes all modules and components of the machine. First the robot will be initialized. After the robot is in home position the loadports and the process modules will be initialized parallel.

Online Visualization Functions

	After restart, the button is only operable if: all modules are connected to the devices they monitor, no emergency stop is active, and the host interface is started (if installed). The button is not operable if initialization is active, emergency stop is active, the scheduler is active, or the operator does not have the "Automatic Mode Operation" right. A LED symbols shows the state of the initialization red = not initialized, or emergency stop is active green = initialized yellow = one module or components is not initialized blue = initialized but one emergency stop alarm messages is still active
Start Scheduler	Starts the scheduler. Control and process jobs sent by the host or created by the Job Creator are started. The button can be pressed if the system is initialized, no emergency stop is active, the scheduler is not active, and the operator has the "Automatic Mode Operation" right.
Suspend Scheduler	Suspends the scheduler. The button can be operated if the scheduler is active and the operator has the right "Automatic Mode Operation" right.
Stop Scheduler	Stops the scheduler. Current control jobs and process jobs are stopped. The button can be operated if the scheduler is active and the operator has the right "Automatic Mode Operation" right.

Online Visualization Functions	
Clear Machine	Cancels all active process jobs and transports the material located in the machine to its origin slot in the origin carrier. The button can be operated if the system is initialized, no emergency stop is active and the operator has the right "Automatic Mode Operation".

Table 27 – Online Visualization Functions

7.4 Loadport Visualization

7.4.1 General

The visualization of a loadport module shows the status of the possibly loaded cassette and the contained substrates.

Depending on the tool configuration, communication with the load port is as follows:

- Brooks loadport: TCP/IP interface
- TDK loadport: serial RS232 interface
- Fortrend loadport: TCP/IP-interface

7.4.2 Module State

This section of the Loadport view shows an overview of different states related to the Loadport.

Module State	
Component states	
Ctrl Connection	This LED symbol shows the state of the communication between the visualization of the module and its control.  blinking Communication interrupted
	 Communication established
Mapping	This LED symbol shows the state of the mapping process

		Mapping is not in process
		Mapping is in process
Eq Connection	This LED symbol shows whether the control is connected to the load port device.	
		Load port device is not connected
		Load port device is connected
Initialized	This LED symbol shows whether the Loadport is initialized.	
	 blinking	Loadport is not initialized
		Loadport is initialized
Warnings	This LED symbol signals active warnings.	
		no warning
	 blinking	Warning active
Alarms	This LED symbol signals active alarms.	
		no alarm
	 blinking	One or more alarms are active.
Cassette Present	This LED symbol shows whether a cassette is currently present or not.	
		Cassette not present
		Cassette present
General Status Models		
EPT Status	This text box shows the actual EPT state of the module. For more information see SEMI E116.	

RAM Status	This text box shows the actual RAM state of the module. For more information see SEMI E10.	
Mode	This text box shows the actual state of the operating mode model.	
	Production	Loadport is in production mode, no manual access possible.
	Maintenance	Loadport is in maintenance mode, no production process possible.
	Disabled	Loadport is disabled and will be ignored by TC control (neither automatic nor manual operation is possible).
Access Mode	This text box shows the actual state of the SEMI E87 access mode model.	
	Manual	The loadport will be loaded manually.
	Auto	The loadport will be loaded over the AMHS via the E84 Interface.
State Models with Hostinterface	When the Hostinterface is activated, the actual values of additional SEMI E87 state models are shown.	
Reservation State	This text box shows the actual state of the SEMI E87 reservation state model.	
	Not Reserved	This state is active when there is no reservation existing for the loadport.
	Reserved	This state is active when there is a reservation for further activity on the loadport.

Transfer State	This text box shows the actual state of the SEMI E87 transfer state model.	
	Out of Service	The state is active, when the load port is set to out of service.
	Ready To Load	This state is active when there is no carrier on the loadport.
	Transfer Blocked	This state is active when a carrier is loaded and is processing.
	Ready To Unload	This state is active when the loadport is available for unloading of a carrier from the loadport to material handling equipment. The material of the carrier was processed, the carrier is undocked and not clamped.
Association State	This text box shows the actual state of the SEMI E87 Loadport/Carrier association state model.	
	Not Associated	There is no carrier association present for the loadport.
	Associated	A Carrier ID has been associated with the loadport. The loadport is not available for new carrier association.
E84	The background color code of this text box indicates the actual state of the E84 Handoff Available Signal.	
	gray	The E84 Handoff Available Signal is switched off.

	green	The E84 Handoff Available Signal is switched on.
RFID	The background color code of this text box indicates the actual state of the RFID Reader connection.	
	gray	The RFID reader for the Carrier ID is not connected.
	green	The RFID reader for the Carrier ID is connected.

Table 28 – Information Box Module State Loadport

7.4.3 Sequence

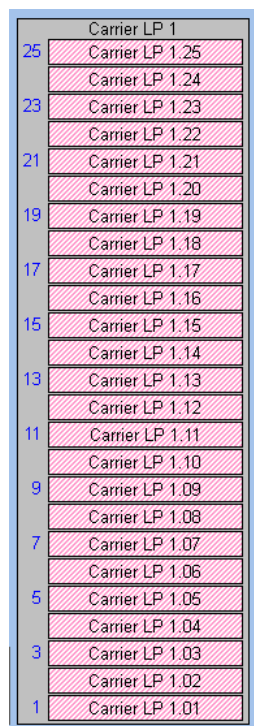
This section of the loadport view shows the actual command the loadport is currently executing and its state.

Sequence	
Sequence	Shows the actual command the loadport ist executing
State	Shows the state of the command the loadport is executing

Table 29 – Information Box Sequence Loadport

7.4.4 Carrier

The list in the lower right corner shows the state of the substrates in a present carrier. The background color of each substrate indicates its actual state. For more detailed information about these states see ➔ [Substrate Information](#).



Carrier LP 1	
25	Carrier LP 1.25
	Carrier LP 1.24
23	Carrier LP 1.23
	Carrier LP 1.22
21	Carrier LP 1.21
	Carrier LP 1.20
19	Carrier LP 1.19
	Carrier LP 1.18
17	Carrier LP 1.17
	Carrier LP 1.16
15	Carrier LP 1.15
	Carrier LP 1.14
13	Carrier LP 1.13
	Carrier LP 1.12
11	Carrier LP 1.11
	Carrier LP 1.10
9	Carrier LP 1.09
	Carrier LP 1.08
7	Carrier LP 1.07
	Carrier LP 1.06
5	Carrier LP 1.05
	Carrier LP 1.04
3	Carrier LP 1.03
	Carrier LP 1.02
1	Carrier LP 1.01

Figure 26 – Load port visualization: Wafer List

This list is not visible if no carrier is present. Clicking on the frame of this scheme or on the 3D model opens a 3D view of the carrier and the substrates inside. A click on one of the substrates opens the dialog with detailed substrate information (see ➔ [Substrate Information](#)).

FOUP Details Modules

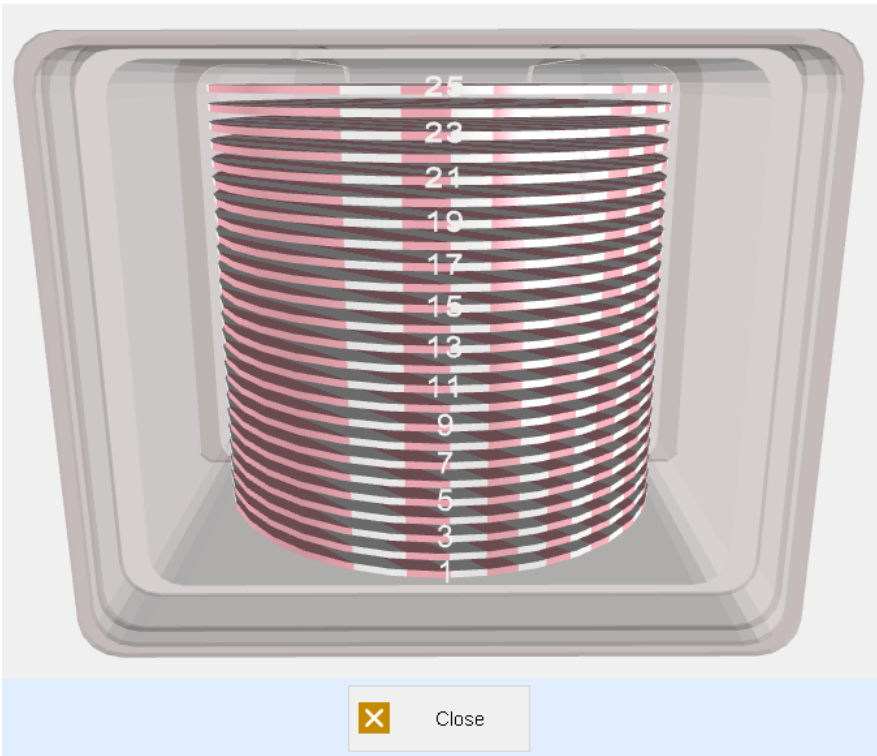


Figure 27 – Load port visualization:Carrier Details

7.4.5 Functions

Function	Description
Carrier Management...	This button opens the command dialog of Carrier Management for this Loadport.
Stop Module	This button stops the module. All running Loadport, E84 or RFID-Reader commands will be stopped.

Table 30 – Loadport Functions

7.4.5.1 Carrier Management

This dialog provides commands to manage a carrier if the Loadport is in state “Online Local” or “Offline”.

Functions
Carrier

Proceed With Carrier	This button is used to continue with the current carrier. It is assumed that the carrier ID is correct and all slots have the correct wafers.
Cancel Carrier	This button cancels carrier processing.
Carrier Recreate	This button recreates the current carrier.
Service Status	
Out of Service	Used to tell the host system the Load Port is out of service.
In Service	Used to tell the host system the Load Port is ready for process ("In Service").
Access Mode	
Manual	Enables manual load operation for operator.
Automatic	Activates automatic loading for this Load Port by an AHMS. Manual load operations are disabled if the automatic access mode is active.

Table 31 – Carrier Management

7.4.6 Manual Mapping

The dialog for manual mapping will be opened after loading a carrier, when the loadport mapping mode is set to **Manual** or the option **Use Manual Mapping Adjustment** is activated.

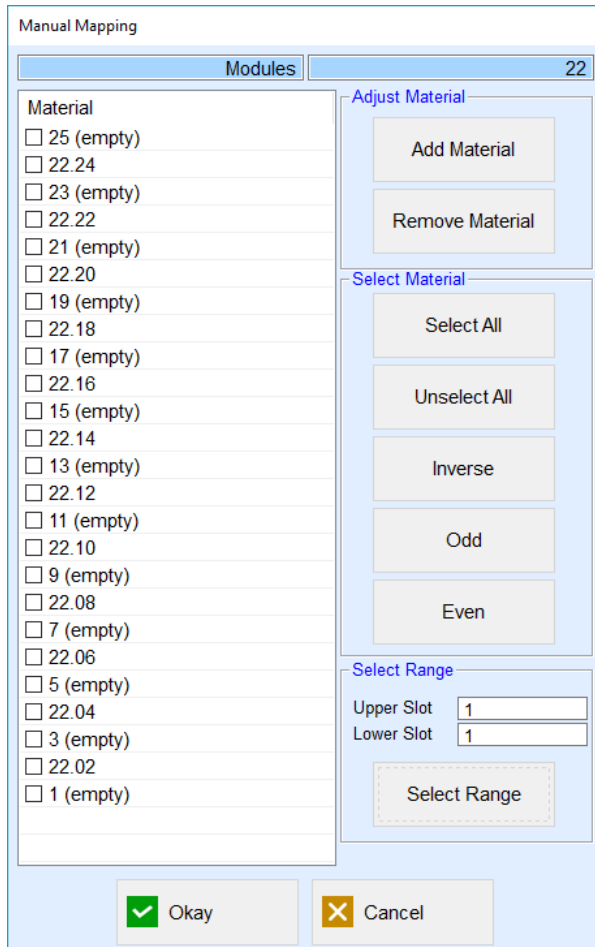


Figure 28 – Loadport Visualization: Manual Mapping Dialog

Manual Mapping	
Material	In this section the slots of the carrier are shown with the current occupancy state of the slots. A check box left from the slot number displays the current selection state of the slot.
Add Material	Adds to the selected slot/slots a material.
Remove Material	Removes from the selected slot/slots a material.
Select Material	

Select All	Selects all slots of the carrier.
Unselect All	Resets the actual slot selection.
Inverse	Invert the actual slot selection. Selected slots will be marked as unselected and unselected slots will be marked as selected.
Odd	Selects every slot with a odd number (1, 3, 5 ...).
Even	Selects every slot with a even number (2, 4, 6 ...).
Select Range	
Upper Slot	In this field the upper slot for the range selection can be defined.
Lower Slot	In this field the lower slot for the range selection can be defined.
Select Range	With this buttons all slots will be selected according to the settings in the field Upper Slot and Lower Slot .
Finish Mapping	
Okay	This button closes the dialog and the mapping process will be finished. After the mapping validation the Job Creator Dialog will be opened.
Cancel	This button closes the dialog without any modification.

Table 32 – Loadport: Manual Mapping Functions

7.4.7 Job Creator Dialog

After mapping of the load port the **Job Creator Dialog** is shown if the machine is not controlled by a SECS/GEM Host. In this dialog substrates for processing can be selected and the automatic control job execution can be started.

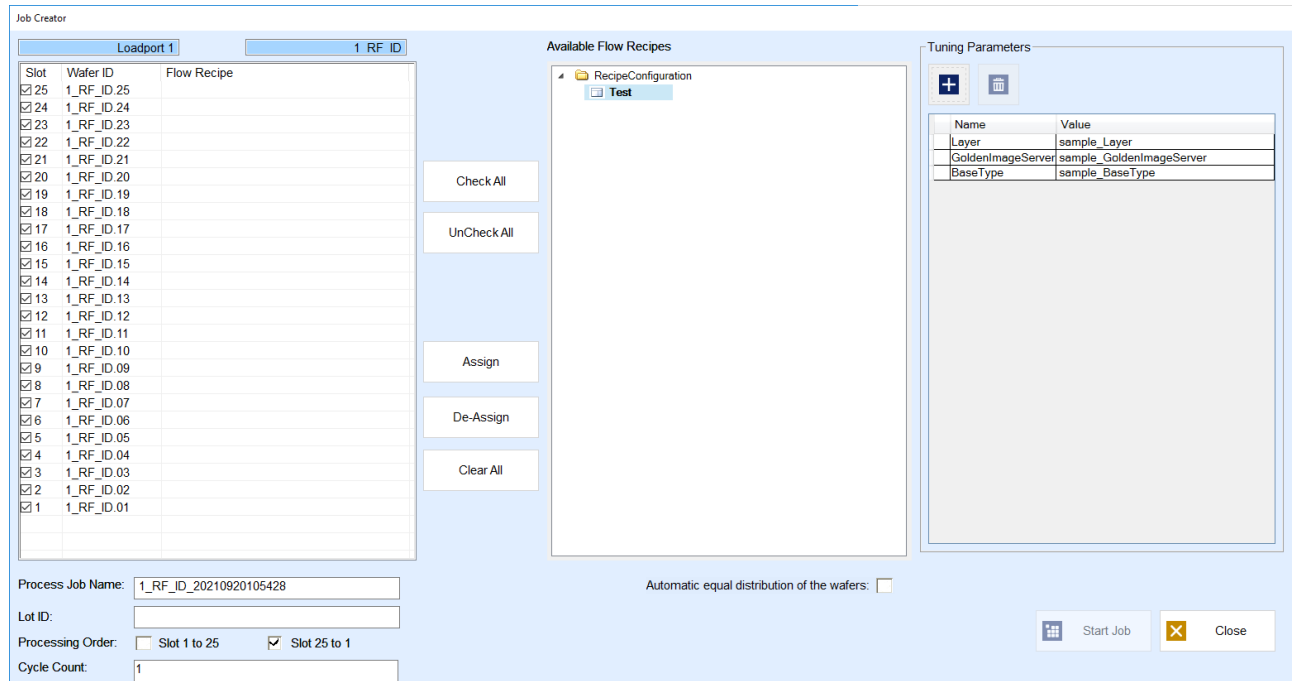


Figure 29 – Job Creator Dialog

The user must enter following information:

Information	Description
Carrier Information	This list-like area shows all process-relevant information about each slot and the loaded wafer of a carrier.
Selection Box	This check box selects the slot (wafer) for the next user operation in the Job Creator dialog. Empty slots are disabled.
Slot number	Right from the check box the slot number is displayed.
Wafer ID	This column shows the wafer ID when the slot is occupied.

Flow Recipe	This columns shows the assigned flow recipe to the wafer in the slot.
Available Flow Recipes	All available flow recipes are displayed in a recipe tree. The operator selects a recipe from this tree. After pressing Assign , the selected recipe is assigned to the selected substrates.
Commands	
Check All	This button selects all occupied slots for further processing.
UnCheck All	This button cancels the selection for all occupied slots for further processing.
Assign	This button assigns the selected recipe in the recipe tree to the selected slots (wafers). Existing assignments are overwritten.
De-Assign	Existing assignments of the selected slots (wafers) are removed.
Clear All	Removes all assignments and cancels the selection off all slots.
Processing Parameter	
Process Job Name	Defines the name of the process job, to generate. This value can be edited. The default name is the RFID-Tag of the cassette placed.
Lot ID	Defines the Lot-ID of the material in the carrier.
Processing Order	Defines the processing order of the wafers. Slot 1 to 25: The wafers will be processed in an ascending order, starting on the slot one. Slot 25 to 1: The wafers will be processed in an descending order, starting on the slot 25. The default processing order is defined in Machine Parameters .

Cycle Count	Defines the number of executions of the process job. If the cycle count is higher than one, the process jobs will be repeated until the cycle count is reached.
Automatic equal distribution of the wafers	Defines whether the automatic equal distribution should be applied to the processing order of the wafers.

Table 33 – Job Creator Dialog

After the recipe assignment the user can press the **Start Job** button. The software will create a new process job and, if necessary, a parent control job. The job will be started automatically.

If the mapping result indicates substrates in state cross or double slotted, an error message will be displayed. As long as the error message has not been acknowledged, the load port module will be in state error and can not be used.

Tuning Parameters

Tuning Parameters belongs to the tuning parameters of a job. They are transferred to the process module each time a wafer is started. Tuning parameters can be added or removed. The column Name provides a combobox which allows only selection of not already used names. The value of the tuning parameter can be edited in column Value. If current value is equals to the default value a "D" will be displayed in the row. The parameter "TuningParametersConfigPath" can be used to define a file with allowed tuning parameters.

7.4.8 Stop Process Job

It is possible to stop running process jobs, by clicking the StopProcess Job... button. Now the operator can select one of the currently running process jobs and afterwards click the Stop Job... button. Jobs can only be stopped at Stop Process Job dialog of their source or target loadport.

The StopProcess Job... button is disabled if the user does not have the right "Allow stop job".

7.4.9 Carrier Type Detection

The selection of a carrier type is only available for the tool configuration **Rembrandt/DaVinci**.

When the carrier is placed on the load port the carrier type will be detected automatically. The automatically carrier type (200 mm or 300 mm FOUP) detection is based on the coding of the info pads on the carrier. If the carrier type is unknown an error message will be shown "Unknown carrier type" and the processing of the carrier is blocked.

If the Parameter "ManualCarrierTypeValidation" (see [Configuration General](#)) is set to "True", the automatic detected carrier types need to be validated by the operator. If the carrier type is known and the Host Communication Mode is "Online Local" or "Offline", a message "Carrier placed on load port. Choose the carrier type ..." and the carrier type selection dialog will be shown. There the operator has to confirm the detected carrier type.

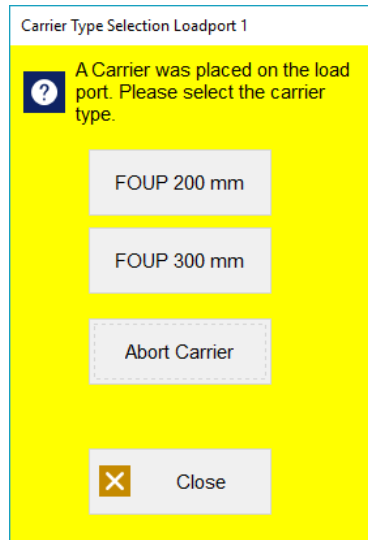


Figure 30 – Loadport Visualization: Carrier Type Selection

The operator can choose one of the following options:

- **FOUP 200 mm:** If the automatically detected carrier type is a 200 mm FOUP the carrier will be created. If not, an error message will be shown: "Wrong carrier type selected". The processing of the carrier will be blocked.
- **FOUP 300 mm:** If the automatically detected carrier type is a 300 mm FOUP the carrier will be created. If not, an error message will be shown: "Wrong carrier type selected". The processing of the carrier will be blocked.
- **Abort Carrier:** No carrier will be created. The processing of the carrier will be blocked.
- **Close:** Closes the dialog. The operator can confirm the carrier type in the message system.

In "Online Remote Mode" only 300 mm FOUPs can be processed. The processing of 200 mm FOUPs will be blocked. The assignment of the coding pads to the FOUP type can be done in the configuration file (see [Configuration General](#)).

7.5 Transport Module Visualization

7.5.1 General

The visualization of the transfer module shows a schematic view of the robot and the status of the wafers on the robot arms. If the robot is clicked with the left mouse button then a window will be opened which allows manual control and parameterization of the component (see [Transfer Module Manual Operation](#)). If the aligner is installed the overall state and material information of the aligner will be displayed in the screen. If the aligner is clicked with the left mouse button then a window will be opened which allows manual control and parameterization of the component (see [Aligner Manual Operation](#)).



Figure 31 – Transfer Module Visualization

Color	State
Blue	Basic state
Green	The transfer module is executing a command.
Yellow	At least one warning assigned to this module is active.
Red	At least one error assigned to this module is active.

Table 34 – Background color of Transfer Module

Color	State
Blue	Basic state
Green	Aligner is executing a command.
Yellow	At least one warning assigned to this module is active.

Red	At least one error assigned to this module is active.
-----	---

Table 35 – Background color of Aligner Module

7.5.2 Module State

The status of the transport module is displayed in a box in the right upper corner.

Module State		
Component states		
Communication	This LED symbol shows the state of the communication between the visualization of the module and its control.	
	 blinking	Communication interrupted
		Communication established
Active	This LED symbol shows the Transfer Module active state.	
		Transfer Module is not active.
		Transfer Module is executing a transfer command.
Warnings	This LED symbol signals active warnings.	
		no Warning
	 blinking	Warning active
Alarms	This LED symbol signals active alarms.	
		no Alarm
	 blinking	Alarm active

Table 36 – Information Box Module State Transfer Module

If the status information box of the module is clicked with the left mouse button then a window is opened where it is possible to change the operating mode and parameterization of the transport module.

7.5.3 Robot State

Status		
Connected	This LED symbol shows whether the robot is connected.	
		Robot is not connected.
		Robot is connected.
Initialized	This LED symbol indicates whether the robot is initialized	
		Robot is not initialized. Running a Initialize command is required before command executing is possible.
		Robot is initialized and ready for use.

Table 37 – Information Box Robot States

7.5.4 Wafer Information Robot

For the transport module an information window is visible which contains information about the wafer presence status, and wafer information for every installed blade.

TM - Wafer Information Upper Blade

Occupation State
Wafer ID
Process Job ID
Flow Recipe

TM - Wafer Information Lower Blade

Occupation State
Wafer ID
Process Job ID
Flow Recipe

Figure 32 – Wafer Information Robot

Wafer Information Robot		
Occupation state	Displays the actual occupation state of the upper or lower blade of the robot.	
	Empty	No wafer is detected by the robot on the current blade.
	Wafer detected	The robot sensors detects a wafer on the current blade.
Wafer ID	Displays the ID of the wafer which is currently on the blade. If no wafer is present the field will be empty.	
Process Job ID	Displays the ID of process job for the wafer which is currently on the blade. If no wafer is present the field will be empty.	
Flow Recipe	Displays the name of flow recipe for the wafer which is currently on the blade. If no wafer is present the field will be empty.	
Background color	If the wafer present status from the robot and the material tracking system is different, the background color will be changed to "Red". If its equal the standard background color will be shown.	

Table 38 – Wafer Information Transfer Module

7.5.5 Aligner State

Status		
Connected	This LED symbol shows whether the aligner is connected.	
		Aligner is not connected.
		Aligner is connected.
Initialized	This LED symbol indicates whether the robot is initialized	

		Aligner is not initialized. Running a Initialize command is required before command executing is possible.
		Aligner is initialized and ready for use.

Table 39 – Information Box Aligner States

7.5.6 Wafer Information Aligner

An information window shows the wafer detection state of the aligner and the material tracking system.

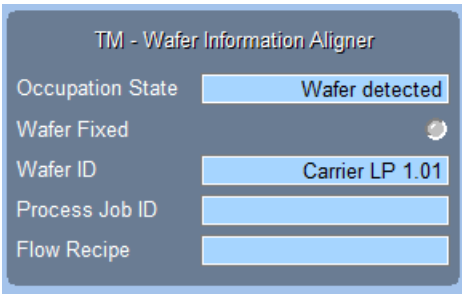


Figure 33 – Wafer Information Aligner

Wafer Information Aligner		
Occupation state	Displays the actual occupation state of the aligner	
	Empty	There is no wafer detected from the aligner.
	Wafer detected	The sensors on the aligner detects a wafer.
Wafer Fixed	This LED symbol indicates whether the wafer is fixed (clamped) on the aligner.	
		The wafer is not fixed.
		The wafer is fixed.
Wafer ID	Displays the ID of the wafer which is currently on the aligner. If no wafer is present the field will be empty.	

Process Job ID	Displays the ID of process job for the wafer which is currently on the aligner. If no wafer is present the field will be empty.
Flow Recipe	Displays the name of flow recipe for the wafer which is currently on the aligner. If no wafer is present the field will be empty.
Background color	If the wafer present status from the aligner and the material tracking system is different, the background color will be changed to "Red". If its equal the standard background color will be shown.

Table 40 – Information Box Wafer state aligner

7.5.7 Sequence

This section of the transport module view shows the actual command the transport module is currently executing and its state.

Sequence	
Sequence	Shows the actual command the transport module is executing (e.g. "Initialization", "Pick").
State	Shows the state of the command the transport module is executing.

Table 41 – Information Box Sequence Transfer Module

7.5.8 Stop Module

In the left bottom of the screen there is a button to Stop Module. If this button is clicked then the robot motion is aborted. An alarm Stop requested is generated. As long as the alarm is unacknowledged the transport module is in error state and not used by the scheduler.

8

Production Dialog

8 Production Dialog

The ToolCommander Production screen supports initial operation assistance and system diagnostics. Inconsistent material tracking information can be adjusted over this dialog.

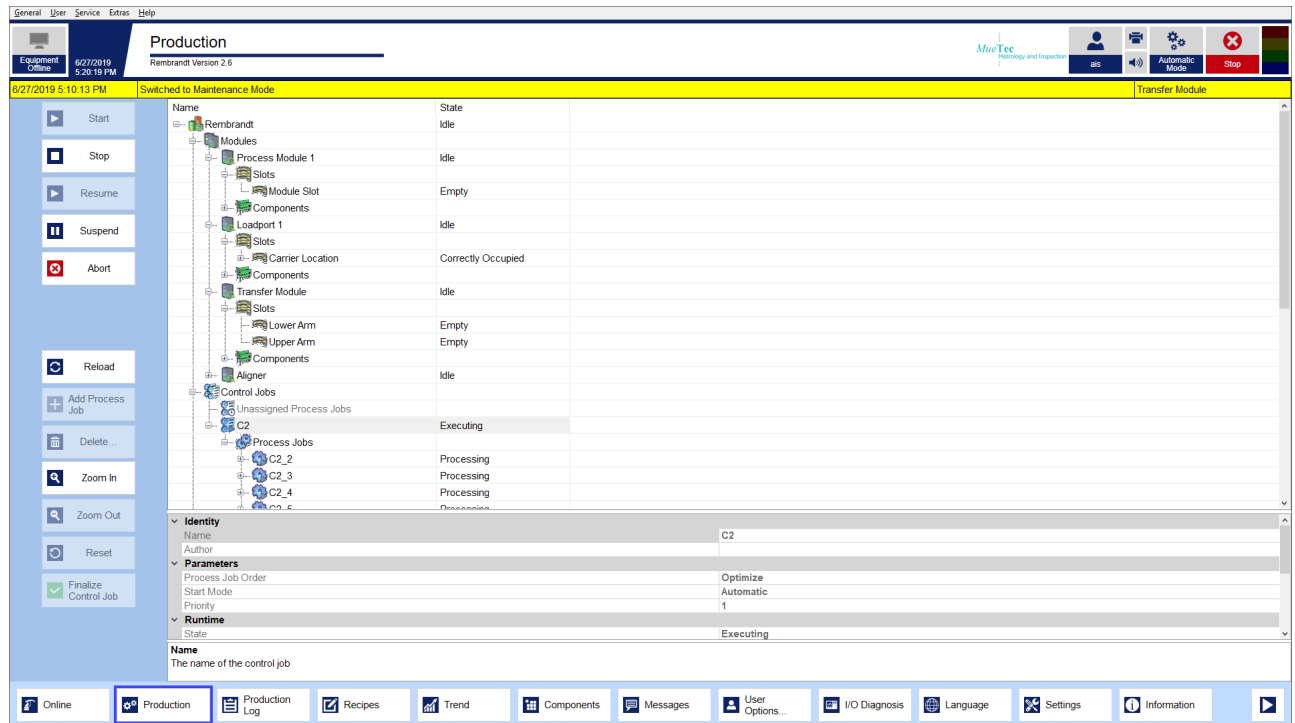


Figure 34 – Production Dialog

The functionality of the production dialog is described in detail in the ToolCommander® - User manual chapter 3.

9

Components

9 Components

9.1 Concept

The control system of the machine is hierarchically structured into modules. Each module can include several components. A component usually has its own device parameters, manual operation and messages which are sent to the message system of the ToolCommander® framework. Different components offer the possibility to indicate their analog values in a separate online trend window. All components used within this machine are described in the following chapters.

The parameters of the components can be only modified by users with the right "Change Parameters". The function of the manual operation of the components are only available for users with the right "Manual Operation".

9.2 Substrate Information

The substrate component is used to display detail information for a specific substrate. This dialog appears whenever the operator clicks on substrate information shown in a module detail view.

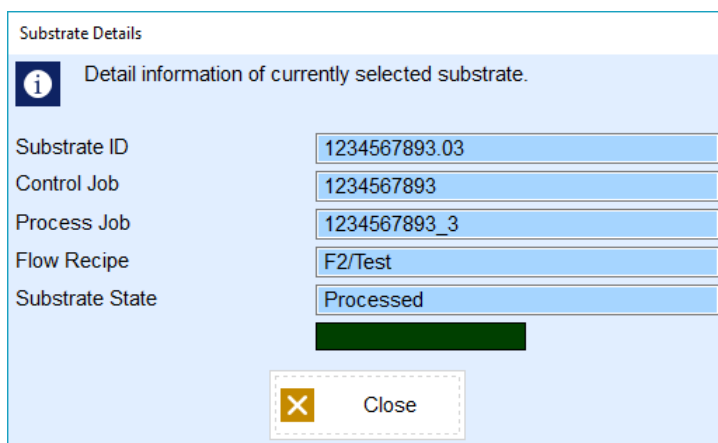


Figure 35 – Dialog Substrate Details

Title	Description
Wafer ID	This textbox shows the unique ID of the selected substrate.
Control Job	This textbox shows the name of the control job the substrate belongs to. For more information about control and process jobs see ToolCommander® User Documentation.

Process Job	This textbox shows the name of the process job for the selected substrate. For more information about control and process jobs see ToolCommander® User Documentation.
Flow Recipe	This textbox shows the name of the flow recipe that is used for the selected substrate. For more information about flow jobs and recipes see ToolCommander® User Documentation.
Substrate Status	This textbox shows the substrate status. The colored field below the text frame indicates the state corresponding to the actual substrate state. For more information see ToolCommander® User Documentation.

Table 42 – Details Substrate Information

The state of a substrate will be displayed as following:

State	Color, Pattern	Color, Pattern (Text)
UNDEFINED or without process job		gray filled
NEEDS PROCESSING		magenta/ white striped
IN PROCESS		white filled
PROCESSED		darkgreen filled
REJECTED		orange filled
STOPPED		turquoise filled
ABORTED		red filled
LOST		red/ yellow striped

SKIPPED		blue filled
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Table 43 – Visualization of Material Process States

9.3 Aligner

9.3.1 Description

The Aligner is controlled by a Yaskawa SR100 controller. The control system communicates with the aligner via TCP/IP. In automatic mode, the alignment angle is taken from the process module recipe. If an ID reader is installed, the wafer is first aligned with the reading angle (see machine [Parameters](#)), after reading the wafer is aligned with the original angle and then aligned with the angle specified in the process module recipe.

9.3.2 Parameters

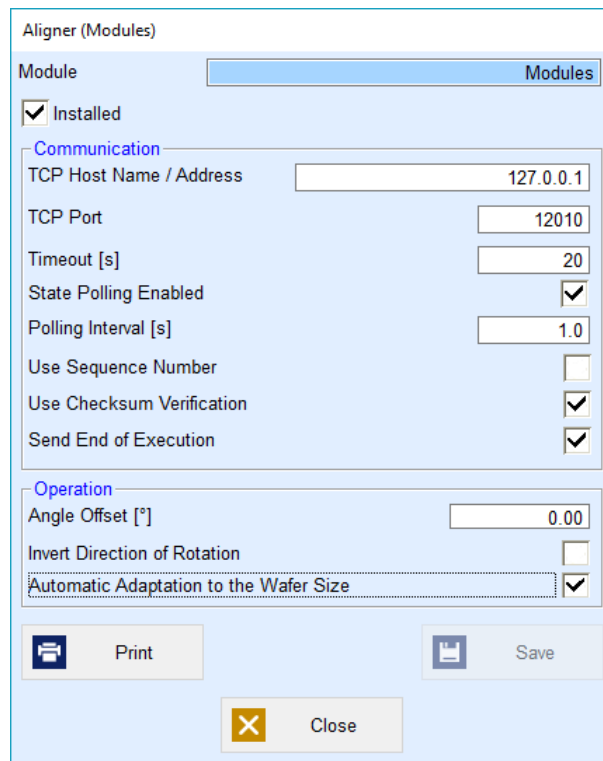


Figure 36 – Aligner: Parameters

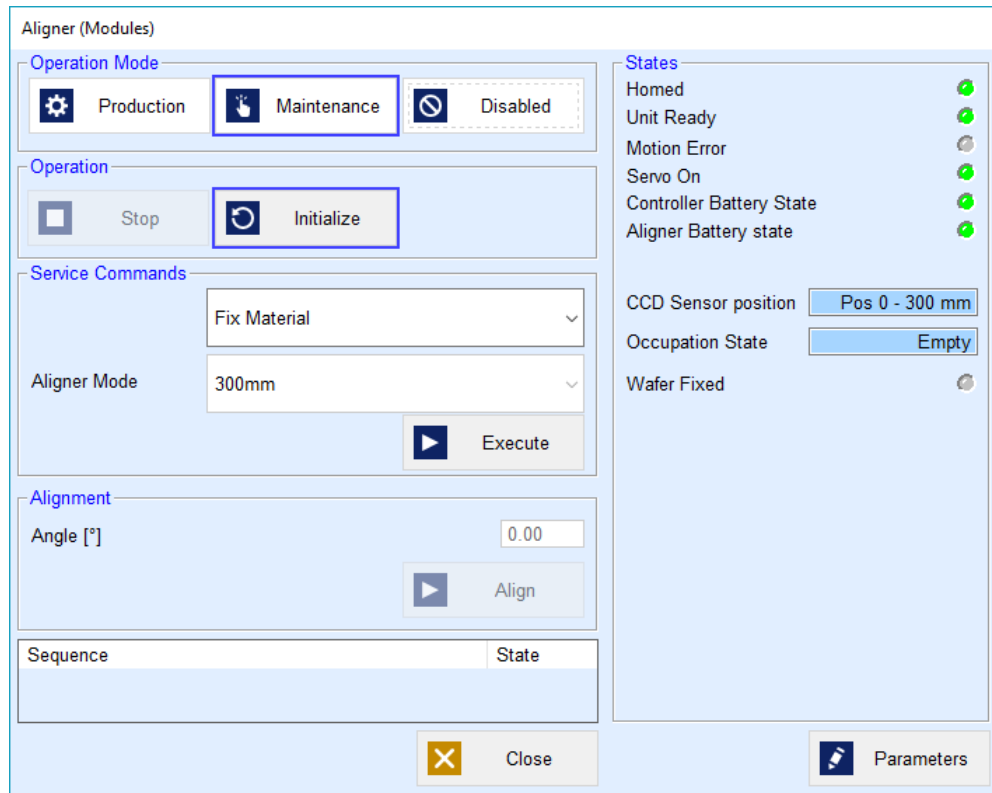
Parameter	Description	Name
Installed	Activates or deactivates the aligner for the system.	selection [yes / no]
Communication		

TCP Host Name/ Address	TCP/ IP address for the communication with the aligner.	string [0..255 characters]
TCP Port	Defines the port for the communication with the aligner.	range [0..65535]
Timeout [s]	Defines a timeout for communication. If there is no answer for a request (command) within this time an error is generated.	range [0.1.. 3600.0]
State Polling Enabled	Determines whether the device driver should send status queries to the aligner hardware at a defined interval.	selection [yes / no]
Polling Interval [s]	Defines the interval at which the status command is to be sent to the aligner hardware.	range [0.1.. 10.0]
Use Sequence Number	Defines whether sequence numbers are to be used for command telegrams to the aligner or not. This parameter should be enabled, when the device is configured to use sequence numbers	selection [yes / no]
Use Checksum Verification	Defines whether check sum validations are activated or not. This parameter should be enabled, when the device is configured to use telegrams with check sum.	selection [yes / no]

Send End of Execution	Defines whether a command finished from the aligner should be acknowledged from the controller or not. If this parameter is set to false the aligner will remain in busy state after sending an alignment or control command.	selection [yes / no]
Operation		
Angle Offset	An angle offset in degree which is used during the alignment. This value is add to every alignment angle, to the default alignment angle for reading of 200 mm and 300 mm wafer, and the alignment angle defined in the module recipe.	range [-360.00.. 360.00]
Invert Direction of Rotation	Defines whether the alignment should be performed inversely. This value is used for the calculation of the alignment setpoint for the reading angle of 200 mm and 300 mm wafer, and the alignment angle defined in the module recipe.	selection [yes, no]
Automatic Adaption to the Wafer Size	Defines whether the CCD-Sensor position of the aligner can be modified automatically. If the value is set to "True" the CCD-Sensor position will be adapted automatically to the size of the present wafer before the wafer is placed on the aligner. If the value is set to "False" the CCD-Sensor position will not be changed.	selection [yes, no]

Table 44 – Parameters Aligner

9.3.3 Manual Operation



Aligner (Modules)

Operation Mode

Production Maintenance Disabled

Operation

Stop Initialize

Service Commands

Fix Material

Aligner Mode 300mm

Execute

Alignment

Angle [°] 0.00

Align

States

Homed Unit Ready Motion Error Servo On Controller Battery State Aligner Battery state

CCD Sensor position Pos 0 - 300 mm

Occupation State Empty

Wafer Fixed

Sequence State

Close Parameters

Figure 37 – Aligner: Manual Operation

Manual Operation

Operation

Stop	Stops the current operation.
Initialize	This button brings the aligner in the home position. Running this command is required before any other command executing is possible.

Service Commands

Fix Material	This service command clamp the wafer.
Release Material	This service command release the wafer.
Set mode	This service command adjust the CCD sensor position on the aligner to the selected wafer size (200 or 300 mm).

Aligner Mode	If the command Set mode is selected, the wafer size (200 or 300 mm) can be defined over this selection box. The wafer size will be used as additional parameter for the service command execution.	
Execute	This button is used to execute the currently selected service command.	
Alignment		
Angle [°]	Specifies the angle which is used for the next alignment command.	
Align	This button aligns the wafer present on aligner.	
Sequence		
Sequences	Shows the actual command the aligner is executing	
State	Shows the state of the command the aligner is executing.	
States		
Homed	This LED symbol indicates the initialization state of the aligner.	
	 blinking	The aligner was not referenced. The execution of an initialize command is required before further commands can be executed.
		The aligner is referenced and ready for use.
Unit Ready	This LED symbol indicates whether the aligner is ready.	
		The aligner is not ready. It is executing a command.
		The aligner's ready. A new command can be executed.
Motion Error	This LED symbol indicates whether the aligner has detected an error.	

		The aligner is in error.
		The Aligner has no error.
Servo On	This LED symbol indicates whether the servo drive is switched on.	
		The servo drive is switched off.
		The servo drive is switched on.
Controller Battery State	This LED symbol indicates the battery status of the Aligner controller.	
		The battery is empty.
		The battery is full.
Aligner Battery State	This LED symbol indicates the battery status of the Aligner.	
		The battery is empty.
		The battery is full.
CCD Sensor Position	This text field displays the current CCD sensor position. The sensor has two positions "Pos 0 - 300 mm" or "Pos 1-200 mm".	
Occupation State	The text field indicates whether the aligner is currently empty or there is a wafer on the aligner. Is a wafer of the aligner the wafer id will be shown. If there is no wafer the text "Empty" will be shown.	
Wafer fixed	This LED symbol indicates whether the wafer is fixed (clamped) on the aligner.	
		The wafer is not fixed.
		The wafer is fixed (clamped).

Table 45 – Manual Operation Aligner

9.3.4 Messages

Alarms are only activated if the device is installed.

Exception	Reaction, recovery
Error during read/write of database [xxx]	A database error occurred during a read/ write process. • Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Connection lost	Connection to Aligner has been lost. If the error is active the transport module is blocked (active transports will be finished). • Retry: restarts communication. • Abort: aborts the wafer. Aligner is not blocked and usable for future (if necessary to bypass the aligner the user has to set the "Ignore Aligning" parameter) • Ignore: wafer is marked as aligned and the process continues (Aligner will not be disabled for further processing).
Communication timeout	This error is raised if a command timed out. If the error is active the transport module is blocked (active transports will be finished). • Acknowledge: restarts communication
A timeout occurred during command execution	The system raises this error if an transfer command timed out. If the error is active the transport module is blocked. • Retry: repeats the transfer command. • Abort: aborts the transfer of the wafer. The robot is not blocked and usable for future.

	• Ignore: wafer is marked as transferred and the process continues (robot will not be disabled for further processing)
Command execution failed. Command='[%Command %]' was responded with level 1 warning = '[%ErrorDescription%]'	The aligner responds to a command (%Command%) with a warning level 1 error (Communication related error). The failure code and the text are displayed in the %ErrorDescription%. • Abort: aborts the command. The aligner is not blocked and usable for future.
Command execution failed. Command='[%Command %]' was responded with level 2 error = '[%ErrorDescription%]'	The aligner responds to a command (%Command%) with a warning level 2 error. The failure code and the text are displayed in the %ErrorDescription%. • Retry: repeats the command. • Abort: aborts the command. The aligner is not blocked and usable for future. • Ignore: command is marked as executed. The process continues (aligner will not be disabled for further processing)
Command execution failed. Command='[%Command %]' was responded with level 1 error = '[%ErrorDescription%]'	The aligner responds to a command (%Command%) with a level 1 error. The failure code and the text are displayed in the %ErrorDescription%. • Retry: repeats the command. • Abort: aborts the command. The aligner is not blocked and usable for future. • Ignore: command is marked as executed. The process continues (aligner will not be disabled for further processing)
Command execution failed. Command='[%Command %]' was responded with level 2 error = '[%ErrorDescription%]'	The aligner responds to a command (%Command%) with a level 2 error. The failure code and the text are displayed in the %ErrorDescription%.

	• Abort: aborts the command. The aligner is not blocked and usable for future.
Command execution failed. Command='[%Command%]' was responded with fatal error ='[%ErrorDescription%]'	The aligner responds to a command (%Command%) with a fatal error. The failure code and the text are displayed in the %ErrorDescription%. • Abort: aborts the command. The aligner is not blocked and usable for future.
Devices not initialized	The aligner devices is not initiatialized.
Aligner sensor detected no material, but the tracking system has a material on aligner	The material presence sensor on the aligner detects material, but the tracking system does not know any material on the aligner. To solve this error create material in the ↗ Production Dialog .
Aligner sensor detected no material, but the tracking system has a material on aligner	The material presence sensor on the aligner does not detect material, but the tracking system has material on the aligner. To solve this error delete the material in the ↗ Production Dialog .

Table 46 – Messages Aligner

9.3.5 Interlocks

The alignment is not possible when:

- the machine is not initialized.
- one of the emergency off button is pressed
- or the wafer is not fixed.

9.4 External Carrier ID Reader

9.4.1 Description

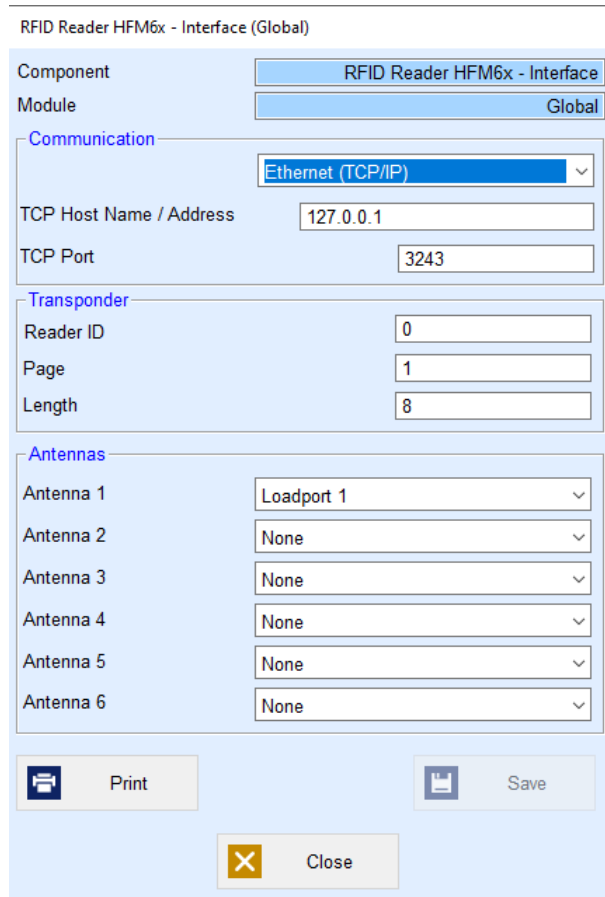
The following 2 external readers can currently be used to read an RF-ID reader from the carrier. They are defined in the system configurator.

Hermos HF80 (RFID Reader HFM6x): This ID reader has 6 antennas, each of which is positioned at a load port.

Hermos LFM4x: This ID reader has 4 antennas, each of which is positioned at a load port.

9.4.2 Parameters

Parameter RFID Reader HFM6x



RFID Reader HFM6x - Interface (Global)

Component: RFID Reader HFM6x - Interface

Module: Global

Communication

Ethernet (TCP/IP)

TCP Host Name / Address: 127.0.0.1

TCP Port: 3243

Transponder

Reader ID: 0

Page: 1

Length: 8

Antennas

Antenna 1: Loadport 1

Antenna 2: None

Antenna 3: None

Antenna 4: None

Antenna 5: None

Antenna 6: None

Print Save Close

Figure 38 – RFID Reader HFM6x - Interface: Parameters

Name	Description	Range
Communication type	Defines the type of communication.	selection [Undefined, Ethernet (TCP/IP), Serial (COM) 1]
TCP Host Name / Address	The TCP/IP address of the RFID Reader HFM6x - Interface. This option is only available if the RFID Reader HFM6x - Interface communicates via TCP/IP.	string [0..255 characters]

Name	Description	Range
TCP Port	The TCP/IP port number. This option is only available if the RFID Reader HFM6x - Interface communicates via TCP/IP.	range [0..65535]
COM Port	COM Port (Serial Port Number) where the RFID Reader HFM6x - Interface is connected to. This option is only available if the RFID Reader HFM6x - Interface communicates via Serial interface.	range [0..2147483647]
Reader ID	Defines the device ID of the RFID Reader HFM6x - Interface .	range [0..9]
Page	Defines the page number to read.	range [1..99]
Length	Defines the number of characters to read from the page.	range [1..100]
Antenna 1-6	Defines the assignment of an antenna to a loadport.	

Table 47 – Parameters RFID Reader HFM6x - Interface

Parameter Hermos LFM4x

RFID Reader LFM 4x (Global)

Component

RFID Reader LFM 4x

Module

Global

Communication

TCP Host Name/ Address:

127.0.0.1

TCP Port:

3242

Antennas

Antenna 1:

Loadport 1

Antenna 2:

Loadport 2

Antenna 3:

-

Antenna 4:

-

 Print

 Save

 Close

Figure 39 – RFID Reader LFM4x - Interface: Parameter

Name	Beschreibung	Bereich
TCP Hostname / Adresse	The TCP/IP address of the RFID Reader LFM4x - Interface. This option is only available if the RFID Reader LFM4x - Interface communicates via TCP/IP.	Zeichenkette [0..255 Zeichen]
TCP Port	The TCP/IP port number. This option is only available if the RFID Reader LFM4x - Interface communicates via TCP/IP.	Bereich [0..65535]

Name	Beschreibung	Bereich
Antenne 1-4	Defines the assignment of an antenna to a load port. Note. Antenna 1 addresses the number 0 on the interface. This value can be moved using the "AntennaCountStartsWith" setting on the Hermos Reader in the MuetecApplication.ini.	

Table 48 – Parameter Komponente RFID Reader Hermos LFM4x

9.5 Fan Filter Unit (FFU)

9.5.1 Description

The values for rotational speed, error signals and pressures of the Fan Filter Unit are transmitted by a B&R PLC, which communicates via Modbus TCP/IP with the control system. The component monitors the conditions at EFEM and measurement platform.

9.5.2 Parameters

FFU_EFEM (Global)

Component
FFU_EFEM

Module
Global

☒ Installed

Limit Control

	Alarm High	Warn High	Warn Low	Alarm Low
Diff pressure limit [Pa]	500.0	400.0	100.0	50.0

Fan Control

Number of Fans	3
Fan speed [rpm]	30
Fan speed tolerance [%]	10.00
Fan speed delay [sec]	300
Fan speed tol. time [sec]	300

Number of Filters
2

Print

Save

Close

Figure 40 – FFU: Parameters

Name	Description	Range
Installed	Activates or deactivates the component for FFU-Monitoring.	selection [yes/no]
Diff pressure limit [Pa]: Alarm High	Defines the uppermost bound for difference pressure. If the difference pressure becomes lower, an alarm is created.	range [0..500]
Diff pressure limit [Pa]: Warn high	Defines the upper bound for difference pressure warnings. If the difference pressure becomes lower, a warning is created.	range [0..500]
Diff pressure limit [Pa]: Warn Low	Defines the lower bound for difference pressure warnings. If the difference pressure becomes lower, a warning is created.	range [0..500]
Diff pressure limit [Pa]: Alarm low	Defines the lowermost bound for difference pressure. If the difference pressure becomes lower, an alarm is created.	range [0..500]
Number of Fans	Number of fans at FFU. According to the fan count the corresponding input/output channels for rotational speed will be monitored. If the value is 0, no channels will be monitored.	range [0..4]
Fan speed [rpm]	Defines the set point for rotational speed. This value will be transmitted to the fan control when the machine starts.	range [0..2600]

Name	Description	Range
Fan speed tolerance [%]	Defines the tolerance for fan speed. If the speed is out of borders, an alarm will be generated. Refer to Fan speed tol. time.	range [0..100]
Fan speed delay [min:s]	The tolerance supervision is delayed by this time. Allows fan to speed up.	range [0:00...60:00]
Fan speed tol. time [min:s]	If the fan speed is out of tolerance for longer than defined by Fan speed tol time, an alarm will be generated.	range [0:00...60:00]
Number of filters	The number of filters in FFU. Depending on this value the corresponding input/output channels for differential pressures are monitored. If the value is 0, no pressures will be monitored.	range [0..4]

Table 49 – Parameters Component Fan Filter Unit

9.5.3 Manual Operation

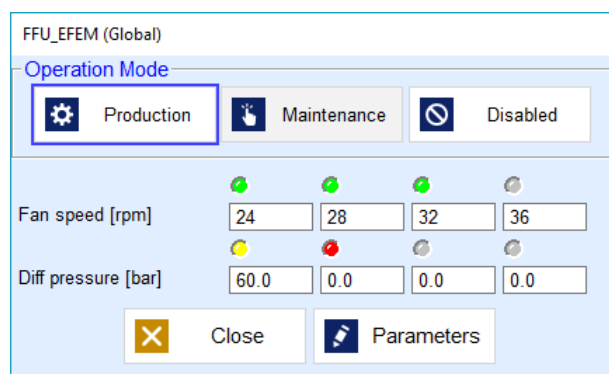


Figure 41 – FFU: Manual Operation

Name	Description	Type
States		
Fan speed 1..4 [rpm]	Shows the current rotational speed of every fan according to the configured number of fans. The state of tolerance monitoring is displayed separately for every fan and value.	value and LED symbol
Diff pressure [bar]	Shows the filters actual differential pressure according to the configured number of filters. The state of differential pressure monitoring is displayed separately for every filter and value.	value and LED symbol

Table 50 – Manual Operation Component Fan Filter Unit

9.5.4 Messages

Exception	Reaction, recovery
Error during read/ write of database [xxx]	- A database error occurred during a read/ write process. - Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Rotational speed tolerance warning fan [1..4]	The rotational speed of fan [1..4] is out of bounds. "Fan speed tol. time" is not over. No recovery action available

Exception	Reaction, recovery
Rotational speed tolerance error fan [1..4]	The rotational speed of fan [1..4] was longer than "fan speed tol. time" out of borders. • No recovery action available
Differential pressure on filter [1..4] lower than "Diff pressure limit [Pa]: Warn Low"	The differential pressure on filter [1..4] was lower than the lower bound for warnings. • No recovery action available
Differential pressure on filter [1..4] higher than "Diff pressure limit [Pa]: Warn High"	The differential pressure on filter [1..4] was higher than the higher bound for warnings. • No recovery action available
Differential pressure on filter [1..4] lower than "Diff pressure limit [Pa]: Alarm Low"	The differential pressure on filter [1..4] was lower than the lower bound for alarms. As long as this alarm is not committed, the module is blocked and no command can be executed. • Retry: Commits alarm and restarts monitoring. • Cancel: Commits alarm and blocks module until differential pressure on filter [1..4] is higher than lowermost bound again. • Ignore: Commits alarm and unblocks module.
Differential pressure on filter [1..4] higher than "Diff pressure limit [Pa]: Alarm High"	The differential pressure on filter [1..4] was higher than the higher bound for alarms. As long as this alarm is not committed, the module is blocked and no command can be executed. • Retry: Commits alarm and restarts monitoring. • Cancel: Commits alarm and blocks module until differential pressure on filter [1..4] is lower than uppermost bound again. • Ignore: Commits alarm and unblocks module.

Table 51 – Exceptions component Fan Filter Unit

9.5.5 Interlocks

- The scheduler stops processing wafers if the component Fan Filter Unit shows any uncommitted error.
- The component cannot be used if the controller is not connected to the PLC.

9.6 Host Interface

The control system has a host interface for 300mm and GEM services.

If the machine is to be operated with the host interface activated, the corresponding parameter (see below) must be set. After activation, the host interface application starts on the control PC of the system.

During shutdown of the TC Control the host interface application will be shut down automatically.

If the tool operates in control state “Online Remote” internal interlocks for production operation of the tool (jobs) will be enabled. This means the user cannot create or modify any control or process jobs. Additionally carrier management/ handling (e.g. cancel carrier, proceed with carrier, dock, undock) cannot be operated locally (only from host).

In “Online Local”, “Offline” or if the host interface is disabled these interlocks are deactivated. That means the according functionality for carrier and job management can be operated locally at the tool.

For more information about this service see the “Host Interface” manual.

9.6.1 Parameters

Host Interface (Global)

Function Selection

 Operation

 Parameters

Component

Host Interface

Module

Global

☒ Enable

Communication

Communication Mode

Server

PPM Send Mode

Unformatted

Host Address

127.0.0.1

TCP/IP Port

5000

HSMS Parameter T3 [s]

45

HSMS Parameter T5 [s]

10

HSMS Parameter T6 [s]

5

HSMS Parameter T7 [s]

10

HSMS Parameter T8 [s]

5

Soft Start Timeout [s]

30

Loadports

Bypass Read ID

☐

PPM

Allow Override Flow Recipes

☐

Allow Override Module Recipes

☐

 Print

 Save

 Close

Figure 42 – Host Interface: Parameters

Parameter	Description	Range
Common		
Enable	Enables or disables the host interface. When disabled no host communication is available and no interaction between host and Equipment will be done.	selection [yes, no]
Communication		

Communication Mode	Specifies the communication mode either "Server" or "Client".	selection [Server, Client]
Host Name/ Address	TCP/ IP address for communication. TCP/ IP address or name is not required for communication mode "Server".	string [0..255 characters]
TCP/ IP Port	TCP/ IP port for communication.	range [1.. 65535]
HSMS Parameter T3 [s]	Specifies HSMS parameter T3. For more information see SEMI E5.	range [1.. 120]
HSMS Parameter T5 [s]	Specifies HSMS parameter T5. For more information see SEMI E5.	range [1.. 120]
HSMS Parameter T6 [s]	Specifies HSMS parameter T6. For more information see SEMI E5.	range [1.. 120]
HSMS Parameter T7 [s]	Specifies HSMS parameter T7. For more information see SEMI E5.	range [1.. 120]
HSMS Parameter T8 [s]	Specifies HSMS parameter T8. For more information see SEMI E5.	range [1.. 120]
Soft Start Timeout [s]	Specifies the soft start timeout.	range [1.. 36,000]
Loadports		
Bypass Read ID	Host will not send command to read the carrier ID when activated.	selection [yes, no]
PPM		
Allow Override Flow Recipes	Host can overwrite flow recipes.	selection [yes, no]
Allow Override Module Recipes	Host can overwrite the module recipes.	selection [yes, no]

Table 52 – Parameters Host Interface

9.6.2 Manual Operation

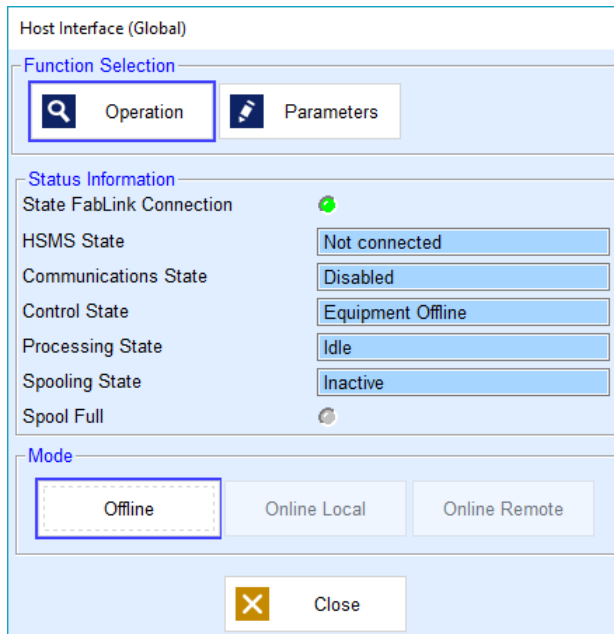


Figure 43 – Host Interface: Manual Operation

Host Interface Manual Operation		
Status Information		
State FabLink Connection	This LED symbol shows the actual FabLink (host interface application) connection state.	
		No connection.
		Connection established.
HSMS State	Shows the actual HSMS state in a readable text format.	
Communications State	Shows the actual communication state in a readable text format.	
Control State	Shows the actual control state in a readable text format.	
Processing State	Shows the actual processing state in a readable text format.	
Spooling State	Shows the actual spooling state in a readable text format.	
Spool full	This LED symbol shows the spool state.	

		Internal spool buffer is ok.
		Internal spool buffer is full.
Mode		
The three buttons in this group can only be operated by users having the user right "Control Fab Host Interface".		
Offline	This button closes the host communication.	
Online Local	This button changes the host communication to online local. Host will be informed about all changes on equipment side. Remote access from host is not possible.	
Online Remote	This button changes the host communication to online remote. Host will be informed about all changes on equipment side and controls the machine remotely.	

Table 53 – Manual Operation Host Interface

9.7 ID Reader

9.7.1 Description

This component communicates with the ID reader. A maximum of two ID readers can be installed. One for the ID on the back side of the material and one for the ID on the front side of the material.

For the tool configuration **Argos** the frame ID (Keyence SR-1000) is read.

For the tool configuration **Rembrandt/DaVinci** the frame ID (HTT WID 120) is read.

A read mode parameter can be used to select whether both ID readers, one ID reader for reading the material ID, or no ID is to be read.

9.7.2 Parameters

ID Reader (Global)

Component

Module

	Front side	Back side
Installed	☒	☐
IP address	127.0.0.1	
Port	20	22
Timeout [ms]	5000	5000

 Print
  Save

 Close

Figure 44 – ID Reader: Parameters

Parameter	Description	Range
Front side		
Installed	Activates or deactivates the substrate ID reader for the substrate frontside	selection [yes / no]
TCP/IP Address	The TCP/IP address of the substrate ID reader.	string [0..255 characters]
TCP/IP Port	The TCP/IP port number.	Range [0..65535]
Timeout [ms]	Defines the communication timeout in milliseconds for the read command. If the read command is not answered in the specified time a error will be displayed.	Range [0..35791394]
Back side		

Installed	Activates or deactivates the substrate ID reader for the substrate backside.	selection [yes / no]
TCP/IP Address	The TCP/IP address of the substrate ID reader.	string [0..255 characters]
TCP/IP Port	The TCP/IP port number.	Range [0..65535]
Timeout [ms]	Defines the communication timeout in milliseconds for the read command. If the read command is not answered in the specified time a error will be displayed.	Range [0..35791394]

Table 54 – Parameters ID Reader

9.7.3 Manual Operation

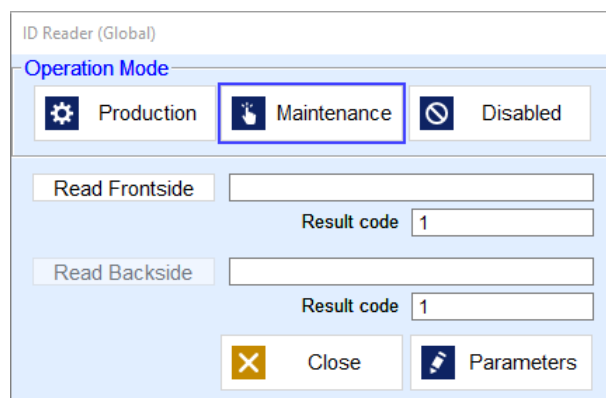


Figure 45 – ID Reader: Manual Operation

Name	Description	Type
Read Frontside	Sends the read command to the front side reader. The reading result is displayed in the text frame right from the button. The result code will be displayed below.	Button (Event)

Name	Description	Type
	If the front side reader is not installed this button is disabled.	
Read Backside	Sends the read command to the back side reader. The reading result is displayed in the text frame right from the button. The result code will be displayed below. If the back side reader is not installed this button is disabled.	Button (Event)

Table 55 – Manual Operation Component ID Reader

9.7.4 Messages

Following table contains the messages and the description of their recovery actions which the component sends to the message server of the ToolCommander® framework. If the component is not installed this messages will not be send.

Exception	Reaction, recovery
Error during read/ write of database [xxx]	A database error occurred during a read/ write process. • Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Invalid IP address configured	The configured IP address string is empty or not valid. • Retry: restarts communication. • Abort: aborts the reading.
The ID reader returned an error code instead of the substrate ID	The ID reader is either not connected or could not read a wafer ID.

	• Retry: retry to read the ID. • Abort: aborts the reading. The substrate ID reader is not blocked and can be used in the future.
--	---

Table 56 – Messages ID Reader

When receiving substrate ID reader specific error message see the corresponding manual for more information.

9.7.5 Interlocks

9.8 Light Tower

9.8.1 Description

The component Light Tower is used to set the signals for the light tower for the tool state according to the configuration.

9.8.2 Parameters

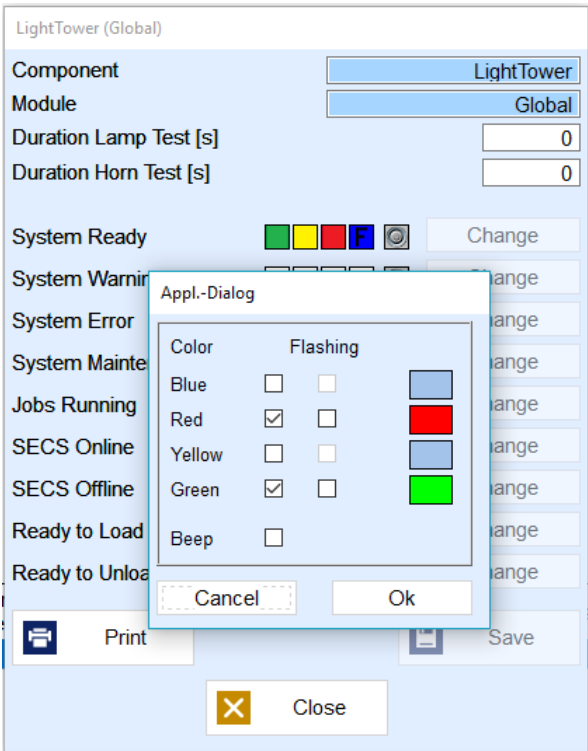


Figure 46 – Parameters Light Tower

For each tool state a combination of light tower signals can be assigned.

Name	Description	Range
Duration Lamp Test [s]	Defines the duration of the lamp test in seconds. The lamp test can be started in the ➔ manual operation dialog of the light tower.	range [0..60]
Duration Horn Test [s]	Defines the duration of the horn test in seconds. The horn test can be started in the ➔ manual operation dialog of the light tower.	range [0..60]
System Ready	Shows the combination of light tower signals when the system is in ready state.	
System Warning	Shows the combination of light tower signals, when the system is in warning.	
System Error	Shows the combination of light tower signals, when the system is in error.	
System Maintenance	Shows the combination of light tower signals, when the system is in maintenance.	
Jobs Running	Shows the combination of light tower signals, when a process jobs is active.	
SECS Offline	Shows the combination of light tower signals, when the SECS connection to an host is in offline mode.	

Name	Description	Range
SECS Online	Shows the combination of light tower signals, when the SECS connection to an host is in online remote mode.	
Ready to Load	Shows the combination of light tower signals, when at least one loadport signals a ready to load.	
Ready to Unload	Shows the combination of light tower signals, when at least one loadport signals a ready to unload.	

Table 57 – Parameters Component Light Tower

The combination of the light tower signals are displayed from left to right in following order:

Field	Description
1. Color field: Green	The field will be white when the signal for green was not selected. If the signal is selected, the field color will be changed to green. If the flashing mode is selected, the letter "F" will be shown in the field.
2. Color field: Yellow	The field will be white when the signal for yellow was not selected. If the signal is selected, the field color will be changed to yellow. If the flashing mode is selected, the letter "F" will be shown in the field.
3. Color field: Red	The field will be white when the signal for red was not selected. If the signal is selected, the field color will be changed to red. If the flashing mode is selected, the letter "F" will be shown in the field.

Field	Description
4. Color field: Blue	The field will be white when the signal for blue was not selected. If the signal is selected, the field color will be changed to blue. If the flashing mode is selected, the letter "F" will be shown in the field.
5. Symbol Horn	The symbol will be marked as disabled when the signal for the horn was not selected. If the signal is selected, the symbol will be marked as enabled.

Table 58 – Parameters Component Light Tower: Signal Combination

The combination of light tower signals can be modified by clicking the **Change** button. A dialog for modification will be shown.

Name	Description	Range
For every color (Blue, Red, Yellow, Green there exist a checkbox for the on/off settings and flashing settings of the signal tower channel.		
Checkbox On/Off	Defines if the signal should be active or not active.	selection [yes, no]
Checkbox Flashing	Determines whether the signal should be in static or flashing mode.	selection [yes, no]
Beep	Defines if the beep signal should be active or not active.	selection [yes, no]

Table 59 – Parameters Component Light Tower: Signal Combination Dialog

9.8.3 Manual Operation

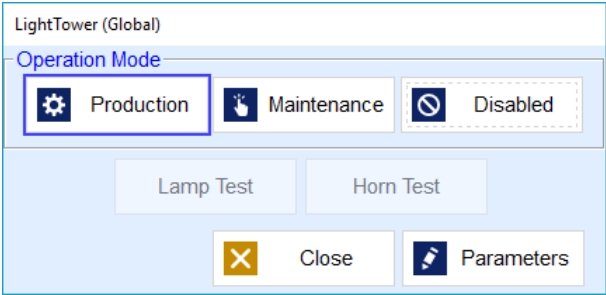


Figure 47 – Light Tower: Manual Operation

Name	Description	Type
Operations		
Lamp Test	Starts the lamp test on the signal tower for the configured time.	Button (Event)
Horn Test	Starts the horn test on the signal tower for the configured time.	Button (Event)

Table 60 – Manual Operation Component Light Tower

9.8.4 Messages

Exception	Reaction, recovery
Error during read/ write of database [xxx]	A database error occurred during a read/ write process. • Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.

Table 61 – Messages Light Tower

9.8.5 Interlocks

The component cannot be used if the connection to the PLC is interrupted.

9.9 Loadport

9.9.1 Description

The component Loadport controls the load port module.

Depending on the machine version, the component communicates with:

- TDK TAS300 Loadports via RS232
- Brooks Vision LEAP TM via TCP/IP
- Fortrend PLM200W Loadports via TCP/IP
- MueTec specialized XML Loadports via TCP/IP

If the module component is not in production mode the scheduler will not be used the module for process job execution.

9.9.2 Mapping

With a TDK and MueTec specialized XML loadport the slot map will be detected by the robot. For other loadports the mapping is performed by the loadport according to the configuration in the parameter dialog (Automatic Mapping, Manual Mapping etc.).

9.9.3 Parameters

Loadport 1 (Modules)

Module

Modules

☒ Installed

Loadport Parameter

E84 Parameter

RFID Reader Parameter

Communication

COM Port

11

Timeout [s]

10000

Device ID

1

Operation

Mapping Mode

Auto Mapping

Use Default Carrier Type

☒

Default Carrier Type

Mapper Installed

☐

Automatic Load

☐

Use Manual Mapping Adjustment

☐

 Print

 Save

 Close

Figure 48 – Loadport: Parameters General (loadport with RS232 communication)

Loadport 1 (Modules)

Module

Modules

☒ Installed

Loadport Parameter

E84 Parameter

RFID Reader Parameter

Communication

TCP Host Name / Address

127.0.0.1

TCP Port

4567

Timeout [s]

10000

Device ID

1

Operation

Mapping Mode

Auto Mapping

Use Default Carrier Type

☒

Default Carrier Type

FOUP 200

Mapper Installed

☐

Automatic Load

☐

Use Manual Mapping Adjustment

☐

 Print

 Save

 Close

Figure 49 – Loadport: Parameters General (loadport with TCP/IP communication)

Loadport Parameter	Description	Range
Installed	Activates or deactivates the loadport completely in the system control.	selection [yes/no]
Communication		
COM Port	COM Port (Serial Port Number) where the loadport is connected to.	range [0..2147483647]

	This option is only available if the loadport communicates via RS232.	
TCP Hostname/Address	The host name or the IP address of the loadport. This option is only available if the loadport communicates via TCP/IP.	string [0..255 characters]
TCP Port	The TCP/IP Port of the loadport. This option is only available if the loadport communicates via TCP/IP.	range [0..2147483647]
Timeout [s]	Defines a communication timeout in seconds. If the loadport does not respond to a request within this time, a corresponding alarm message is generated.	range [0..2147483647]
Device ID	Defines the Device ID to address the loadport.	string [0..255 characters]
Operation		
Mapping Mode	Defines the mapping mode of the loadport.	selection [No Mapping, Auto Mapping, Manual Mapping]

Use Default Carrier Type	Determines whether the default carrier type should be used when setting up a carrier or not. If this value is activated, the carrier with the preset parameter Default Carrier Type is created after setting up a carrier. If the value is deactivated, the carrier type is automatically recognized using the info pad information on the carrier. If the tool is a bridge tool, carrier type recognition is always automatic and the parameter cannot be changed.	selection [yes/no]
Default Carrier Type	Defines the carrier type to be selected when a carrier has been loaded and the loadport has no carrier type detection. Depending on the carrier type, the material type for the substrates is generated after mapping (e.g. wafer 200 mm, wafer 300 mm...).	selection [FOUP 200, FOUP 300, FOUP Frames 380]
Mapper installed	Determines whether the loadport can map carriers or not. If not activated, the loadport does not map a carrier.	selection [yes/no]
Automatic Load	When activated the system loads the carrier automatical after it is docked.	selection [yes/no]
Use Manual Mapping Adjustment	When activated a dialog for manual adjustment of the carrier map will be displayed after automatic mapping.	selection [yes/no]

Table 62 – Parameters Loadport - General

Loadport 1 (Modules)

Module

Modules

☒ Installed

Loadport Parameter

E84 Parameter

RFID Reader Parameter

E84 Timeouts

Timeout TP1 [s]

1

Timeout TP2 [s]

1

Timeout TP3 [s]

60

Timeout TP4 [s]

60

Timeout TP5 [s]

1

Timeout TP6 [s]

2



Print



Save



Close

Figure 50 – Loadport: Parameters E84

E84 Parameter	Description	Range
Timeout TP1 [s]	TR_REQ signal did not turn ON within specified time. This parameter will be send to the loadport.	range [1..999]
Timeout TP2 [s]	BUSY signal did not turn ON within specified time. This parameter will be send to the loadport.	range [1..999]

Timeout TP3 [s]	Carrier was not detected/ removed within specific time. This parameter will be send to the loadport.	range [1..999]
Timeout TP4 [s]	BUSY signal did not turn OFF within specified time. This parameter will be send to the loadport.	range [1..999]
Timeout TP5 [s]	VALID signal did not turn OFF within specified time. This parameter will be send to the loadport.	range [1..999]
Timeout TP6 [s]	VALID signal did not turn ON within specified time. This is only used for continuous mode. This parameter will be send to the loadport.	range [1..999]

Table 63 – Parameters Loadport- E84 Parameters

Loadport 1 (Modules)

Module

☒ Installed

Loadport Parameter | E84 Parameter | RFID Reader Parameter

Auto Read ID ☒

Use Default Carrier ID ☐

Default Carrier ID

 Print  Save

 Close

Figure 51 – Loadport: Parameters RFID Reader

RF ID Reader Parameter	Description	Range
Auto Read ID	When activated the system starts the reading of the carrier id after the carrier was loaded.	selection [yes/no]
Use Default Carrier ID	When activated the default carrier id will be used when the reading of the carrier id failed or no RF Id Reader is installed.	selection [yes/no]
Default Carrier ID	In this field the default carrier ID can be defined.	string [0..255 characters]

Timeout [s]	Defines a timeout for communication. If there is no answer for a request (command) within this time an error is generated. This option is only available if the Loadport has an external RF Id reader.	range [0.1..1000.0]
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Table 64 – Parameters Loadport - RF ID Reader Parameters

9.9.4 Manual Operation

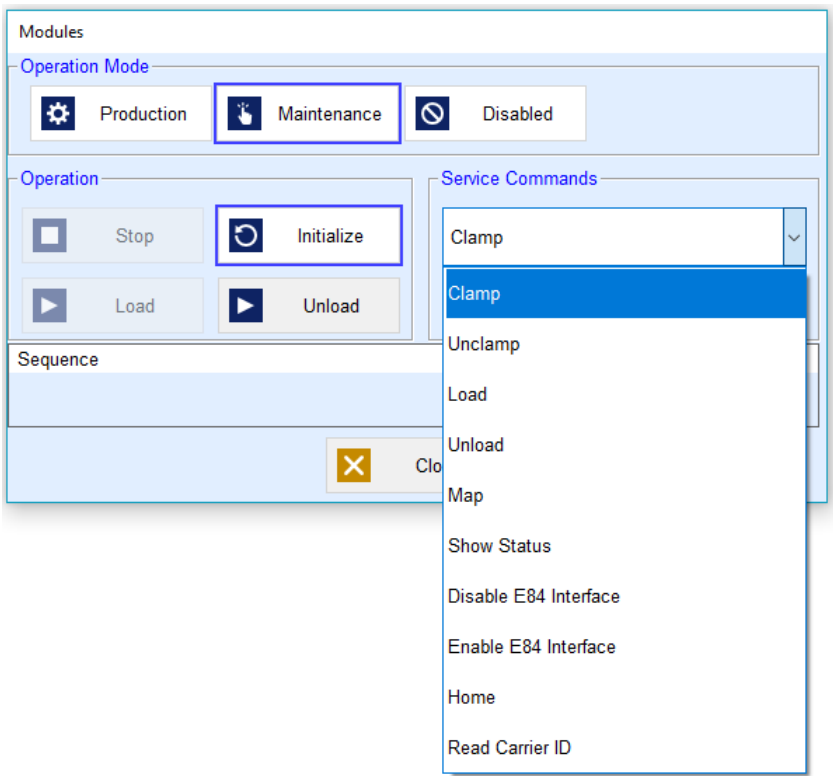


Figure 52 – Loadport: Manual Operation

Name	Description	Type
Operation		
Initialize	Initializes the loadport.	Button [Event]

Name	Description	Type
Load	Loads the current FOUP.	Button [Event]
Unload	Unloads the current FOUP.	Button [Event]
Stop	Stops current activities.	Button [Event]
Service Commands		
Service Commands	This combo box lists all available sequences (commands) of the loadport. The service commands, Clamp, Unclamp, Load, Unload, MAP, Read Carrier ID, HOME, Enable E84, Disable E84, Home can be selected.	Dropdown list [Names of Available Commands / Sequences]
Execute	Starts the execution of the service command selected in the dropdown list.	Button [Event]
State Information		
Sequences	Displays the list of current activities of the Loadport component with the name and status of the sequences (activities).	

Table 65 – Manual Operation Loadport

9.9.5 Messages

9.9.5.1 TC Generated Alarms

Following table contains the messages and the description of their recovery actions which the component sends to the message server of the ToolCommander® framework. If the component is not installed this messages will not be send.

Exception	Reaction, recovery
Error during read/ write of database [xxx]	A database error occurred during a read/ write process.

	• Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Communication timeout	If this error is active Load Port will be blocked. • Retry: Resets communication
No communication response	If the error is active Load Port will be blocked. • Retry: restarts communication
Double slotted wafer or cross loaded wafer detected	If the error is active processing will be stopped. • Retry: remaps the carrier • Ignore: process continues, slots with an invalid wafer state will be disabled for any access. • In maintenance mode: abort
Invalid Mapping Results [Slot 7: mapped state <CorrectlyOccupied>, expected state <Empty>]	This error occurs, when the expected slot map does not fit with the scanned slot map of the carrier. If the error is active handling on the loadport will be stopped. • Remap: reads again the slot map of the carrier • Unload: The carrier will be unloaded.
Invalid Mapping Result: [Crossed: 2]	This error occurs, when the slot map could not be read correctly. If the error is active handling on the loadport will be stopped. • Remap: reads again the slot map of the carrier • Unload: The carrier will be unloaded.
Robot reports error during carrier mapping.	An error occurs on the robot while mapping the carrier. This only affects systems where the slot assignment of the carrier is read by the robot.

	• Remap: reads again the slot map of the carrier • Unload: The carrier will be unloaded.
Nack received: the [%Command%] command is invalid or malformed	The [%Command%] was received by the component, but has invalid parameters. • Abort: cancels the command
Abort received: the device is unable to perform the [%Command%] command at this time because: [%Reason %]	The [%Command%] was validly received by the component, but the component aborted the execution of the command. The abort can be justified by [%Reason%]. • Retry: retries the command • Ignore: cancels the command and proceeds with the next command • Abort: cancels the command
[%Command%] command timed out	No valid response was received to the [%Command%]. • Retry: retries the command • Ignore: cancels the command and proceeds with the next command • Abort: cancels the command

Table 66 – TC Generated Alarms for Loadport

9.9.5.2 Device Generated Alarms

For detailed information about device generated alarms see the Loadport manual.

Note on Fortrend Loadport:

Fortrend Loadports do not report E84 timeouts individually but in groups (TP1/TP2/TP5 and TP3/TP4), so specific alarm for this device is not possible.

- If the dec reports an E84 timeout of the TP1/TP2/ TP5 group, only a TP1 alarm is displayed.

- If the device reports an E84 timeout of the TP3/TP4 group, only a TP3 alarm is displayed.

9.9.6 Interlocks

- Loadport will not respond to host or operator interactions until initialization is complete.
- Loadport will not respond to host or operator interactions if the robot is not in safe position.
- The movement of wafers is only possible if the carrier sensor detects a carrier and the wafer map is known.

9.10 Machine

9.10.1 Description

The component is used to setup global machine parameters for logging, the machine and material type names as well as cluster tool properties and operation of function groups, which concern the entire machine. The component works independently of modules and has no possibility to change the operation mode.

9.10.2 Parameters

Spector 5500 (Global)

Component Spector 5500

Module Global

Machine Name

Sample Application

Parameters

Reading Angle 200mm [°] Default Processing Order: Slot 25 to 1

Reading Angle 300mm [°]

Wafer ID Reader Reading Mode No reading

Set No Reading if not Online Remote ☐

Wafer Process Duration [s] Mapping ☒ Loader ☐ Roboter

Allow Distribution ☒ Equal distribution (step size)

Logging Levels

Load Ports	Verbose	Scheduler	Verbose
TMC	Error	VCTC Server	Error
Robot	Error	Host Interface	Error
Process Modules	Error	IO Device	Error

Logging Data

☒ Move to Folder

Destination Folder

Delete / Move Log Files After Time [h]

Production Log

☒ Delete ☐ Move to Folder

Destination Folder

Delete / Move Production Log After Time [h]

 Print
  Close
  Save

Figure 53 – Machine: Parameters

Parameter	Description	Range
Common		
Machine Name/ Serial Number	The value in this field defines the machine identity.	string
Parameters		

Reading Angle 200 mm [°]	Alignment angle for 200 mm or 150mm wafer for reading the wafer id. If the angle is 0 the wafer will not be aligned before reading the wafer id. This parameter is only used systems with aligner. Note: This setting is only used if the "UseWafer200ReadingPosition" parameter has the value "True". In this case, the wafer is transported to the position of the reader.	range [0.0...360.0]
Reading Angle 300 mm [°]	Alignment angle for 300 mm wafer for reading the wafer id. If the angle is 0 the wafer will not be aligned before reading the wafer id. This parameter is only used systems with aligner. Note: If the angle is greater than 0 and the value of the "UseWafer200ReadingPosition" parameter is "False", the wafer is rotated to this setting independent of the wafer size. In this case, the wafer is not transported to the position of the reader.	range [0.0...360.0]
Wafer Process Duration [s]	Defines the default duration for wafer processing in the process module. This parameter is used for calculation of the progress information in the Operator GUI.	

Allow Distribution	Defines the processing sequence of wafers in multiple process chambers. If this option is activated, the wafers of a cassette are processed in several process modules. If this option is deactivated, the wafers of a cassette are only processed in one process module.	selection [yes, no]
Default Processing Order	Defines the default processing order for the wafer processing started over the Job Creator Dialog .	selection [Slot 1 to 25, Slot 25 to 1]
Equal distribution (step size)	Defines the step size used for the equal distribution of wafers in jobs when the Equal Distribution option is enabled in the Job Creator Dialog. The order of the wafers is defined based on the step size as follows: Each n (step size) wafers is selected and placed at the beginning of the processing order and the remaining wafers are appended.	
Wafer ID Reader Reading Mode	Over this combobox the read mode is defined. Following read modes can be selected: • No Read: The w ID will not be read • Front side: The ID on the front side of the substrate will be read. • Back side: The ID on the back side of the substrate will be read.	selection [0 - No Read, 1 - Front side, 2- Back side, 3 - Both sides mandatory, 4 - Both sides optional]

	- Both sides mandatory: The ID on the back side and the ID on the front side will be read. - Both sides optional: the ID on the back side or the ID on the front side will be read.	
Set Not Reading Mode if not Online Remote	If True, the software resets the ReadMode to "No Reading" when the control state the control state "OnlineRemote" is left.	
Logging Levels		
These values define the logging level of the different components (e.g. Load Ports, TMC, Robot ...)	Logfiles are generated for all modules. The content of these logfiles can be controlled by setting the logging levels. Only log messages with the same or a higher logging level will be written. Log entries with a lower logging level will be ignored. When using 'Verbose' all messages will be logged. The default folder for logfiles is "C:\AIS\ToolCommander\VCTC_InternalLogging\". All logfiles are written in TraceViewer (*.svclog) format.	selection [Critical, Error, Information, Verbose]
Logging Data		
Move to Folder	If this option is set the TC will move the logfiles to the defined folder.	selection [yes, no]
Destination Folder	The path in this field defines the destination folder where the logfiles have to be moved.	string [no limits]
Delete/ Move Log Files After Time [h]	After this time the TC will move or delete logfiles.	range [24...1000]

Production Log		
Delete	If this option is set the TC will delete the production log files.	
Move to Folder	If this option is set the TC will move the production log files to the defined folder.	
Destination Folder	The path in this field defines the destination folder where the production log files have to be moved.	string [no limits]
Delete/ Production Log Files After Time [h]	After this time the TC will move or delete production log files.	range [24...1000]

Table 67 – Parameters Machine Component

9.10.3 Manual Operation

This component does not have manual operations.

9.10.4 Messages

Message	Reaction, recovery
Error during read/write of database [xxx]	Retry: Acknowledges the alarm and repeats the last database action. "[xxx]" contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.

Table 68 – Messages Machine Component

9.11 Media Supply

9.11.1 Description

This component monitors the main air pressure and the main vacuum in the EFEM and on the measurement platform.

9.11.2 Parameters

MediaSupply_EFEM (Global)

Component

Module

☒ Installed

Limit Control

	Alarm High	Warn High	Warn Low	Alarm Low
Air pressure limit [bar]	0.0	0.0	0.0	0.0
Vacuum limit [mbar]	0.0	0.0	0.0	0.0

 Print  Save  Close

Figure 54 – Media Supply: Parameters

Name	Description	Range
Installed	Enables or disables media monitoring.	selection [yes/no]
Air pressure limit [bar]: Alarm High	Defines the uppermost acceptable value for main pressure. If the main pressure gets higher, an alarm will be created.	range [0..10]
Air pressure limit [bar]: Warn High	Defines the highest acceptable value for main pressure. If the main pressure gets higher, a warning will be created.	range [0..10]
Air pressure limit [bar]: Warn Low	Defines the lowest acceptable value for main pressure. If the main pressure gets lower, a warning will be created.	range [0..10]
Air pressure limit [bar]: Alarm Low	Defines the lowermost acceptable value for main pressure. If the main pressure gets lower, an alarm will be created.	range [0..10]

Name	Description	Range
Vacuum limit [mbar]: Alarm High	Defines the uppermost acceptable value for main vacuum pressure. If the pressure gets higher, an alarm will be generated.	range [-1000..0]
Vacuum limit [mbar]: Warn High	Defines the highest acceptable value for main vacuum pressure. If the pressure gets higher, a warning will be generated.	range [-1000..0]
Vacuum limit [mbar]: Warn Low	Defines the lowest acceptable value for main vacuum pressure. If the pressure gets lower, a warning will be generated.	range [-1000..0]
Vacuum limit [mbar]: Alarm Low	Defines the lowermost acceptable value for main vacuum pressure. If the pressure gets lower, an alarm will be generated.	range [-1000..0]

Table 69 – Parameters Component Media Supply

9.11.3 Manual Operation

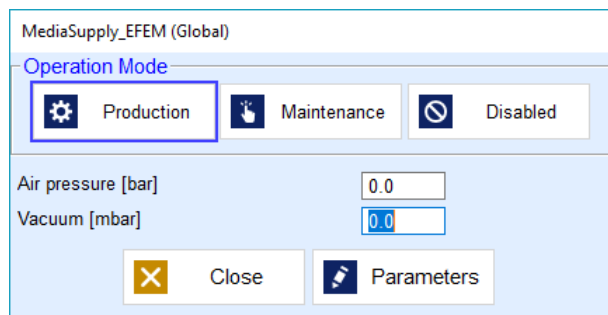


Figure 55 – Media Supply: Manual Operation

Name	Description	Type
States		

Name	Description	Type
Air pressure [bar]	Shows the current value of main pressure.	Value
Vacuum [mbar]	Shows the current value of main vacuum.	Value

Table 70 – Manual Operation Component Media Supply

9.11.4 Messages

Exception	Reaction, recovery
Error during read/ write of database [xxx]	A database error occurred during a read/ write process. • Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Main pressure lower than "Air pressure limit [bar]: Warn Low"	The main pressure was lower than the lower bound for warnings. • No recovery action available
Main pressure higher than "Air pressure limit [bar]: Warn High"	The main pressure was higher than the higher bound for warnings. • No recovery action available
Main pressure lower than "Air pressure limit [bar]: Alarm Low"	The main pressure was lower than the lower bound for alarms. As long as this alarm is not committed, the module is blocked and no command can be executed. • Retry: Commits alarm and restarts monitoring. • Cancel: Commits alarm and blocks module until main pressure is higher than lowermost bound again.

Exception	Reaction, recovery
	• Ignore: Commits alarm and unblocks module.
Main pressure higher than "Air pressure limit [bar]: Alarm High"	The main pressure was higher than the higher bound for alarms. As long as this alarm is not committed, the module is blocked and no command can be executed. • Retry: Commits alarm and restarts monitoring. • Cancel: Commits alarm and blocks module until main pressure is lower than uppermost bound again. • Ignore: Commits alarm and unblocks module.
Main vacuum pressure lower than "Vacuum limit [mbar]: Warn Low"	The main vacuum pressure was lower than the lower bound for warnings. • No recovery action available
Main vacuum pressure higher than "Vacuum limit [mbar]: Warn High"	The main vacuum pressure was higher than the higher bound for warnings. • No recovery action available
Main vacuum pressure lower than "Vacuum limit [mbar]: Alarm Low"	The main vacuum pressure was lower than the lower bound for alarms. As long as this alarm is not committed, the module is blocked and no command can be executed. • Retry: Commits alarm and restarts monitoring. • Cancel: Commits alarm and blocks module until main vacuum pressure is higher than lowermost bound again. • Ignore: Commits alarm and unblocks module.

Exception	Reaction, recovery
Main vacuum pressure higher than "Vacuum limit [mbar]: Alarm High"	The main vacuum pressure was higher than the higher bound for alarms. As long as this alarm is not committed, the module is blocked and no command can be executed. • Retry: Commits alarm and restarts monitoring. • Cancel: Commits alarm and blocks module until main vacuum pressure is lower than uppermost bound again. • Ignore: Commits alarm and unblocks module.

Table 71 – Exceptions Component Media Supply

9.11.5 Interlocks

- The component cannot be used if the PLC is not connected to the control system.
- If the EFEM Media Supply component detects an vacuum error the handling on the robot will be blocked.

9.12 Performance Counters

This component is used for watching the free disk space, CPU usage and free memory on a PC. For more detailed information about "Performance Counters" see ToolCommander® - User manual chapter 7.2.

9.13 Process Module

9.13.1 Description

The Process Module component is connected via TCP/IP via XML protocol to the ToolCommander®. The following functions are used by the system control:

- Material transfer between process module and robot
- Recipe start for processing
- Transmission of status values

9.13.2 Parameters

Process Module 1 (Modules)

Module

Modules

☒ Installed

Communication

Port

12001

Print

Save

Close

Figure 56 – Process Module: Parameters

Parameter	Description	Range
Installed	Activates or deactivates the process module completely in the system control.	selection [yes, no]
Communication		
Port	Activates or deactivates the process module completely in the system control.	range [0..9999]

Table 72 – Parameters Component Process Module

9.13.3 Manual Operation

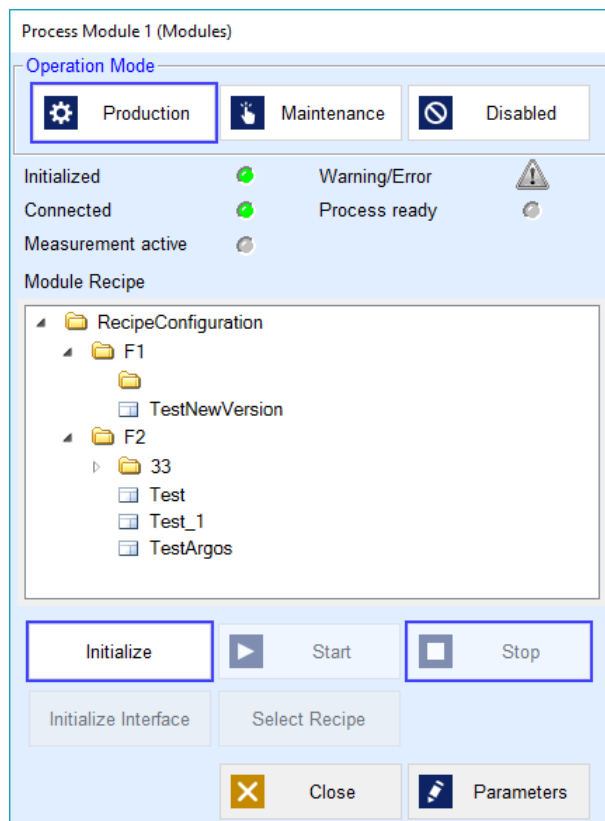


Figure 57 – Process Module: Manual Operation

Manual Operation		
States		
Initializes	This LED symbol indicates the initialization status of the process module.	
	 flashing	The process module was not initialized.
		The process module is initialized.
Measurement active	This LED symbol indicates whether the measurement is active in the process module.	
		No measurement is active.
		The measurement is active.

Warning/Error	This LED symbol indicates the error status of the process module.	
		The process module is not in error.
		The process module has at least one warning, but no error.
		The process module has at least one error.
Process ready	This LED symbol indicates whether the process module is ready for processing.	
		The process module is not ready for processing.
		The process module is ready for processing.
Recipe selection		
Module Recipe	This tree view contains all defined module recipes from which a recipe can be selected for inspection in the process module.	
Operation		
Initialize	Initializes the process module.	
Start	Starts the measurement in the process module.	
Stop	Interrupts the measurement in the process module.	
Initialize Interface	Initializes the interface to the process module. Triggers the service function "Initialize" of the XML interface.	
Select Recipe	Sends the module recipe selected in the recipe tree to the process module.	

Table 73 – Manual Operation Process Module

9.13.4 Messages

Exception	Reaction, recovery
Exceptions system control	
Error during read/write of database [xxx]	A database error occurred during a read/ write process. • Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Connection interrupted	The connection to the process module interface was interrupted, communication is no longer possible. As long as the error is active, no further commands are executed and the module is blocked. • Retry: Acknowledges the alarm and then attempts are made to re-establish the connection.
No response to command received	A command was not answered within a parameterized time. • Retry: Acknowledges the alarm and executes the associated command again. • Abort: Acknowledges the alarm and terminates the execution of the command.
Exceptions Process Modul	
Hardware alarms are displayed accordingly and can be confirmed in the controller. The definition of the alarms and the associated error acknowledgement options are reported by the process module via XML interface to the ToolCommander®.	

Table 74 – Messages Component Process Module

9.13.5 Interlocks

- Manual operations are only possible when PM is in maintenance mode.
- During Emergency Stop a process on the PM can not be started.

9.14 Service Lamp

The service lamp of the measuring platform can be switched on and off via the component. The signal for the lamp is set or reset via a B&R PLC that is coupled to the system via Modbus TCP/IP. A digital input signal indicates whether the service lamp is on.

In automatic mode (online remote with the host), a warning is displayed when the lamp is switched on.

9.14.1 Parameters

This component does not have any parameter.

9.14.2 Manual Operation

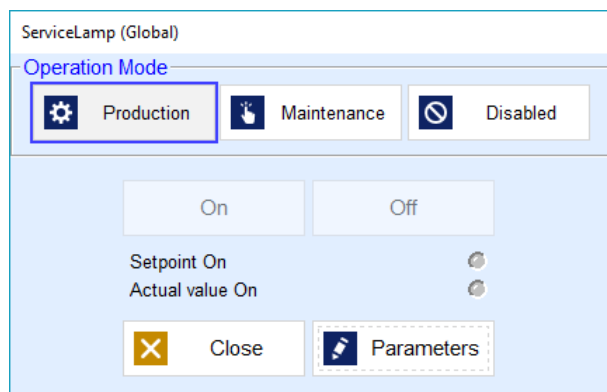


Figure 58 – Service Lamp: Manual Operation

Name	Description	Type
Operations		
On	Sets the signal of the service lamp to "1" (on).	Button [Event]
Off	Sets the signal of the service lamp to "0" (off).	Button [Event]
State Information		

Name	Description	Type
Setpoint On	An LED symbols shows the actual setpoint.	color change
		gray = switched off
		green = switched on
Actual Value On	An LED symbols shows the feedback signal from the PLC.	color change
		gray = switched off
		green = switched on

Table 75 – Manual Operation Service Lamp

9.14.3 Messages

This component does not have any message.

9.14.4 Interlocks

The component cannot be used if the connection to the PLC is interrupted.

9.15 Transfer Module

9.15.1 Description

The component Transfer Module is the component which is used to control the transport module. If the module component is not in production mode the scheduler will not use the module for transport job execution.

The tool communicates with the Yaskawa SR100 and MueTec specialized XML controller via TCP/IP.

9.15.2 Initialization

Only after restarting the controller does the Yaskawa SR100 robot check whether it still has material on its arms. Therefore, the controller of the robot is restarted after an emergency stop or after the failure of the servo drive before the robot is initialized. After reset of the controller a parametrized time period is waited (parameter: controller reset timeout) before the robot is initialized. During the control reset, a dialog is displayed with information about the current progress of the initialization and the status of the robot.

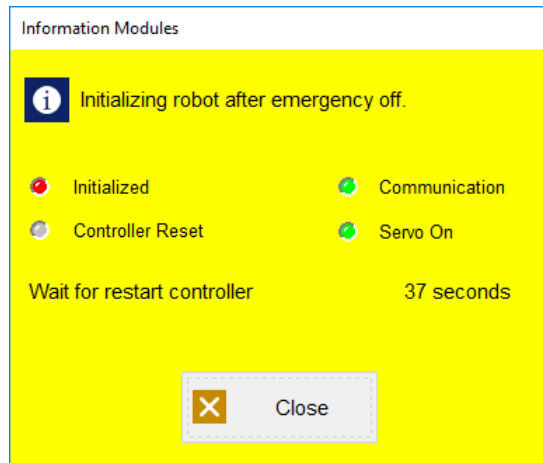


Figure 59 – Transfer Modul: Initialization after emergency stop

If no emergency stop was active or if the servo drive of the robot is not off, the robot is initialized without restarting the controller.

9.15.3 Parameters

Transfer Module1 (Modules)

Module Transfer Module1

Communication

TCP/IP Address 127.0.0.1

TCP Port 12020

Command Timeout [s] 120.0

State/Position Polling Enabled ☒

Polling Interval [s] 0.1

Use Sequence Number ☐

Use Checksum Verification ☐

Send End of Execution ☒

Read Mapping Reference Data ☐

Speed Level Level 1 - Fast

Hand Configuration

Lower Blade Installed ☐ Material Type Large Masks (> 6 Inch)

Upper Blade Installed ☒ Material Type Small Mask (6 Inch)

Controller Reset Timeout [s] 60

Placement Correction ☐

Mapper Installed ☒

Mapping Type ☒ Reflective ☐ Transmitted

Mapping Angle Loadport 1 [°] 0.0

Mapping Angle Loadport 2 [°] 0.0

Mapping Angle Loadport 3 [°] 0.0

 Speed Parameter ...

 Print  Close  Save

Figure 60 – Transfer Module: Parameters

Transfer Module2 (Modules)

Module

Communication

TCP/IP Address

TCP Port

Command Timeout [s]

Hand Configuration

Used Arm

 Print
  Close
  Save

Figure 61 – MueTec specialized XML Transfer Module: Parameters

Parameter	Description	Range
Communication		
TCP/ IP Host Name/ Address	TCP/ IP address for the communication with the robot.	string [0..255 characters]
TCP Port	Defines the port for the communication with the robot.	range [0..65535]
Command Timeout [s]	Defines a timeout for communication. If there is no answer for a request (command) within this time an error is generated.	range [0.1.. 3600.0]
State/Position Polling Enabled	Determines whether the device driver should send status and position queries to the robot hardware at a defined interval.	selection [yes / no]
Polling Interval [s]	Defines the interval at which the status/ command is to be sent to the robot hardware.	range [0.1.. 10.0]

Use Sequence Number	Defines whether sequence numbers are to be used for command telegrams to the robot or not. This parameter should be enabled, when the device is configured to use sequence numbers	selection [yes / no]
Use Checksum Verification	Defines whether check sum validations are activated or not. This parameter should be enabled, when the device is configured to use telegrams with check sum.	selection [yes / no]
Send End of Execution	Defines whether a command finished event from the robot should be acknowledged from the controller or not. If this parameter is set to false the robot will remain in busy state after sending a transfer or control command.	selection [yes / no]
Read Mapping Reference Data	Determines whether the reference data is read from the device and written to the log file after mapping.	selection [yes / no]
Speed Level	Defines the speed level for the robot movement.	selection [Level 1 - High, Level 2 - Middle, Level 3 - Low]
Hand Configuration		
Lower Blade Installed	Defines whether hand 1, the lower blade on the robot is installed.	selection [yes, no]
Upper Blade Installed	Defines whether hand 2, the upper blade on the robot is installed.	selection [yes, no]

Lower Blade Material Type	Defines the material type of the lower blade. This feature is only available for machines of type Spector 5500 .	selection [Small Masks 6 inch, Large Masks > 6 inch]
Upper Blade Material Type	Defines the material type of the upper blade. This feature is only available for machines of type Spector 5500 .	selection [Small Masks 6 inch, Large Masks > 6 inch]
Used Arm	Indicates which arm is being used. This feature is only available for machines of type Argos 150 HT .	display [Upper Arm, Lower Arm]
Settings		
Controller Reset Timeout [s]	Defines the time span in seconds how long the initialization routine waits after resetting the controller with initialization of the robot.	range [1.. 200]
Placement Correction	Defines whether the parameter "PlacementCorrectionParameter" from StationConfig is to be used.	selection [yes, no]
Mapping Type	Defines the mapper used on the robot. If "Transmitted Mapper" is selected, no wafer may be on the arm of the robot during the mapping procedure.	selection [Reflective, Transmitted]
Mapper Installed	Defines whether a mapper is installed or not. If this option is activated the robot will be used for mapping.	selection [yes, no]

Mapping Angle Loadport 1 [°]	Defines the mapping angle in ° for the mapping of the loadport 1. This parameter is only available when the mapping is done by the robot.	range [-20.0.. 20.0]
Mapping Angle Loadport 2 [°]	Defines the mapping angle in ° for the mapping of the loadport 2. This parameter is only available when the mapping is done by the robot.	range [-20.0.. 20.0]
Mapping Angle Loadport 3 [°]	Defines the mapping angle in ° for the mapping of the loadport 3. This parameter is only available when the mapping is done by the robot.	range [-20.0.. 20.0]
Speed Parameter ...	This buttons opens an dialog where the speed parameters of the robot can be reviewed and modified.	

Table 76 – Parameters Transfer Module

9.15.3.1 Speed Parameters

In the dialog Speed parameters the speed parameters of the robot can be modified. Changes can only be made by authorized service personnel.

Transfer Module (Modules): Speedparameter

Parameter ID	Description	Value	Robot Value	Default Value	Unit
R0000	Rotation axis transfer speed(level 1) for host (w/o wafer)	44	0	330	deg/sec
R0001	Rotation axis transfer speed(level 2) for host (w/o wafer)	33	0	165	deg/sec
R0002	Rotation axis transfer speed(level 3) for host (w/o wafer)	213	0	33	deg/sec
R0005	Rotation axis transfer speed(level 1) for host (w/ wafer)	330	0	330	deg/sec
R0006	Rotation axis transfer speed(level 2) for host (w/ wafer)	165	0	165	deg/sec
R0007	Rotation axis transfer speed(level 3) for host (w/ wafer)	24	0	33	deg/sec
R0010	Rotation axis transfer low speed(level 1) for host	25	0	33	deg/sec
R0011	Rotation axis transfer low speed(level 2) for host	26	0	33	deg/sec
R0012	Rotation axis transfer low speed(level 3) for host	27	0	33	deg/sec
R0015	Rotation axis Home speed(level 1) for host	28	0	33	deg/sec
R0016	Rotation axis Home speed(level 2) for host	29	0	33	deg/sec
R0017	Rotation axis Home speed(level 3) for host	30	0	33	deg/sec
R0020	Rotation axis Speed(level 1) in low speed area for host	40	0	16.5	deg/sec
R0021	Rotation axis Speed(level 2) in low speed area for host	16.5	0	16.5	deg/sec
R0022	Rotation axis Speed(level 3) in low speed area for host	16.5	0	16.5	deg/sec
R0030	Rotation axis transfer speed(low) for T.P (w/o wafer)	2.623	0	2.623	deg/sec
R0031	Rotation axis transfer speed(middle) for T.P (w/o wafer)	5.247	0	5.247	deg/sec
R0032	Rotation axis transfer speed(high) for T.P (w/o wafer)	10.493	0	10.493	deg/sec
R0033	Rotation axis transfer speed(low) for T.P (w/ wafer)	2.623	0	2.623	deg/sec
R0034	Rotation axis transfer speed(middle) for T.P (w/ wafer)	5.247	0	5.247	deg/sec
R0035	Rotation axis transfer speed(high) for T.P (w/ wafer)	10.493	0	10.493	deg/sec
R0036	Rotation axis Transfer speed(low) for T.P	2.623	0	2.623	deg/sec
R0037	Rotation axis Transfer speed(middle) for T.P	2.623	0	2.623	deg/sec
R0038	Rotation axis Transfer speed(high) for T.P	2.623	0	2.623	deg/sec
R0039	Rotation axis Home speed(low) for T.P	2.623	0	2.623	deg/sec
R0040	Rotation axis Home(middle) speed for T.P	2.623	0	2.623	deg/sec
R0041	Rotation axis Home speed(high) for T.P	2.623	0	2.623	deg/sec
R0042	Rotation axis Speed(low) in low speed area	1.311	0	1.311	deg/sec
R0043	Rotation axis Speed(middle) in low speed area	1.311	0	1.311	deg/sec

Figure 62 – Dialog Speed Parameters

Name	Description	Range
Send to Robot	If this button is pressed, the actual values of the speed parameters set up in the column <i>Value</i> will be send to the robot. Attention: The servo drives will be switched off during the parameter update. The button is enabled, when the user has the right "Change parameters", the robot is connected, and when there is no active transfer handling and no wafer one the robot.	

Name	Description	Range
Load Robot Values	If this button is pressed, the actual speed values from the robot are read out. They are displayed in the Robot Value column. The button is enabled, when the user has the right "Change parameters" and the robot is connected.	
Use Robot Values	When this button is pressed, the value in the Value column is overwritten with the value in the Robot Value column. Changes will be only stored in the database when the Save button was pressed.	
The speed parameter table has following columns:		
Parameter ID	In this column the robot internal parameter id e.g. "R0200" is displayed.	
Description	In this column the description of the robot internal parameter is displayed. See parameter table of the Yaskawa SR100 Controller for further details.	
Value	In the column the parameter value can be modified. Changes will be only stored in the database when the Save button was pressed.	

Name	Description	Range
	The value can be only modified when the user has the right "Change parameters".	
Unit	In this column the unit of the parameter is displayed.	mm/sec, deg/sec
Default Value	In this column the default value of the parameter is displayed. See parameter table of the Yaskawa SR100 Controller for further details.	
Actions		
Close	This button closes the dialog without saving the changes.	
Save	This saves the changes. The button is enabled when the user has the maintenance right.	

Table 77 – Parameters Component Roboter- Speed Parameters

9.15.4 Manual Operation

Transfer Module (Modules)

Operation Mode

☒ Production
 ☒ Maintenance
 ☐ Disabled

Commands

Command: Transfer Material End Effector: Lower Arm
 Source Module: Loadport 1 Target Module: Process Module 1
 Source Slot: Slot 1 Target Slot: Module Slot

Service Commands

Sequence: _____ State: _____

States

Homed ☒
 Safe Position ☒
 On Home Position ☒
 Unit Ready ☒
 Motion or Position Error ☐
 Servo On ☒
 Controller Battery State ☒
 Manipulator Battery state ☒
 Controller Reset ☐

Last Stage: Station: Home Slot:

Occupation Upper Blade: Empty

Occupation Lower Blade: Empty

Robot Commands

Figure 63 – Transfer Module: Manual Operation

Transfer Module2 (Modules)

Operation Mode

☒ Production
 ☒ Maintenance
 ☐ Disabled

Commands

Command: Map Carrier End Effector: _____
 Source Module: _____ Target Module: _____
 Source Slot: _____ Target Slot: _____

Service Commands

Sequence: _____ State: _____

States

Initialized ☒
 Safe Position ☒
 Active ☐

Last Stage: _____

Occupation Upper Blade: Empty

Robot Commands

<VctcComponentRequest ComponentName="Transfe

Figure 64 – MueTec specialized XML Transfer Module: Manual Operation

Manual Operation

Commands

Command	This combo box specifies the command that will be executed when the user clicks on "Execute" button.	
	• Pick Material Necessary parameters are end effector, source module and slot.	Picks wafer from the specified module and slot using the specified arm.
	• Place Material Necessary parameters are end effector, target module and slot.	Places wafer to the specified station and slot using the specified arm.
	• Transfer Material Necessary parameters are end effector, source station, source slot, target station and target slot.	Transfers a wafer from the specified source station and slot to the specified target station and slot using the specified arm.
	• Exchange Necessary parameters are end effector, target module and target slot	Picks with the free arm a wafer from the selected target module and target slot. The wafer from the selected arm is placed after pick on the selected target module and slot.
	• Home	Activates the home procedure of the robot.
	• Map Carrier	The mapping command can be executed on the selected target module. This command is only available when the robot supports the mapping of the loadport.

	• Move Material	Corrects the material tracking mismatch. If a sensor detects material but in the material tracking system does not know a wafer at this position, the wafer can be moved without handling.
Source Module	This combo box selects the source module.	
Source Slot	This combo box selects the source slot. The number of slots depends on the selected source module.	
End Effector	This combo box selects the arm to be used for next command.	
Target Module	This combo box selects the target module.	
Target Slot	This combo box selects the target slot. The number of slots depends on the selected target module.	
Execute Command	This button starts the selected command using selected values for End Effector, modules and slots.	
States		
Homed	This LED symbol indicates robot home state.	
	 blinking	The robot was not referenced. The execution of an initialize command is required before further commands can be executed.
		Robot is homed and ready for use.
Safe Position	This LED symbol indicates whether the robot is in a safe position.	
		Robot is not in a safe position.
		Robot is in a safe position.

On Home Position	This LED symbol indicates whether the robot is in on home position.	
		Robot is not on home position.
		Robot is on home position.
Unit Ready	This LED symbol indicates whether the robot is ready.	
		Robot is not ready.
		Robot is ready.
Motion or Position Error	This LED symbol indicates whether the robot detected an error.	
		Robot is in error state.
		No error detected.
Servo On	This LED symbol indicates whether the servo is switched on.	
		Servo is switched off.
		Servo is switched on.
Controller Battery state	This LED symbol indicates the controller battery state.	
		State not OK.
		State OK.
Manipulator Battery state	This LED symbol indicates the manipulator battery state.	
		State not OK.
		State OK.
Controller Reset	This LED symbol indicates the actual state of the output signal for the external controller reset.	

		The output signal of the controller reset channel is set to low.
		The output signal of the controller reset channel is set to high.
Last Stage	This text field shows the station and the slot of the last transfer command.	
Occupation Upper Blade	The textbox indicated whether the upper blade is currently empty or has a wafer detected.	
Occupation Lower Blade	The textbox indicated whether the lower blade is currently empty or has a wafer detected.	
Active	This LED symbol indicates whether a command is currently active.	
		No command is active
		A command is active.
Service Commands		
Initialize	This button brings the robot in the Home position. Running this command is required before any other command execution is possible.	
Reset Controller	This button triggers the reset signal of the controller for two seconds. The transfer module need to be initialized after controller reset.	
Sequence		
Sequence	Shows the actual command the robot is executing.	
State	Shows the state of the command the robot is executing.	
Robot Commands	This button executes the entered command and displays it in the area below if the command is valid. This button is available only in the maintenance mode.	

Table 78 – Manual Operation Transfer Module

9.15.5 Messages

Exception	Reaction, recovery
Error during read/write of database [xxx]	A database error occurred during a read/ write process. • Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Connection lost	Connection to robot has been lost. If the error is active the transport module is blocked (active transports will be finished). • Retry: restarts communication.
Communication timeout	This error is raised if a command timed out. If the error is active the transport module is blocked (active transports will be finished). • Acknowledge: restarts communication
A timeout occurred during command execution	The system raises this error if an transfer command timed out. If the error is active the transport module is blocked. • Retry: repeats the transfer command. • Abort: aborts the transfer of the wafer. The robot is not blocked and usable for future. • Ignore: wafer is marked as transferred and the process continues (robot will not be disabled for further processing)

Command execution failed. Command='[%Command%]' was responded with level 1 warning = ' [%ErrorDescription%]'	The robot responds to a command (%Command%) with a warning level 1 error (Communication related error). The failure code and the text are displayed in the %ErrorDescription%. • Abort: aborts the command. The robot is not blocked and usable for future.
Command execution failed. Command='[%Command%]' was responded with level 2 warning = ' [%ErrorDescription%]'	The robot responds to a command (%Command%) with a warning level 2 error. The failure code and the text are displayed in the %ErrorDescription%. • Retry: repeats the command. • Abort: aborts the command. The robot is not blocked and usable for future. • Ignore: command is marked as executed. The process continues (robot will not be disabled for further processing)
Command execution failed. Command='[%Command%]' was responded with level 1 error = ' [%ErrorDescription%]'	The robot responds to a command (%Command%) with a level 1 error. The failure code and the text are displayed in the %ErrorDescription%. • Retry: repeats the command. • Abort: aborts the command. The robot is not blocked and usable for future. • Ignore: command is marked as executed. The process continues (robot will not be disabled for further processing)
Command execution failed. Command='[%Command%]' was responded with level 2 error = ' [%ErrorDescription%]'	The robot responds to a command (%Command%) with a level 2 error. The failure code and the text are displayed in the %ErrorDescription%. • Abort: aborts the command. The robot is not blocked and usable for future.

Command execution failed. Command='[%Command%]' was responded with fatal error =[%ErrorDescription%]	The robot responds to a command (%Command%) with a fatal error. The failure code and the text are displayed in the %ErrorDescription%. • Abort: aborts the command. The robot is not blocked and usable for future.
An unknown sequence [%SequenceName%] was requested for execution	The transfer module can not execute the sequence with the name %SequenceName%, because it does not exist. The sequence will be aborted.
Creating the sequence [%SequenceName%] failed due to invalid parameters	The transfer module can not execute the sequence with the name %SequenceName%, because the parameters are invalid. The sequence will be aborted.
Timeout while locking stations [%StationName%]	The transfer module can not execute the transfer sequence because the station %StationName% can not be locked for the transfer.
Source slot [%SlotNumber%] at module [%ModuleName%] is not occupied	The transfer module can not execute a pick, exchange or transfer sequence because the material tracking could not detect a material at the source slot (%SlotNumber%) of the module (%ModuleName%).
Target slot [%SlotNumber%] at module [%ModuleName%] is not empty	The transfer module can not execute a place or transfer sequence because the material tracking detects a material on the target slot (%SlotNumber%) of the module (%ModuleName%).
The lower arm of the robot is disabled	The transfer module can not execute a transfer sequence with the lower hand because the it is disabled.
The upper arm of the robot is disabled	The transfer module can not execute a transfer sequence with the upper hand because the it is disabled.

Material not compatible to target slot	The transfer module can not execute a transfer sequence because the target slot is not configured for the actual material.
Robot is not in a safe position. The robot needs to be homed to continue.	The transfer module can not execute a transfer sequence because the robot is not in a safe position • Home: Sets the robot to the home position (safe position) and continues the transfer sequence.
No wafer detected after pick.	After pick, the sensors on the robot does not detect a wafer. • Acknowledge: Clears the error.
Wafer detected after place.	After place, the sensors on the robot still detect a wafer. • Acknowledge: Clears the error.
Creating the sequence [%SequenceName%] failed because the station id for robot communication could not be determined.	The transfer module can not execute the sequence with the name %SequenceName%, because a station id for the station could not be found.
Creating the sequence [%SequenceName%] failed because the station id for robot communication could not be determined due to multiple involved material types.	The transfer module can not execute the sequence with the name %SequenceName%, because a station id for the station could not be found for mixed materials.
Sensor detected material on [%Arm%], but no material is known in tracking system.	The robot sensors detect material on one hand (%Arm%), but the tracking system is not aware of any material. The transport is blocked. To solve this error create material in the Production Dialog .
Sensor detected no material on [%Arm%], but the tracking system has a material on this position.	The robot sensors does not detect a material on hand (%Arm%), but material is known in the tracking system. The transport is blocked. To solve this error delete the material in the Production Dialog .

Other robot arm [%Arm%] is not empty.	The transfer module can not execute a exchange sequence, because the pick hand (%Arm%) is not empty.
Robot Initializing	This message is shown when the initializing sequence of the robot is active.
Error while locking stations [%StationName%]	The transfer module cannot execute a transfer sequence because the station (%StationName%) cannot be locked. • Acknowledge: The sequence will be aborted.
Please ensure that the tool is safe to release handling interlocks and that material information in the ToolCommander is correct. Triggering the 'Release' action will resume operation of the tool.	This message is shown when a sequence fails and the error was acknowledged. • Release: Clears the error. The next transfer sequences can be executed.
Mapping of loadport [%LoadPortName%] is not completed, material transport not possible.	The transfer module can not execute a transfer sequence to or from a loadport module (%LoadPortName%), because the slot map is unknown.
Robot is not initialized.	The robot is not initialized. The transfer module can not execute any transfer sequence. To solve this error start the initializing sequence on the robot.
Process in Module [%ModuleName%] was aborted.	A transfer sequence was aborted by a module (%ModuleName%).
A wafer detected at source slot after pick.	After a pick sequence the material sensor still detects material at the source slot. • Acknowledge: Clears the error. The sequence will be aborted.
No wafer detected at target slot after place.	After a place sequence the material sensor does not detect material at the target slot. • Acknowledge: Clears the error. The sequence will be aborted.

Wafer absence error lower arm. Robot is blocked for wafer transfer. Please correct the position of the wafer manually and release the robot.

During a pick sequence, the material sensor on the lower end effector loses the material presence information. The position of the wafer can only be corrected by the operator in the machine.

- Release Robot: Opens a dialog for the transfer release of the robot. The chapter [Transfer Modul Wafer Absence Error](#) describes in details the functions of this dialog

Wafer absence error upper arm. Robot is blocked for wafer transfer. Please correct the position of the wafer manually and release the robot.

During a pick sequence, the material sensor on the upper end effector loses the material presence information. The position of the wafer can only be corrected by the operator in the machine.

- Release Robot: Opens a dialog for the transfer release of the robot. The chapter [Transfer Modul Wafer Absence Error](#) describes in details the functions of this dialog

Nack received: the [%Command%] command is invalid or malformed

The [%Command%] was received by the component, but has invalid parameters.

- Abort: cancels the command

Abort received: the device is unable to perform the [%Command%] command at this time because: [%Reason%]

The [%Command%] was validly received by the component, but the component aborted the execution of the command. The abort can be justified by [%Reason%].

- Retry: retries the command
- Ignore: cancels the command and proceeds with the next command
- Abort: cancels the command

[%Command%] command timed out

No valid response was received to the [%Command%].

- Retry: retries the command

	• Ignore: cancels the command and proceeds with the next command • Abort: cancels the command
--	---

Table 79 – Messages Transfer Module

9.15.6 Interlocks

- Transports with transfer module are only possible when no Emergency Off is active.
- If the robot is not initialized, the loadports cannot be initialized.
- Transports are only possible if there is material in the machine.
- If the EFEM Media Supply component detects an vacuum error the handling on the robot will be blocked.
- Transport with the transport module is locked to or from a module if the source or destination module is in an error state, locked, not in automatic mode, or not initialized.

9.15.7 Wafer Absence Error

If the robot reports a "wafer absence error" (error code 0x2BA0 for the lower arm, error code =0x2BB0 for the upper arm) during the removal of a wafer from the load port, the message "Wafer absence error upper arm" or "Wafer absence error lower arm" is displayed in the message system. This error can only be corrected by manually positioning the wafer on the arm or in the carrier. As long as the error is not corrected, the robot is locked for transport.

Via the error recovery action "Release Robot" a dialog for the transport release of the robot is displayed. The robot can only be released by an operator with the right "manual operation".

Transport Release Robot

i Transfer blocked after wafer absence error. Correct the material position manually inside the machine and release the robot.

Robot State Information

☒ Initialized
 ☐ Unit Ready

Occupation Upper Blade

Occupation Lower Blade

Robot Commands

Module

End Effector

☐

Material in Error

Wafer ID

Source Position

Current Position

Target End Effector

Figure 65 – Transfer Modul: Dialog Transport Release Robot

The following functions are available in the dialog for releasing the robot for transport:

Robot State Information		
Initialized	This LED symbol indicates initialized state of the robot.	
		The robot is initialized.
		The robot is not initialized.
Unit Ready	This LED symbol indicates whether the robot is ready.	
		Robot is not ready.
		Robot is ready.

Occupation Upper Blade	The textbox indicated whether the upper blade is currently empty or has a wafer detected
Occupation Lower Blade	The textbox indicated whether the lower blade is currently empty or has a wafer detected
Robot Commands	
Module	Über eine Auswahlliste kann ein Loadport ausgewählt werden.
End Effector	Über eine Auswahlliste kann ein End-Effektor ausgewählt werden.
Move Endeffector to Get Ready Position	This button sends the command to move the robot arm to the robot. The selected end effector is moved to the Get-Ready position of slot 25 of the selected loadport. To the left of the button, an LED icon indicates the current status of the command execution.
	 No command is executed or the command is terminated.
	 The command is executed.
Material in Error	
Wafer-ID	Display the wafer ID of the material where the wafer absence error occurred.
Source Position	The original position of the material in the error. Display of the load port and the slot from which the material should be removed.
Current Position	The current position of the material in the error.
Target End Effector	The robot arm with which the material should be removed from the carrier.
Material On Source	This button can be used to set the material in the material tracking to its original position. If the current position corresponds to the original position, the button cannot be used.

Material On Robot	Use this button to set the material in the material tracking to the position of the target end effector. If the current position corresponds to the position of the target end effector, the button cannot be operated.
Release	This button resets the material error and releases the robot for transport.
Release	This button closes the dialog.

Table 80 – Dialog Transport Release Robot

9.16 Camera Aligner

9.16.1 Description

This component communicates with the Camera Aligner. For the tool configuration **Spector 5500** the material offsets are read via a Keyence-XGX reading device. For the tool configuration **Argos 150 HT**, a measurement is performed via the MueTec specialized XML Camera Aligner.

9.16.2 Parameters

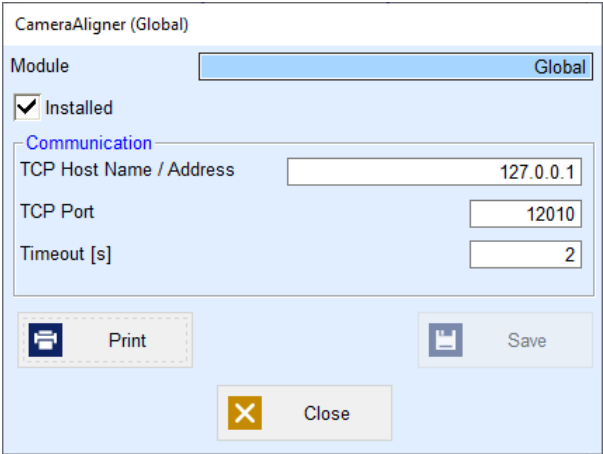


Figure 66 – Camera-Aligner: Parameters

Parameter	Description	Range
Installed	Activates or deactivates the Camera-Aligner.	selection [yes / no]

TCP Host Name/Address	The TCP/IP address of the Camera-Aligner.	string [0..255 characters]
TCP Port	Die TCP/IP-Port number.	range [0..65535]
Timeout [s]	Defines the communication timeout in milliseconds. If a command is not answered in the specified time a error will be displayed.	range [0..35791394]

Table 81 – Parameters Camera-Aligner

9.16.3 Manual Operation

CameraAligner (Global)

Operation Mode

☒ Production
 ☒ Maintenance
 ☐ Disabled

Operation

Material Offsets

End Effector: Upper Arm

Material Name: Mask 6 Inch Slot1

Reading Results

Name	Value
X-Offset	70.20000
Y-Offset	80.20000
Theta-Offset	60.20000

Sequence: Read Material Offsets State: Active

Figure 67 – Camera-Aligner: Manual Operation

Manual Operation

Operation

Stop	Stops the current operation.
Initialize	This button initialize the camera aligner.
Material Offsets	
End Effector	Selection of the end effector with material which should be measured. If the end effector contains material the name of the material will be displayed in the field "Material Name".
Material Name	Displays the name of the material on the selected end effector.
Read Material-Offsets	Triggers the reading of the material offsets.
Read results	Displays the read material offsets.
Sequence	
Sequences	Shows the actual command the camera aligner is executing.
State	Shows the state of the command the camera aligner is executing.

Table 82 – Manual operation Camera-Aligner

9.16.4 Messages

Following table contains the messages and the description of their recovery actions which the component sends to the message server of the ToolCommander® framework. If the component is not installed this messages will not be send.

Exception	Reaction, recovery
Error during read/ write of database [xxx]	A database error occurred during a read/ write process. • Retry: restarts the last database operation.

	Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Invalid IP address configured	The configured IP address string is empty or not valid. • Retry: restarts communication. • Abort: aborts the reading.
A timeout occurred during command execution	The system raises this error if a command timed out. • Retry: repeats the command. • Abort: aborts the command execution. • Ignore: continues with the next command
Command [%Command%]. Received unknown error code = [n] .	The system raises this error if a command was responded by the device with an unknown error code. • Abort: aborts the command execution.
Command [%Command%] failed. Error code not set	The system raises this error if a command was responded by the device with an error, but no error code was set. • Abort: aborts the command execution.
Nack received: the [%Command%] command is invalid or malformed	The [%Command%] was received by the component, but has invalid parameters. • Abort: cancels the command
Abort received: the device is unable to perform the [%Command%] command at this time because: [%Reason%]	The [%Command%] was validly received by the component, but the component aborted the execution of the command. The abort can be justified by [%Reason%]. • Retry: retries the command • Ignore: cancels the command and proceeds with the next command

	• Abort: cancels the command
[%Command%] command timed out	No valid response was received to the [%Command%]. • Retry: retries the command • Ignore: cancels the command and proceeds with the next command • Abort: cancels the command

Table 83 – Messages Camera-Aligner

9.16.5 Interlocks

- The component cannot be operated if no camera aligner is installed and if the camera aligner device is not connected.

9.17 Flipper

9.17.1 Description

Diese Komponente kommuniziert mit dem Flipper.

Bei der Maschinenausprägung **Spector 5500** werden Masken mit dem Flipper-Gerät der Firma Fortrend, gedreht.

9.17.2 Parameters

Flipper (Global)

Module

Global

☒ Installed

Communication

TCP Host Name / Address

127.0.0.1

TCP Port

12011

Timeout [s]

1

Print

Save

Close

Figure 68 – Flipper: Parameters

Parameter	Description	Range
-----------	-------------	-------

Installed	Activates or deactivates the Flipper.	selection [yes / no]
TCP Host Name/Address	The TCP/IP address of the Flipper.	string [0..255 characters]
TCP Port	The TCP/IP-Port number.	range [0..65535]
Timeout [s]	Defines the communication timeout in milliseconds. If a command is not answered in the specified time a error will be displayed.	range [0..35791394]

Table 84 – Parameters Flipper

9.17.3 Manual Operation

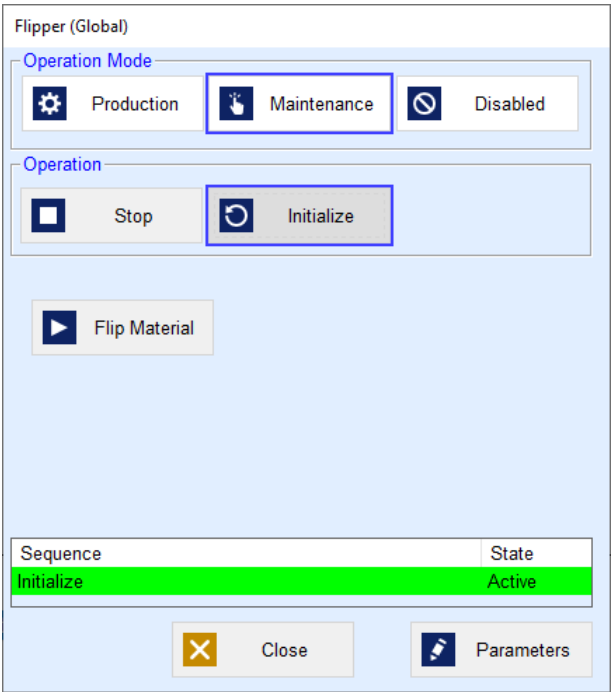


Figure 69 – Flipper: Manual Operation

Manual Operation	
Operation	
Stop	Stops the current operation.

Initialize	This button initialize the flipper device.
Flip Material	Triggers the flipping of the material.
Sequence	
Sequences	Shows the actual command the flipper is executing
State	Shows the state of the command the flipper is executing.

Table 85 – Manual operation Flipper

9.17.4 Messages

Following table contains the messages and the description of their recovery actions which the component sends to the message server of the ToolCommander® framework. If the component is not installed this messages will not be send.

Exception	Reaction, recovery
Error during read/ write of database [xxx]	A database error occurred during a read/ write process. • Retry: restarts the last database operation. Text [xxx] contains an additional SQL-Server message.
No parameters found	The component was not parametrized yet.
Invalid IP address configured	The configured IP address string is empty or not valid. • Retry: restarts communication. • Abort: aborts the reading.
A timeout occurred during command execution	The system raises this error if a command timed out. • Retry: repeats the command. • Abort: aborts the command, and the following commands.

- | | |
|--|---|
| | <ul style="list-style-type: none">• Ignore: continues with the next command |
|--|---|

Table 86 – Messages Flipper

9.17.5 Interlocks

- The component cannot be operated if no flipper device is installed and if the flipper device is not connected.
- The transfer system cannot place material at the flipper station if the flipper is not in a safe position.

10

Process Module Recipes

10 Process Module Recipes

10.1 General

The module recipes are stored as recipe files in a drive. The root directory of the recipes can be configured (see [Configuration General](#)). The administration of the recipe files takes place via the ToolCommander®. The metadata (creator, editor, modification date, directory structure) are stored in the file "RecipeConfiguration" in the root directory of the recipes. The process module recipes are saved as XML files in a subdirectory structure that can be changed by the user.

The recipe editor developed by MueTec and the dialog for comparing recipes are integrated into the process module editor.

10.2 Recipe Editor

The process module recipe editor is divided into three areas:

- **All Recipes** (top right): Displays the recipe tree with the directory structure and the recipe files.
- **Recipe Versions** (bottom right): Display a list with the version history of a selected recipe.
- **Button sidebar** (left): Buttons with functions for managing recipe files and editing recipes.

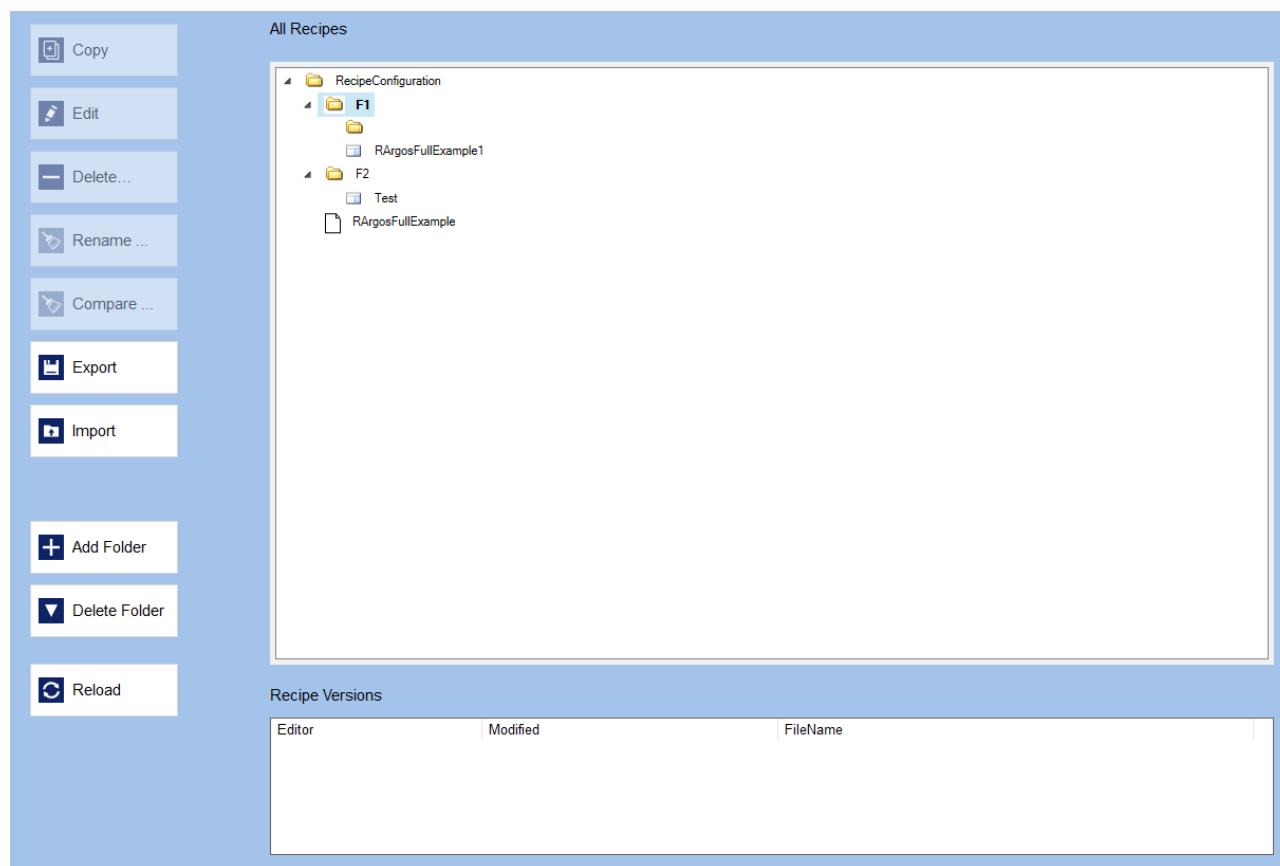


Figure 70 – Visualization Module Recipe Editor

Note: The name of the recipe + the path structure must not be longer than 90 characters.

Functions Module Recipe Editor	
Copy	This button is used to copy an existing recipe as base for a new recipe. The button can only be operated if a recipe is selected in the recipe tree.
Edit	This button opens a dialog for editing the recipe. The recipe editor is implemented by the MueTec (for further details see the operation manual from MueTec). The button can only be operated if a recipe is selected in the recipe tree.
Delete	This button is used to delete a recipe. The button can only be operated if a recipe is selected in the recipe tree.

Functions Module Recipe Editor

Rename

This button is used to rename a recipe. The button can only be operated if a recipe is selected in the recipe tree.

Compare

This button displays a dialog for comparing two recipe versions. The recipe comparison dialog is implemented by the MueTec (for further details see the operation manual from MueTec).

The button can only be operated if two versions of a recipe are selected in the recipe version history.

Export

This button can be used to export a recipe or a folder. If a recipe is selected, only the recipe is exported. If a directory is selected, all subdirectories and all recipes under the selected directory and subdirectories are exported.

The export path is defined in the MueTecApplication.ini file under the "VisuApplication" section in the "RecipeExportPath" parameter.

During the export itself, a new directory is created below the export path. The name of the directory is determined by the date and time of the export. The directory always contains the file "RecipeConfiguration.xml". This contains the structure of the exported data. This file must be selected for the import. The corresponding recipes are located directly next to it under the "RecipeConfiguration" directory.

Import

This button allows you to import previously exported recipes and directories into the recipe system. After pressing the button, a dialogue opens in which the recipe structure to be imported can be selected. The "RecipeConfiguration.xml" file should always be loaded from an export for the import.

Functions Module Recipe Editor	
Add Folder	This button is used to create a new directory. The button can only be operated if a directory is selected in the recipe tree.
Delete Folder	This button is used to delete a directory. The button can only be operated if a directory is selected in the recipe tree.
Reload	After pressing the button, the recipe view is reloaded. If there is a red exclamation mark next to the button, it is recommended to update the view. Note: After editing the recipes from this dialogue, the recipe view is automatically updated. The red exclamation mark only appears if recipes have been added, changed or deleted by the host or the automatic synchronisation of the recipe files.

Table 87 – Module Recipe Editor functions

